

Supplementary Information for: How do extreme weather events affect fishing operations? Knowledge engineering a risk analysis with Swedish Baltic Sea fishers

Contents

Elicitation of parameters from fishers. Interviews treatment and integral text	2
Scoring	2
Interpretation	3
Interviews - integral text (English)	9
Interviewee: I 1	9
Interviewee: I 2	25
Interviewee: I 3	39
Interviewee: I 4	54
Interviewee: I 5	69
Interviewee: I 6	85
Interviewee: I 7	100
Treatment of fisheries statistics available in European and Swedish repositories	117
Data sources by fishing style	117
Fishing gears	119
Fishing target	122
Economic data	125
Conditional probability tables	132
Network structure	133
Algebraic methods for Ranked nodes	134
Decision nodes	135
Chance nodes - Operational	136
Chance nodes - User	145
Chance nodes - Economic	150
Chance nodes - Individual	156

Chance nodes - Societal	161
Risk	166

Elicitation of parameters from fishers. Interviews treatment and integral text

Scoring

The interview notes were reviewed to identify quantitative scores applying following criteria (in this ordered):

- *C1 (score available)*: When a clear and unique self-declared score was provided, this was used as a score. In case the interviewee provided two scores (e.g.: 4 or 5), it was used the average (4.5). For those answers were alternative were provided for gear/target, create separate scoring.
- *C2 (score to be interpreted)*: Answers that were providing meaningful text but not a clear number, or that were considered worth of further attention, were scored as 999.
- *C3 (does not apply)*: Answers that were missing because the event/condition does not apply were scored as -999. When the “does not apply” was identified in more than 50% of the answers, the combination of fishing style-event was excluded from further analysis.
- *C4 (can't answer)*: Answers that were missing because people did not have a personal experience to refer but refers to an event that is relevant (i.e.: if during a storm you never go out, you might don't know about how catchability changes) were scored as -998.
- *C5 (garbled)*: Answers that were not coherent with the question were scored -997

Table S.1: coding system used to categorise answers according to their interpretation

Score	Type	Description
x	C 1	Number (x) available
999	C 2	To be interpreted
-999	C 3	Does not apply
-998	C 4	Cant answer
-997	C 5	Garbled

The questionnaires originated more than 500 scores. Of these, 267 enabled clear quantitative scoring, whereas 105 answers indicated clear information but no quantitative score and required interpretation. A total of 158 answers indicated that the combination of extreme events and questions was not relevant (123) for that question or there was not enough information to answer (35). Only 5 answers were considered incoherent with the question (e.g.: Supplementary Table A.165), and these were excluded from the analysis.

Table S.2: number of answers by interpretation category

Score	Frequency
x	274
-999	118
999	102
-998	38
-997	4

Interpretation

A structured coding approach was applied to those answers scored as 999 using both human expertise and artificial intelligence assistance.

- Text for unclear answers was revised to extract the main keywords (i.e.: most people say; I think it does not affect so much etc.), and to identify whether the answer was needed to support narrative or to provide quantitative information.
- The keywords were pooled based on the expected type of answer (e.g.:). Three researchers independently assigned quantitative scores to textual descriptions based on predefined LIKERT scale. These expert based ratings were averaged to create a consensus dataset, which was used to establish interpretive patterns linking descriptive terms to quantitative values.
- Answers without keywords needed for quantitative information were reviewed with Artificial Intelligence. We employed OpenAI’s GPT-5 (ChatGPT, September 2025 version) as an assistant to generalize these patterns to new responses. The model was not trained on external data but was guided by the team’s own scoring framework. For each new description, the model produced a suggested numerical score along with a written rationale. All outputs were reviewed and validated by the researchers to ensure consistency with the established coding scheme
- Answers without keywords that were needed to support narrative (i.e.: management) were left unscored.

Expert scoring

Expert scoring of keywords produced the following results, which were used to assign scores to 87 answers:

Table S.3: scores assigned by experts to answers not providing an explicit scoring

question	coding	answer	Exp1	Exp2	Exp3	mean	sd
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	hard	2	2	2	2	0.0
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	easier	3	4	4	4	0.6
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	to some extent	3	4	3	3	0.6
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	no	1	1	1	1	0.0
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	not difficult	4	3	5	4	1.0
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	it is possible	3	4	4	4	0.6
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	probably not entirely impossible	2	2	2	2	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	you can’t [do something]	1	1	1	1	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	yes but [in a different way]	3	3	3	3	0.0

is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	it has a big negative impact	1	1	2	1	0.6
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	still possible [to do something]	3	3	3	3	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	most do not [do something]	no info	2	2	2	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	impossible [to do something]	1	1	1	1	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	nothing can be done	1	1	1	1	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	we can do little	2 or 4, depending on Swedish wording (we can do little vs we can do a little)	2	3	2	0.7
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	I would try to	no info	3	4	4	0.7
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	we wait	no info	1	??	1	NA
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	sometimes	3	3	3	3	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	i'll do it anyway	no info	4	4	4	0.0
is it [something] different when [event] happens	1-5; from 1 – very similar, to 5 – very different	very similar	1	1	1	1	0.0
is it [something] different when [event] happens	1-5; from 1 – very similar, to 5 – very different	small difference	2	2	2	2	0.0
is it [something] different when [event] happens	1-5; from 1 – very similar, to 5 – very different	it may be different	4	4	3	4	0.6
is it [something] different when [event] happens	1-5; from 1 – very similar, to 5 – very different	it has an effect	no info on direction	3	4	4	0.7
is it [something] different when [event] happens	1-5; from 1 – very similar, to 5 – very different	it affects	no info on direction	3	4	4	0.7
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	it is harder to do	2	2	2	2	0.0
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	there is a marginal negative effect	2	2	2	2	0.0
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	it is more difficult	2	2	2	2	0.0

does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	it get worse	2	2	2	2	0.0
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	there is a little negative impact	2	2	2	2	0.0
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	not so big impact	3	2	3	3	0.6
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	a little worrisome	no info	2	2	2	0.0
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	it certainly has a negative effect	2	1	1	1	0.6
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	is good	4	4	4	4	0.0
how much is [something] important	1-5; from 1 – not important at all, to 5 – very important	fairly important	3	3	3	3	0.0
how much is [something] important	1-5; from 1 – not important at all, to 5 – very important	not a strong link	2	2	2	2	0.0
how much is [something] important	1-5; from 1 – not important at all, to 5 – very important	it has a real significance	5	4	5	5	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	it has great effect	5	4	5	5	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	it is big effect	5	4	5	5	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	it may have an effect	3	2	3	3	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	it has big impact	5	4	5	5	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	Quite small	2	2	2	2	0.0
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	not so much	2	2	2	2	0.0
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	not really	1	1	1	1	0.0
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	it does	3	2	3	3	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	quite a lot	4	4	4	4	0.0

does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	some	2	3	3	3	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	some, but less than other	2	2	2	2	0.0
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	greatly	4	5	4	4	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	most do not	2	2	2	2	0.0
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	no	1	1	1	1	0.0
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	improve	4	3	4	4	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	a lot	4	5	5	5	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	it's strong	5	5	4	5	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	pretty much	4	4	4	4	0.0
How much concern for your personal safety	0-2; 0 – no concern, 1 – personal injuries, 2 – life-threatening	big concern	no info bc difference between 1 & 2 is type of concern, not level alone	2	2	2	0.0
How much concern for your personal safety	0-2; 0 – no concern, 1 – personal injuries, 2 – life-threatening	some risk	no info bc difference between 1 & 2 is type of concern, not level alone	1	1	1	0.0
How much concern for your personal safety	0-2; 0 – no concern, 1 – personal injuries, 2 – life-threatening	worrisome	no info bc difference between 1 & 2 is type of concern, not level alone	1	1	1	0.0
does [something] allows you to change location	0-2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible	sometimes, but not always	1	1	1	1	0.0
does [something] allows you to change location	0-2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible	it allows a lot of freedom	2	2	2	2	0.0

AI scoring

This option was applied to a limited number of responses.

Q1

Question: How much do you think they vary among the group of people you represent? Possible answers: 1–5; from 1 – not at all, to 5 – very much

Observed Answer: it is so personal, some do not care at all, they just drink beer all day; we discuss things like this all the time, most are probably quite stressed by such situations

Score: 4

Why: The reply indicates clear heterogeneity — a minority who “do not care at all” vs a majority who are “quite stressed.” That is meaningful variation but phrased so that most are similar (stressed) with a minority different. In your training examples, minority exceptions vs majority agreement tended to map to a high-but-not-maximum value. Confidence: medium–high.

Q2

Question: How much do you think they vary among the group of people you represent? Possible answers: 1–5; from 1 – not at all, to 5 – very much

Observed Answer: There are some who have rights but don’t practice at all, and then some who practice like I do, just now and then. Then there are those who have leased out their permits, and in those cases, there is somebody who practices quite intensively

Score: 5

Why: The answer describes the full spectrum (no practice → occasional → very intensive). That explicit broad distribution matches your examples that consistently received top-end variability scores. Confidence: high. Q2 Question: How likely is it that you explore new techniques or strategies? Possible answers: [1–5; from 1 – very unlikely, to 5 – very likely] *Observed Answer:* Would probably try to find solutions; activity would certainly change, and then people would follow that change; there is probably some variation among people depending on what one is specialized in *Score:* 4 *Why:* The speaker says they “would probably try” and expects change and follow-up by others — language that in your dataset equates to a positive likelihood but not the absolute “definitely” (which you typically scored a 5). The note about variation reduces absolute certainty slightly. Confidence: medium–high.

Q3

Question: If your income from activity A decreases, can you complement or replace it? Possible answers: [1–5; from 1 – very unlikely, to 5 – very likely]

Observed Answer: It would be very hard to replace activity A it with other income. Most have other jobs in winter, but A is such a significant part of income

Score: 1

Why: The phrase “very hard to replace” and “A is such a significant part” matches wording in your dataset that mapped to the low end (very unlikely) — even though there are seasonal other jobs, the respondent explicitly indicates replacement is essentially not feasible. Confidence: high.

Q4

Question: How easy is it to avoid an extreme weather event by going to another location?

Possible answers: [1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid] *Observed Answer:* I Stop my activity *Score:* 1 *Why:* The respondent says they stop the activity instead of relocating to avoid the event — that implies relocation is not used/feasible. In your training set, answers that indicate “we don’t/stop” were scored at the difficult end (1). Confidence: medium — short reply is slightly ambiguous about whether “stop” means temporarily pause or that relocation is impossible, but both interpretations point to very low ease of avoiding by moving.

Q5 *Question:* If the following type of extreme event occurs, how likely is it that you go out and do the activity? Possible answers: [1–5; from 1 – very unlikely, to 5 – very likely] *Observed Answer:* we wait because the product can spoil *Score:* 1 *Why:* “We wait” because of spoilage risk indicates they do not go out during the event — i.e., very unlikely. In your dataset, similar phrasing (“we wait / won’t go because...”) mapped to the lowest likelihood. *Confidence:* high.

Interviews - integral text (English)

Legend:

Score	Type	Description
x	C 1	Number (x) available
999	C 2	To be interpreted
-999	C 3	Does not apply
-998	C 4	Cant answer
-997	C 5	Garbled

Interviewee: I 1

focused on the waters around stockholm, both in the archipelago and in lakes. he guides mainly for trout, pike, and perch, and makes a living from his business. on average, they guide about 100 days per year. the guiding business is seasonal and follows the fishâ€™s behaviour, activity, and migrations. they fish in lakes, streams and coastal areas. they can use boats in coastal areas and lakes. it is more common in spring and summer. in winter they guide ice fishing. if there is no ice they can switch to spinning but this is unpredictable. the most fished species are sea trout, salmon, pike, and perch. they primarily practice catch and release fishing. because fishing guide is mostly a touristic activity, it is more active in the weekend, and their flexibility is linked to customers flexibility. how they adapt: fishing guide adapts all the time according to the conditions and the clients need. they are flexible for place, species and gear, but its ability to adapt to conditions depends on clientâ€™s flexibility. he notices a decline in demand during extreme hot weather, and for security reason he cannot guide if he feels danger. in winter he can replace ice fishing with spinning but cannot book this in advance. in summer he can choose between different options depending on the conditions. he can change fish location and target (and accordingly gear) in heatwaves, algal bloom and stormy conditions. in hw he can move the trips to mornings and evenings. in hw and ab he moves from shallow waters for pike to below the thermocline for sea trout; in stormy conditions he goes the other way around.

Q 2 - go_fishing

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1-5; from 1 – very unlikely, to 5 – very likely]

Table S.5: Interviewee: I 1 Q 2

	Event	Answer	Score
a	Heatwave in the winter season	Very different, depends on what is booked. Sometimes ice fishing is booked and then we have to cancel all bookings, sometimes open water fishing is possible. Lose ice fishing customers and gain spin fishing ones; 1 ice fishing (spin fishing customers can't be booked in advance in winter, you can't count on a heatwave in winter)	NA
b	Heatwave in the summer season	4. Some fishing is closed if it gets too hot, e.g., river fishing. It's difficult to fish if it gets extremely hot, but I try to adapt; so far I haven't canceled coastal fishing, I move it to mornings & evenings. But, for example, Nyköpingsån was closed because the water was too warm. Also depends on how hot it is. It happened in 2018, depends on what happens with our fish	4

c	Severe icing (in the winter season)	1. Haven't experienced severe icing, only seen it on YouTube; don't know if it happens on sportfishing boats; it has to be safe, not dangerous; if it's icing like on the Great Lakes, then the boat breaks	1
d	Algal bloom (in the summer season)	1. Usually happens offshore; can't handle it if I have a trip, e.g., from Sandhamn, but then we can go inland when there's algal bloom	1
e	Extreme cloudburst (in the summer season)	1, dangerous	1
f	Gale	5Gale is normal (sometimes we cancel), from strong gale and upwards we stay home (18 19 m/s average and 25 m/s in gusts is not possible)	5
g	Storm	1	1
h	Are there any of these in combination that affect how likely it is that you go out and fish?	A lot is about safety, combinations become dangerous, e.g., strong icing and strong winds, I avoid this as much as possible; would also get fewer requests, since fishing should be enjoyable	NA

Multiple options identified

Table S.6: Interviewee: I 1 Q 2

	Event	gear	Score (given)
a	Heatwave in the winter season	ice fishing	1
a	Heatwave in the winter season	rod	5

Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1-5; from 1 – much worse, 3 – no difference, to 5 – much better]

Table S.7: Interviewee: I 1 Q 3

	Event	Answer	Score
a	Heatwave in the winter season	1 ice fishing, there is no catchability if there are no fish; however, the fish like when it gets warmer, so if it's spin fishing then 5	NA
b	Heatwave in the summer season	2	2
c	Severe icing (in the winter season)	No, can't conduct fishing; if it is possible to fish then it becomes 2.	2
d	Algal bloom (in the summer season)	2, the fish move from that area; also becomes difficult fishing (fished during an algal bloom this summer, had to go all the way into the bay because where we intended to fish we only got weeds on the hook).	2

e	Extreme cloudburst (in the summer season)	Don't go out in cloudbursts; fish react to pressure changes; if there is low pressure during a downpour, then they stop biting, but if it rises after the downpour then it can be good; so I can't say	-998
f	Gale	: 2, a gale doesn't matter for the fishing itself, other than that the gear is hard to use	2
g	Storm	Don't go out in a storm	-998

Multiple options identified

Table S.8: Interviewee: I 1 Q 3

	Event	gear	Score (given)
a	Heatwave in the winter season	ice fishing	1
a	Heatwave in the winter season	rod	5

Q 4 - catch_assemblage

Do you catch a different composition of species during such extreme weather events?

[1–5; from 1 – very similar, to 5 – very different species composition]

Table S.9: Interviewee: I 1 Q 4

	Event	Answer	Score
a	Heatwave in the winter season	: 5 sea trout on open water, instead of perch and pike under the ice	-997
b	Heatwave in the summer season	: 5 redirect the fishing down below the thermocline	-997
c	Severe icing (in the winter season)	No information	-998
d	Algal bloom (in the summer season)	More about changing location (deeper), 5 choose other species (we fish pike & perch, sea trout)	-997
e	Extreme cloudburst (in the summer season)	No other species, more that we don't fish	-998
f	Gale	: 2 sometimes; if it blows too hard and we are supposed to fish trout, then sometimes we have to go in and fish perch & pike instead	2
g	Storm	Cancel	-998

Q 5 - catch_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

Table S.10: Interviewee: I 1 Q 5

	Event	Answer	Score
a	Heatwave in the winter season	3	3
b	Heatwave in the summer season	2, a b it worse	2
c	Severe icing (in the winter season)	3	3
d	Algal bloom (in the summer season)	2	2
e	Extreme cloudburst (in the summer season)	3	3
f	Gale	3	3
g	Storm	3	3

Q 6 - cope_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1–5; from 1 – very unlikely, to 5 – very likely]

We release 98% of the fish; we are more concerned with the survival of the fish, e.g., during a heatwave survival becomes difficult; we don't want to fish pike when it's 30 degrees, then we'd rather fish sea trout that can survive better when we release it, since it goes down to colder depth

Table S.11: Interviewee: I 1 Q 6

	Event	Answer	Score
a	Heatwave in the winter season	NA	-999
b	Heatwave in the summer season	3	3
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	3	3
e	Extreme cloudburst (in the summer season)	NA	-999
f	Gale	NA	-999
g	Storm	NA	-999

Q 7 - lag_effect

Do this extreme weather events have long-term effects on how you fish or catch composition

4. Yes, I do have to change the setup of the fishing. With milder winters, I can't fish or might even take a regular job

Table S.12: Interviewee: I 1 Q 7

	Event	Answer	Score
a	Heatwave in the winter season	NA	4
b	Heatwave in the summer season	NA	-999
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	NA	-999
e	Extreme cloudburst (in the summer season)	NA	-999
f	Gale	NA	-999
g	Storm	NA	-999

Q 8 - unc_2_7

How much do you think the answers to these questions (questions 2–7) vary between fishers in the group you represent? 1–5; from 1 – not at all, to 5 – very much

Table S.13: Interviewee: I 1 Q 8

Answer	Score
5, since I am chairman I have the mandate to represent the others	1.5

Q 9 - personal_safety

How much concern for your personal safety do these extreme weather events bring?

[0–2; 0 – no concern, 1 – personal injuries, 2 – life-threatening

Table S.14: Interviewee: I 1 Q 9

	Event	Answer	Score
a	Heatwave in the winter season	1 I do it for ice fishing; if no one falls in now, it is a miracle	NA

b	Heatwave in the summer season	0	0
c	Severe icing (in the winter season)	2	2
d	Algal bloom (in the summer season)	0	0
e	Extreme cloudburst (in the summer season)	1	1
f	Gale	1 (ordinary gale none, but strong gale)	1
g	Storm	2	2

Multiple options identified

Table S.15: Interviewee: I 1 Q 9

	Event	gear	Score (given)
a	Heatwave in the winter season	ice fishing	1

Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0–3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

Table S.16: Interviewee: I 1 Q 10

	Event	Answer	Score
a	Heatwave in the winter season	0	0
b	Heatwave in the summer season	0	0
c	Severe icing (in the winter season)	don't know, probably 2. (have broken parts on the boat twice because of it)	2
d	Algal bloom (in the summer season)	0	0
e	Extreme cloudburst (in the summer season)	don't know, more if the boat sinks; is that a 3 then, don't know, depends on how big the damage is; in combination with storm then it becomes 3. Usually, you get away from a cloudburst. There is a risk; maybe 2	2
f	Gale	1	1
g	Storm	3	2

Q 11 - minor_damage_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

Table S.17: Interviewee: I 1 Q 11

Answer	Score - low (given)	Score - up (given)	Unit
10 20k kr	10000	20000	sek_unit

Q 12 - major_damage_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

Table S.18: Interviewee: I 1 Q 12

Answer	Score - low (given)	Score - up (given)	Unit
20 600k kr	20000	6e+06	sek_unit

Q 13 - change_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0–2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

Change all the time according to the conditions; place, species and gear

Table S.19: Interviewee: I 1 Q 13

	Event	Answer	Score
a	Heatwave in the winter season	NA	-999
b	Heatwave in the summer season	NA	-999
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	NA	-999
e	Extreme cloudburst (in the summer season)	NA	-999
f	Gale	NA	-999
g	Storm	NA	-999

Q 14 - unc_9_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

Table S.20: Interviewee: I 1 Q 14

Answer	Score
great similarity between boat guides (1)	1

Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

Easy for me to switch but ask what the customers want (e.g., Americans who are booked for ice fishing). Large differences from the normal are a problem, with bookings we do not know what we shall book (e.g., closed Nyköpingsån fishing due to heatwave). The customers also have to be flexible. 2 for those that are not dangerous; 1 for what is dangerous (cannot conduct activities)

Table S.21: Interviewee: I 1 Q 15

	Event	Answer	Score
a	Heatwave in the winter season	NA	1
b	Heatwave in the summer season	NA	2
c	Severe icing (in the winter season)	NA	1
d	Algal bloom (in the summer season)	NA	2
e	Extreme cloudburst (in the summer season)	NA	1
f	Gale	NA	1
g	Storm	NA	1

Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

Table S.22: Interviewee: I 1 Q 16

Answer	Score
--------	-------

0 , if you take the extreme weather event to its peak, extreme cloudburst or storm, then the boat does not help; same if it is extreme heatwave, then we cannot go out 38 42 °C; in 2018 we could not book the fishing. Another parameter: our activity also depends on demand, and demand is made impossible during many of these extreme weather events	0
---	---

Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.23: Interviewee: I 1 Q 17

Answer	Score
2	2

Q 18 - management__space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.24: Interviewee: I 1 Q 18

Answer	Score
Sport fishing has great mobility; the accessibility to water. 2.	2

Q 19 - management__species

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.25: Interviewee: I 1 Q 19

Answer	Score
2 Yes	2

Q 20 - unc__15__19

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

Table S.26: Interviewee: I 1 Q 20

--

Answer	Score
Probably varies, yes it probably varies between fishers: some have a home harbor and are not so flexible; does that apply to management too do you think? don't know, generally not so much variation there	2

Q 21 - stress

How much stress does the risk of the following events cause? A is for injuries, b for gear and c for condition [1–5; from 1 – really not stressful, to 5 – very stressful]

Table S.27: Interviewee: I 1 Q 21

Type	Answer	Score
a Injuries	5	5
b Damages to gear	5	5
c Fish condition	4 (one does not want to fish if the fish are not doing well, it is also bad promotion for me as a fishing guide)	4

Q 22 - unc_21

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

Table S.28: Interviewee: I 1 Q 22

Answer	Score
it is so personal, some do not care at all, they just drink beer all day; we discuss things like this all the time, most are probably quite stressed by such situations	4

Q 23 - job_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.29: Interviewee: I 1 Q 23

Answer	Score
I have to work, so I have no choice; everyone has the opportunity to work with something else who has two legs and two arms; I only guide, I have nothing I can lean back on. Is it 3? I would have to.	3

Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.30: Interviewee: I 1 Q 24

Answer	Score
Would probably try to find solutions; sport fishing itself would certainly change, and then fishing guides would follow that change; there is probably some variation among guides depending on what one is specialized in	4

Q 25 - credit

How easy is it to gain access to credits, financing or loans for investments

[1–5; from 1 – very difficult, to 5 – very easy]

Table S.31: Interviewee: I 1 Q 25

Answer	Score
1. there are no possibilities; completely self financed. The bank would just laugh at one. The businesses are too small, and no one invests in this. Our whole industry is like that	1

Q 26 - unc_23_25

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

Table S.32: Interviewee: I 1 Q 26

Answer	Score
see answer above; the question was not asked, because the answers had already been made.	2

Q 27 - knowledge_transmission

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1–5; from 1 – not important at all, to 5 – very important]

Table S.33: Interviewee: I 1 Q 27

Answer	Score
the world does not end without fishing guides; but we are ambassadors for how the development of fishing continues; it matters a lot how we handle fish & fishing in front of the general sport fishing mass, which is very large, 20% of Sweden's population; 5. important for a positive handling of fish and restoration of habitat. We are the only voice for the fish in the water. Important that there is a small scale tourist fishing industry, for a positive development for the fish	5

Q 28 - benefit_local_community

How much does your local community benefit from your fishing

[1–5; from 1 – not important at all, to 5 – very important]

Table S.34: Interviewee: I 1 Q 28

Answer	Score
quite a lot; 3; influences so that more sport fishing equipment is sold in the nearby shops, hotel nights, everything around lunches, dinners, and so on	3

Q 29 - benefit_personal_community

How much does your “community” benefit from your fishing? Explain!

[1–5; from 1 – not important at all, to 5 – very important]

Table S.35: Interviewee: I 1 Q 29

Answer	Score
5. see answer to 27.	5

Q 30 - traditional_food

How important is your fishing for providing traditional food

[1–5; from 1 – not important at all, to 5 – very important]

Table S.36: Interviewee: I 1 Q 30

Answer	Score
1. fish are not cows. We do not fish for food, almost none of us.	1

Q 31 - unc_26_30

How much do you think the answers to the questions about the contribution of fishing to society (questions 27–30) vary between fishers in the group you represent

Table S.37: Interviewee: I 1 Q 31

Answer	Score
great variation 5.	5

Q 32 - outdoor_benefit

How important is it for you to be outdoors (when you fish) for your well-being

[1–5; from 1 – not important at all, to 5 – very important]

Table S.38: Interviewee: I 1 Q 32

Answer	Score
5	5

Q 33 - identity

How much does fishing contribute to your identity and self-image

[1–5; from 1 – very little, to 5 – very much]

Table S.39: Interviewee: I 1 Q 33

Answer	Score
5	5

Q 34 - sense_of_home

How much does fishing contribute to creating a bond to local places

[1–5; from 1 – very little, to 5 – very much]

Table S.40: Interviewee: I 1 Q 34

Answer	Score
5, it creates positive memories; that is how one creates a relationship to places	5

Q 35 - unc_31_34

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

Table S.41: Interviewee: I 1 Q 35

Answer	Score
it is probably very unanimous among sport fishers; that is why one works with this; 1.	1

Q 36 - fdays

Approximately how many days per year do you fish?

[free text description]

Table S.42: Interviewee: I 1 Q 36

Answer	Score
100	100

Q 37 - income

Approximately how large is your gross income from a fishing day on average?

[free text description]

Table S.43: Interviewee: I 1 Q 37

Answer	Score - low (given)	Score - up (given)	Unit
6000 8000 SEK/day and boat, generally the same for everyone	6000	8000	sek_day

Q 38 - season_maint

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

Table S.44: Interviewee: I 1 Q 38

Answer	Score - low (given)	Score - up (given)	Unit
oh, you don't want to know! what I do is so uneconomical; if the engine breaks then several 100,000 SEK go up in smoke, electric motor 50,000 SEK, sonar 50,000 SEK; is it rather 100,000 200,000 than 10,000 50,000 SEK? yes. Large part is expenses for material, applies also to the cars; 5 days a week fishing, 1 service day per week, 1 family day	50000	10000	sek_season

Q 39 - fuel_trip

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

Table S.45: Interviewee: I 1 Q 39

Answer	Score - low (given)	Score - up (given)	Unit
maybe 1000 SEK/fishing day; this is the biggest expense besides food for a day	1000	1000	sek_day

Q 40 - extra_fuel

What are the costs for extra fuel to travel further than normal?

[free text description]

Table S.46: Interviewee: I 1 Q 40

Answer	Score - low (given)	Score - up (given)	Unit
1500 SEK for such a longer travel day	500	500	sek_day

Q 41 - extra_crew

What are the costs for extra crew on board?

[free text description]

Table S.47: Interviewee: I 1 Q 41

Answer	Score - low (given)	Score - up (given)	Unit
I always guide myself; there are some who sometimes take in an extra person; quite expensive, since it is salary and employer's contributions etc. (guess they get 1500 SEK, so in total it becomes 3000 SEK).	2000	3000	sek_day

Q 42 - extra_cleaning_cost

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

Table S.48: Interviewee: I 1 Q 42

Answer	Score - low (given)	Score - up (given)	Unit
a whole workday for service, so it costs what I earn on a fishing day; no, not for algal bloom, but in gale or storm there are many lines that must be fixed; twice I have broken the boat due to icing, the trim motor and the steering, 15,000 SEK per time.	NA	NA	sek_day

Q 43 - extra_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?

[free text description]

Table S.49: Interviewee: I 1 Q 43

Answer	Score - low (given)	Score - up (given)	Unit
the biggest cost is the loss of income. Cancelled fishing days. A tourist usually does not rebook, especially if it is short notice change then I don't have time to get in something else. Guides for corporate events are at 30,000 SEK, or 60,000 SEK when we are out with 6 boats; if you miss that then the whole income disappears; loss of income also arises because people do not book when there is extreme weather.	NA	NA	sek_day

Q 44 - unc_36_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

Table S.50: Interviewee: I 1 Q 44

Answer	Score
significant variation; 5.	5

Q 45 - addition

Anything you want to add?

[free text description]

Table S.51: Interviewee: I 1 Q 45

Answer	Score
There may be disease among the fish during heatwaves, and that can have consequences for fishing for five years ahead; and algal bloom we do not know, but it does not feel good. We are completely dependent on the fish being well. How have perch and pike been doing this summer? It was a bit weak when it was hot, but now they have started again. Probably quite a normal year. One adapts to the new. Sour that one cannot fish in the outer archipelago. It is an enormous difference from how it was when I was little, we would need supportive stocking of the pike stocks. First year with protection. I have not seen so much herring in 15 years! We have had incredibly little fish coming in to spawn earlier; when they last did test fishing at Mörkö, there was an 84 86% decline in spawning herring. Out last week, then there was herring in Furusundsleden, a lot of herring, proper schools. Summer herring. Don't know when I last saw that. Very positive! Will have great impact on the predatory fish stocks. Hope that the seals can move out again, so they don't have to hang around	t Slus

Q 46 - substitution_capacity

If I lose one planned fishing day, I can usually make up for it later without it affecting what I achieve overall

[1–5; from 1 – strongly disagree, to 5 – strongly agree]

Table S.52: Interviewee: I 1 Q 46

Answer	Score
The income is gone for that day. You lose customers. Even if you reschedule the fishing day, you lose a new customer on the rescheduled day.	1

Q 47 - unc_46

How much do you think the answers to these questions (question 46) vary between fishers in the group you represent?

Table S.53: Interviewee: I 1 Q 47

Answer	Score
--------	-------

The variation is small between those who guide full-time, everyone would probably agree. If you guide part-time, a lost fishing day doesn't matter that much. So some variation.

Interviewee: I 2

Operates in the north of bothnian bay (kalix). They fish in coastal waters with gillnets and traps (salmon traps and other traps for perch and bream), and they operate from boat at very short distances from their harbour. The target they mention is perch, bream, herring, salmon. They also run vendace fishery for roe production. The main fishing season is summer (from april to august). In winter they generally do not go out until spring comes. they can do sporadically some ice fishing. How they adapt: structured succession of fishing styles within season. depending on the moment, they want to aim for something and may be forced to change. in the case they are fishing with nets (for pike, perch and herring?) they are flexible. their options to avoid hw and ab are to fish deeper, to avoid storms is to choose a different bay. hw especially affects perch, which die immediately. they notice more breams with hw. salmon traps are very rigidly managed and tied to a unique boat. salmon traps do not suffer from hw, but they do of storm, so if they must haul them up with a bigger boat, they get a penalty. vendace season is also rigid and short.

Q 2 - go_fishing

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1-5; from 1 – very unlikely, to 5 – very likely]

Table S.54: Interviewee: I 2 Q 2

	Event	Answer	Score
a	Heatwave in the winter season	Extremely warm winter we ice fish so little, therefore it is a bit difficult to answer. (2)	2
b	Heatwave in the summer season	(3) Depends a little on what we are fishing	3
c	Severe icing (in the winter season)	1	2
d	Algal bloom (in the summer season)	Doesn't affect us much up here; locally there may be algal bloom in a bay. (4)	4
e	Extreme cloudburst (in the summer season)	3	3
f	Gale	1, not me at least with my small boat	1
g	Storm	1	1
h	Are there any of these in combination that affect how likely it is that you go out and fish?	Not really; it's the wind that is decisive for us	NA

Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better]

Table S.55: Interviewee: I 2 Q 3

	Event	Answer	Score
a	Heatwave in the winter season	3	3
b	Heatwave in the summer season	2, Becomes worse, maybe not for all species, but they move to other areas, so you have to search elsewhere	2
c	Severe icing (in the winter season)	Hypothetical question, since we always have ice in winter; icing can happen in autumn, but we rarely fish then, so can't answer that	-998
d	Algal bloom (in the summer season)	Hard to answer, doesn't occur very often	-998
e	Extreme cloudburst (in the summer season)	3	3
f	Gale	It becomes harder to fish, but does not affect the actual catch; (2). Can affect a little since it's harder to fish	2
g	Storm	In a storm you can't fish, so catchability is zero	1

Q 4 - catch_assemblage

Do you catch a different composition of species during such extreme weather events?

[1–5; from 1 – very similar, to 5 – very different species composition]

Table S.56: Interviewee: I 2 Q 4

	Event	Answer	Score
a	Heatwave in the winter season	1	1
b	Heatwave in the summer season	Somewhat different species if we fish in the same place e.g., a lot of bream; somewhat different (4).	4
c	Severe icing (in the winter season)	NA	-998
d	Algal bloom (in the summer season)	NA	-998
e	Extreme cloudburst (in the summer season)	1	1
f	Gale	1	1
g	Storm	NA	-998

Q 5 - catch_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

Table S.57: Interviewee: I 2 Q 5

	Event	Answer	Score
a	Heatwave in the winter season	3	3
b	Heatwave in the summer season	1 Yes, then it gets quite a lot worse. Perch cannot tolerate lying in nets; they die almost immediately. Salmon in traps is a short fishing period and in deep water, so it doesn't get warm enough to affect them	NA
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	If there were to be strong algal bloom, which hasn't happened so far, then it would certainly affect negatively hypothetical.	2
e	Extreme cloudburst (in the summer season)	NA	-998
f	Gale	NA	-998
g	Storm	NA	-998

Multiple options identified

Table S.58: Interviewee: I 2 Q 5

	Event	target	Score (given)
b	Heatwave in the summer season	perch	1
b	Heatwave in the summer season	salmon	3

Q 6 - cope_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.59: Interviewee: I 2 Q 6

	Event	Answer	Score
a	Heatwave in the winter season	NA	-999
b	Heatwave in the summer season	Fish in deeper waters, maybe a middle value (3).	3

c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	Fish in deeper waters, not difficult to avoid	4
e	Extreme cloudburst (in the summer season)	NA	-999
f	Gale	NA	-999
g	Storm	NA	-999

Q 7 - lag_effect

Do this extreme weather events have long-term effects on how you fish or catch composition 1–5; from 1 – not at all, to 5 – very much:

(3) Wind and heat will have long term impacts

Table S.60: Interviewee: I 2 Q 7

	Event	Answer	Score
a	Heatwave in the winter season	NA	3
b	Heatwave in the summer season	NA	3
c	Severe icing (in the winter season)	NA	-998
d	Algal bloom (in the summer season)	NA	-998
e	Extreme cloudburst (in the summer season)	NA	-998
f	Gale	NA	3
g	Storm	NA	3

Q 8 - unc_2_7

How much do you think the answers to these questions (questions 2–7) vary between fishers in the group you represent? 1–5; from 1 – not at all, to 5 – very much

Table S.61: Interviewee: I 2 Q 8

Answer	Score
1 Very little the conditions are about the same for everyone	1

Q 9 - personal_safety

How much concern for your personal safety do these extreme weather events bring?

[0–2; 0 – no concern, 1 – personal injuries, 2 – life-threatening]

Table S.62: Interviewee: I 2 Q 9

	Event	Answer	Score
a	Heatwave in the winter season	0	0
b	Heatwave in the summer season	0	0
c	Severe icing (in the winter season)	2 boat may sink	2
d	Algal bloom (in the summer season)	0	0
e	Extreme cloudburst (in the summer season)	0	0
f	Gale	0 no major concern	0
g	Storm	2	2

Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0–3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

Table S.63: Interviewee: I 2 Q 10

	Event	Answer	Score
a	Heatwave in the winter season	0) No boat used, the snowmobile can get stuck, but no damage, only extra work	0
b	Heatwave in the summer season	More work cleaning gear from fouling, (1) if that can even be called damage.	0
c	Severe icing (in the winter season)	1	1
d	Algal bloom (in the summer season)	0	1
e	Extreme cloudburst (in the summer season)	(1) If the boat sinks or the engine ends up under water (that's enough for one summer).	1
f	Gale	1	1
g	Storm	(2) Mostly to gear. It very rarely happens that we lose gear completely, but a small risk exists	1.5

Q 11 - minor_damage_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

Table S.64: Interviewee: I 2 Q 11

Answer	Score - low (given)	Score - up (given)	Unit
Tens of thousands of SEK, nets and such	10000	50000	sek_unit

Q 12 - major_damage_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

Table S.65: Interviewee: I 2 Q 12

Answer	Score - low (given)	Score - up (given)	Unit
Several hundred thousand SEK	1e+05	5e+05	sek_unit

Q 13 - change_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0–2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

Table S.66: Interviewee: I 2 Q 13

	Event	Answer	Score
a	Heatwave in the winter season	(1) We might choose areas where travel on ice is easier.	1
b	Heatwave in the summer season	(1) Might change areas a little bit	1
c	Severe icing (in the winter season)	We fish so little during that season that it's hard to say	-999
d	Algal bloom (in the summer season)	0	0
e	Extreme cloudburst (in the summer season)	(0) We wait until the downpour is over and then go out; usually it only lasts a few hours.	0
f	Gale	(2) Then we may not go out at all.	2
g	Storm	(2) Then it's quite certain we won't go out.	2

Q 14 - unc_9_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

Table S.67: Interviewee: I 2 Q 14

Answer	Score
1	1

Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

Table S.68: Interviewee: I 2 Q 15

	Event	Answer	Score
a	Heatwave in the winter season	Not so easy, rather difficult (2).	2
b	Heatwave in the summer season	(2) Quite difficult.	2
c	Severe icing (in the winter season)	can't answer	-999
d	Algal bloom (in the summer season)	Quite easy (4).	4
e	Extreme cloudburst (in the summer season)	Doesn't affect much, so (not scored).	-999
f	Gale	When it's windy, it's windy everywhere, so quite difficult (1) although you can always find some spot.	1
g	Storm	Impossible to fish at all (1).	1

Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

Table S.69: Interviewee: I 2 Q 16

Answer	Score
1	1

Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.70: Interviewee: I 2 Q 17

Answer	Score
We only have one landing harbor, so it requires us to stay in the local area. (0)	0

Q 18 - management_space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.71: Interviewee: I 2 Q 18

Answer	Score
Generally, we have quite a lot of mobility in terms of management; fixed gear locations are accessible fishing spots. Only for salmon fishing the license is tied to a specific boat, so you can't switch to a larger boat. For net fishing it doesn't matter. (Example given: switching to a larger and safer boat than the one tied to the salmon license in order to haul the salmon trap during gale, but then getting a penalty on the license).	999

Q 19 - management_species

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.72: Interviewee: I 2 Q 19

Answer	Score
1 Depends on the fishing method. Salmon is the most restricted by management; it's the hardest to switch between species. Vendace (siklōja) too, because of the short season. At that time, we have no other alternative species to fish.	1

Q 20 - unc_15_19

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

Table S.73: Interviewee: I 2 Q 20

Answer	Score
1 same for everyone	1

Q 21 - stress

How much stress does the risk of the following events cause? A is for injuries, b for gear and c for condition [1–5; from 1 – really not stressful, to 5 – very stressful]

Table S.74: Interviewee: I 2 Q 21

	Type	Answer	Score
a	Injuries	Personal injury during fishing: 3	3
b	Damages to gear	Losing fishing gear: 2 not so stressful since it is quite rare	2
c	Fish condition	Low quality of the fish in the catch: 2 possible to avoid so not very stressful	2

Q 22 - unc_21

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

Table S.75: Interviewee: I 2 Q 22

Answer	Score
4 It varies a lot; some don't dare to go out when it's just a little windy.	5

Q 23 - job_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.76: Interviewee: I 2 Q 23

Answer	Score
Personally: (5). For Bothnian Bay fishers in general: If a large part of the vendace season were stormy, it would be very hard to replace it with other income. Most have other jobs in winter, but fishing is such a significant part of income	1

Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.77: Interviewee: I 2 Q 24

Answer	Score
--------	-------

5 We do it all the time, try to find new products and species to use. 5
People must learn to eat more species

Q 25 - credit

How easy is it to gain access to credits, financing or loans for investments

[1–5; from 1 – very difficult, to 5 – very easy]

Table S.78: Interviewee: I 2 Q 25

Answer	Score
1 If you want to borrow 1 million for a machine, you need to have 1 million in your pocket. Not many fishers have that	1

Q 26 - unc_23_25

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

Table S.79: Interviewee: I 2 Q 26

Answer	Score
5 Very large variation.	5

Q 27 - knowledge_transmission

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1–5; from 1 – not important at all, to 5 – very important]

Table S.80: Interviewee: I 2 Q 27

Answer	Score
5	5

Q 28 - benefit_local_community

How much does your local community benefit from your fishing

[1–5; from 1 – not important at all, to 5 – very important]

Table S.81: Interviewee: I 2 Q 28

Answer	Score
5 Vendace fishing is extremely important for Kalix, not only for jobs but also for the town's image	5

Q 29 - benefit_personal_community

How much does your “community” benefit from your fishing? Explain!

[1–5; from 1 – not important at all, to 5 – very important]

Table S.82: Interviewee: I 2 Q 29

Answer	Score
5	5

Q 30 - traditional_food

How important is your fishing for providing traditional food

[1–5; from 1 – not important at all, to 5 – very important]

Table S.83: Interviewee: I 2 Q 30

Answer	Score
[5] Very important: vendace, salmon, and herring.	5

Q 31 - unc_26_30

How much do you think the answers to the questions about the contribution of fishing to society (questions 27–30) vary between fishers in the group you represent

Table S.84: Interviewee: I 2 Q 31

Answer	Score
[1] No variation. It is very important for everyone, I dare say.	1

Q 32 - outdoor_benefit

How important is it for you to be outdoors (when you fish) for your well-being

[1–5; from 1 – not important at all, to 5 – very important]

Table S.85: Interviewee: I 2 Q 32

Answer	Score
4 important	4

Q 33 - identity

How much does fishing contribute to your identity and self-image

[1–5; from 1 – very little, to 5 – very much]

Table S.86: Interviewee: I 2 Q 33

Answer	Score
--------	-------

Q 34 - sense_of_home

How much does fishing contribute to creating a bond to local places

[1–5; from 1 – very little, to 5 – very much]

Table S.87: Interviewee: I 2 Q 34

Answer	Score
Hard to say for specific places. For example, Kalix archipelago what should I say? We live here, so yes, important.(4)	4

Q 35 - unc_31_34

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

Table S.88: Interviewee: I 2 Q 35

Answer	Score
[1] Not much variation. Most are very proud to be fishers; how much one fishes in practice may vary, but in terms of meaning, there are no major differences	2

Q 36 - fdays

Approximately how many days per year do you fish?

[free text description]

Table S.89: Interviewee: I 2 Q 36

Answer	Score - low (given)	Score - up (given)	Unit
Hard to say. If we have 6 months without ice, we actively fish 1 third of the days. But we keep the gear in the water all ice free days.	50	100	days

Q 37 - income

Approximately how large is your gross income from a fishing day on average?

[free text description]

Table S.90: Interviewee: I 2 Q 37

Answer	Score - low (given)	Score - up (given)	Unit

Impossible to calculate per day. Over 6 months? It's the four weeks of vendace fishing that make up the income. This varies. I haven't broken down exactly what the fishing itself gives, plus there's all the processing afterwards. Then you'd also have to deduct labor costs. Teija calculates roughly to get a value: 1.4 million SEK per year from vendace fishing. Plus some perch and bream. We haven't fished salmon for several seasons — the quota is fished up in 7 days [by those who have good fishing waters near the Torne River], so it's not worth setting the gear, since it takes a week to set and a week to haul.

Q 38 - season_maint

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

Table S.91: Interviewee: I 2 Q 38

Answer	Score - low (given)	Score - up (given)	Unit
I don't remember, it gets lumped together with all other repairs at Guldhaven. We buy 1 boat per season several hundred thousand SEK and a new trawl, 200k SEK, other stuff 100k SEK. Roughly 0.5 million SEK over two years.	1e+05	5e+05	sek_season

Q 39 - fuel_trip

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

Table S.92: Interviewee: I 2 Q 39

Answer	Score - low (given)	Score - up (given)	Unit
Don't dare estimate.	NA	NA	sek_day

Q 40 - extra_fuel

What are the costs for extra fuel to travel further than normal?

[free text description]

Table S.93: Interviewee: I 2 Q 40

Answer	Score - low (given)	Score - up (given)	Unit
Marginal, since we have such short distances	NA	NA	sek_day

Q 41 - extra_crew

What are the costs for extra crew on board?

[free text description]

Table S.94: Interviewee: I 2 Q 41

Answer	Score - low (given)	Score - up (given)	Unit
None for fishing itself. We bring in staff for processing [130 people). Finding staff is the limiting factor in fishing, especially when there is as much vendace as there is now	NA	NA	sek_day

Q 42 - extra_cleaning_cost

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

Table S.95: Interviewee: I 2 Q 42

Answer	Score - low (given)	Score - up (given)	Unit
Cleaning traps takes some time, and it gets worse as the water gets warmer. Cleaning 10 traps took 0.5 h. Now we only have 20 traps in the water because of the perch, but when we fish the most, we have 100 traps. Then it means a lot of driving to clean them every three weeks; otherwise, they stay all season, or we clean once. Built ourselves from mink wire. Clean with a pressure washer. Same with bream traps	3	3	minutes_gear

Q 43 - exytra_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?

[free text description]

Table S.96: Interviewee: I 2 Q 43

Answer	Score - low (given)	Score - up (given)	Unit
Things blow away. Doesn't cost much, but takes time to pick everything up when it's scattered across the bay.	NA	NA	not_reported

Q 44 - unc_36_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

Table S.97: Interviewee: I 2 Q 44

Answer	Score
5	5

Q 45 - addition

Anything you want to add?

[free text description]

Table S.98: Interviewee: I 2 Q 45

Answer	Score
The striped perch are real stripes! They have found more in test fishing. Small ones, now frozen, which she will send to us. 234 perch in the test fishing, 9 33 cm. We have a lot of perch here now, never been so many on rod fishing — in a 1 hour round we get 10 perch of eating size. Why do you think? Now that the vendace stock is so good, they have a lot of food, and they grow well when it's warm. Seems to be good recruitment too, since we caught so many small ones. We had the nets here in Långviken, along a gradient from open sea to inner bay: mostly bream in the innermost net, and mostly perch in the outermost net. The largest perch were farthest out.	- 999

Q 46 - substitution_capacity

If I lose one planned fishing day, I can usually make up for it later without it affecting what I achieve overall

[1–5; from 1 – strongly disagree, to 5 – strongly agree]

Table S.99: Interviewee: I 2 Q 46

Answer	Score
For the vendace fishery, a lost day can have a huge impact (could be a million in lost income), linked to the restricted quotas. For the other small-scale coastal fishery, a lost day is of almost no importance.	5

Q 47 - unc_46

How much do you think the answers to these questions (question 46) vary between fishers in the group you represent?

Table S.100: Interviewee: I 2 Q 47

Answer	Score
You will likely get the same reply from both type of fishers	1

Interviewee: I 3

id3 is professional coastal fishermen in southwest bothnian sea (norrsundet). they fish with gillnets and traps. traps are only used for salmons. the main target is herring, whitefish, salmon, perch, and eel. they

also run a restaurant and have several fish trucks. the herring fishing takes place a few hundred meters from the harbor. they mention whitefish, salmon, perch in the local area, and in the gulf of gÅvle. their average distance travel is about 1 hour (assuming 10 knots it is about 20 km) and the longest distance travel is 3 hours (assuming 10 knots it is max 55 km) the main fishing season is summer (from april to august). in winter they generally do not go out until spring comes. how they adapt: id3 can adapt when fishing with nets to hw by avoiding hottest hours, performing shorter sets and using cooling boxes. can also adapt to hw, ab and gales by changing fishing location. to avoid hw and ab they can go further out, the opposite for gales. in any case they can move on short distances and can cope with gales only in shallow waters, they donâ€™t go deep and with storms they donâ€™t go out at all. their spatial mobility is mainly affected by the boat (12m boat) and infrastructure (they want to go back home) on second instance by management that makes possible to fish in proximity (over a coastline spanning of about 60 km in aerial distance). there is not the same flexibility for salmon traps, which are fixed.

Q 2 - go_fishing

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.101: Interviewee: I 3 Q 2

	Event	Answer	Score
a	Heatwave in the winter season	Depends on ice. Boats are usually hauled up in winter. If forecast shows winter is over, we might launch. Unlikely to put in and haul up again. Might use the small boat if open water, but ice may return. Large boats remain on land. =1 regardless.	-999
b	Heatwave in the summer season	Then we fish, just go further out where it's cooler. =5.	5
c	Severe icing (in the winter season)	5 (icing can happen in spring).	-999
d	Algal bloom (in the summer season)	Rare up here, but still fish. =5.	5
e	Extreme cloudburst (in the summer season)	If gear is already out, we go anyway. If not, we wait. =3.	3
f	Gale	No, we stay home. =1.	1
g	Storm	No, we stay home. =1.	1
h	Are there any of these in combination that affect how likely it is that you go out and fish?	Wind has the greatest impact here. Wind direction matters too (this August mostly W & NW winds, cooling the water – no point fishing perch then, they just stop moving).	NA

Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better]

Table S.102: Interviewee: I 3 Q 3

	Event	Answer	Score
a	Heatwave in the winter season	Not relevant	-999
b	Heatwave in the summer season	Heatwave makes it worse. =2.	2
c	Severe icing (in the winter season)	not relevant	-999
d	Algal bloom (in the summer season)	For perch, it gets worse (=2). Same for herring, due to warmth.	3
e	Extreme cloudburst (in the summer season)	Doesn't affect catchability (only whether we go out). =3.	3
f	Gale	Yes – after a gale from the right direction, herring can blow in, salmon may swim more too (E, NE in summer). =4. From W & NW it gets worse instead.	NA
g	Storm	Same, just stronger wind.	NA

Multiple options identified

Table S.103: Interviewee: I 3 Q 3

	Event	target	Score (given)
d	Algal bloom (in the summer season)	perch	2
d	Algal bloom (in the summer season)	herring	2
f	Gale	herring	4
f	Gale	salmon	4
g	Storm	herring	4
g	Storm	salmon	4

Q 4 - catch_assemblage

Do you catch a different composition of species during such extreme weather events?

[1–5; from 1 – very similar, to 5 – very different species composition]

Table S.104: Interviewee: I 3 Q 4

	Event	Answer	Score
a	Heatwave in the winter season	Not relevant.	-999

b	Heatwave in the summer season	very similar (1)	1
c	Severe icing (in the winter season)	not relevant	-999
d	Algal bloom (in the summer season)	not directly (1)	1
e	Extreme cloudburst (in the summer season)	no (1)	1
f	Gale	1	1
g	Storm	1	1

Q 5 - catch_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

Table S.105: Interviewee: I 3 Q 5

	Event	Answer	Score
a	Heatwave in the winter season	not relevant	-999
b	Heatwave in the summer season	much worse, must get them quickly into cooling boxes (1)	1
c	Severe icing (in the winter season)	not relevant	-999
d	Algal bloom (in the summer season)	no, just from the heat	3
e	Extreme cloudburst (in the summer season)	no (3)	3
f	Gale	no (3)	3
g	Storm	no (3)	3

Q 6 - cope_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.106: Interviewee: I 3 Q 6

	Event	Answer	Score
a	Heatwave in the winter season	NA	-999

b	Heatwave in the summer season	Only b. Go out before midday heat. Early mornings for herring, perch later in day. Adjust by hauling nets earlier. =5.	5
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	NA	-999
e	Extreme cloudburst (in the summer season)	NA	-999
f	Gale	NA	-999
g	Storm	NA	-999

Q 7 - lag_effect

Do this extreme weather events have long-term effects on how you fish or catch composition

1

Table S.107: Interviewee: I 3 Q 7

	Event	Answer	Score
a	Heatwave in the winter season	NA	-999
b	Heatwave in the summer season	NA	1
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	NA	1
e	Extreme cloudburst (in the summer season)	NA	1
f	Gale	NA	1
g	Storm	NA	1

Q 8 - unc_2_7

How much do you think the answers to these questions (questions 2–7) vary between fishers in the group you represent? 1–5; from 1 – not at all, to 5 – very much

Table S.108: Interviewee: I 3 Q 8

Answer	Score
--------	-------

Q 9 - personal_safety

How much concern for your personal safety do these extreme weather events bring?

[0–2; 0 – no concern, 1 – personal injuries, 2 – life-threatening

Table S.109: Interviewee: I 3 Q 9

	Event	Answer	Score
a	Heatwave in the winter season	NA	-999
b	Heatwave in the summer season	0	0
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	0	0
e	Extreme cloudburst (in the summer season)	0	0
f	Gale	rarely go ouyt but sudden events possible	1
g	Storm	don't go out	-998

Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0–3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

Table S.110: Interviewee: I 3 Q 10

	Event	Answer	Score
a	Heatwave in the winter season	NA	-999
b	Heatwave in the summer season	NA	0
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	0, though nets can get dirty.	0
e	Extreme cloudburst (in the summer season)	0	0

f	Gale	Could damage or destroy gear/boat, or cause minor damage	2
g	Storm	Could damage or destroy gear/boat, or cause minor damage	2

Q 11 - minor_damage_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

Table S.111: Interviewee: I 3 Q 11

Answer	Score - low (given)	Score - up (given)	Unit
50,000 SEK/year for nets that break, or propellers when blown onto land/rocks	1	50000	sek_year

Q 12 - major_damage_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

Table S.112: Interviewee: I 3 Q 12

Answer	Score - low (given)	Score - up (given)	Unit
Rare for boats – then we wouldn't be out. But a salmon trap can be destroyed in a summer storm, costing 200,000 SEK	50000	2e+05	sek

Q 13 - change_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0–2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

Table S.113: Interviewee: I 3 Q 13

	Event	Answer	Score
a	Heatwave in the winter season	NA	-999
b	Heatwave in the summer season	Yes, set in other places, change a lot. =2. Salmon gear is fixed at paid spots all summer, so they stay.	2
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	In bays, same as in heatwaves	2

e	Extreme cloudburst (in the summer season)	0	0
f	Gale	2. Move to less windy spots. For herring it's harder, so maybe stay home.	2
g	Storm	Don't go out at all, follow forecasts. Our boat is 12 m, so never in storms	-998

Q 14 - unc_9_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

Table S.114: Interviewee: I 3 Q 14

Answer	Score
Uniform. 1.	1

Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

Table S.115: Interviewee: I 3 Q 15

	Event	Answer	Score
a	Heatwave in the winter season	NA	-999
b	Heatwave in the summer season	Works – go further out (usually warmest inshore). =5.	5
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	If heat lasts long, algal bloom spreads and worsens fishing. =2.	2
e	Extreme cloudburst (in the summer season)	Yes, possible.	4
f	Gale	Yes, but usually means staying home	4
g	Storm	Storms are hard to avoid.	2

Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

Table S.116: Interviewee: I 3 Q 16

Answer	Score
Yes, can move around for herring, whitefish, perch. Only salmon gear is fixed. Our operations use active gear, so we can relocate, but only short distances – catch must be landed quickly. (Older generations even traveled to Trosa/Grisslehamn.)	999

Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.117: Interviewee: I 3 Q 17

Answer	Score
No – no infrastructure elsewhere, must return home. =0.	0

Q 18 - management_space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.118: Interviewee: I 3 Q 18

Answer	Score
Not well informed. Probably possible in Gävle Bay (we have exemptions locally). =1.	1

Q 19 - management_species

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.119: Interviewee: I 3 Q 19

Answer	Score
0. Few commercial species here, license not species-specific	0

Q 20 - unc_15_19

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

Table S.120: Interviewee: I 3 Q 20

Answer	Score
Same situation for all. =1	1

Q 21 - stress

How much stress does the risk of the following events cause? A is for injuries, b for gear and c for condition
[1–5; from 1 – really not stressful, to 5 – very stressful]

Table S.121: Interviewee: I 3 Q 21

Type	Answer	Score
a Injuries	1, not stressed about that.	1
b Damages to gear	Yes – after summer storms worried salmon traps may blow away. =5.	5
c Fish condition	Can happen if winds last days, then quality is lost, fish becomes worthless. Usually not stormy that long. =3.	3

Q 22 - unc_21

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

Table S.122: Interviewee: I 3 Q 22

Answer	Score
Fishers are calm and experienced, know when to go out. =1.	1

Q 23 - job_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.123: Interviewee: I 3 Q 23

Answer	Score
Yes – we already do many other things. Impossible to live solely off fishing on the Norrland coast. Must diversify, requires entrepreneurship. =5	5

Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.124: Interviewee: I 3 Q 24

Answer	Score
Always striving to improve, never want to stop developing. Every time we smoke fish, want it better. Always want renewal. =5	5

Q 25 - credit

How easy is it to gain access to credits, financing or loans for investments

[1–5; from 1 – very difficult, to 5 – very easy]

Table S.125: Interviewee: I 3 Q 25

Answer	Score
1. Not a thriving industry. We've thought about new boats for better work environment, but economically impossible with so little fish	1

Q 26 - unc_23_25

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

Table S.126: Interviewee: I 3 Q 26

Answer	Score
Not all alike – some more ambitious than others, depends on vision. Some are “fair-weather fishers,” others try to make it work year-round. =5.	5

Q 27 - knowledge_transmission

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1–5; from 1 – not important at all, to 5 – very important]

Table S.127: Interviewee: I 3 Q 27

Answer	Score
Very important. We want it to live on. Almost dying out, barely any succession in families. In Gävle municipality soon only us left. =5. That's why I started – to continue and develop what earlier generations built. History carries weight.	5

Q 28 - benefit_local_community

How much does your local community benefit from your fishing

[1–5; from 1 – not important at all, to 5 – very important]

Table S.128: Interviewee: I 3 Q 28

Answer	Score
Very important, =5. We're the largest employer here. In July we had 50 on payroll. Coastal fishing benefits many. Attracts 40–50,000 tourists annually. They fuel up, shop at local stores. Huge significance for the whole community.	5

Q 29 - benefit_personal_community

How much does your “community” benefit from your fishing? Explain!

[1–5; from 1 – not important at all, to 5 – very important]

Table S.129: Interviewee: I 3 Q 29

Answer	Score
5. Doesn't have to be me, but important that I'm the youngest investing – inspires cousins to invest too. Many long-time fishers easily deliver fish to Bergmans.	5

Q 30 - traditional_food

How important is your fishing for providing traditional food

[1–5; from 1 – not important at all, to 5 – very important]

Table S.130: Interviewee: I 3 Q 30

Answer	Score
5. Very important for Swedish food production.	5

Q 31 - unc_26_30

How much do you think the answers to the questions about the contribution of fishing to society (questions 27–30) vary between fishers in the group you represent

Table S.131: Interviewee: I 3 Q 31

Answer	Score
Very similar. That's why they continue, try to recruit new ones. But hard since chances are poor, little money. =1.	1

Q 32 - outdoor_benefit

How important is it for you to be outdoors (when you fish) for your well-being

[1–5; from 1 – not important at all, to 5 – very important]

Table S.132: Interviewee: I 3 Q 32

Answer	Score
--------	-------

Q 33 - identity

How much does fishing contribute to your identity and self-image

[1–5; from 1 – very little, to 5 – very much]

Table S.133: Interviewee: I 3 Q 33

Answer	Score
5. Now it's my whole identity (used to play hockey, now entrepreneurship in fishing defines me).	5

Q 34 - sense_of_home

How much does fishing contribute to creating a bond to local places

[1–5; from 1 – very little, to 5 – very much]

Table S.134: Interviewee: I 3 Q 34

Answer	Score
Very important – many specific places with memories. As important as self-image. =5	5

Q 35 - unc_31_34

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

Table S.135: Interviewee: I 3 Q 35

Answer	Score
Quite similar – everyone loves being outdoors, values the freedom. Calm water, sun shining – few jobs beat that!	2

Q 36 - fdays

Approximately how many days per year do you fish?

[free text description]

Table S.136: Interviewee: I 3 Q 36

Answer	Score - low (given)	Score - up (given)	Unit
Varies, 120–150 days of net fishing.	120	150	days

Q 37 - income

Approximately how large is your gross income from a fishing day on average?

[free text description]

Table S.137: Interviewee: I 3 Q 37

Answer	Score - low (given)	Score - up (given)	Unit
Hard to say. Some days 10–15,000 SEK, other days 0 if seals eat everything	10000	15000	sek_day

Q 38 - season_maint

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

Table S.138: Interviewee: I 3 Q 38

Answer	Score - low (given)	Score - up (given)	Unit
About 200,000 SEK. 100–150,000 in damages. A lot of time spent on repairs, much voluntary work, patching holes in winter	150000	2e+05	sek_season

Q 39 - fuel_trip

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

Table S.139: Interviewee: I 3 Q 39

Answer	Score - low (given)	Score - up (given)	Unit
Several thousand SEK. Longest trips 2–3 hours out: 3,000–5,000 SEK/day. Short trips about half. Sometimes just outside the restaurant here	1500	2500	sek_day

Q 40 - extra_fuel

What are the costs for extra fuel to travel further than normal?

[free text description]

Table S.140: Interviewee: I 3 Q 40

Answer	Score - low (given)	Score - up (given)	Unit

See previous	3000	5000	sek_day
--------------	------	------	---------

Q 41 - extra_crew

What are the costs for extra crew on board?

[free text description]

Table S.141: Interviewee: I 3 Q 41

Answer	Score - low (given)	Score - up (given)	Unit
Rare – usually only if a lot of fish. Then about 200 SEK/hour, unless pensioners help voluntarily.	0	200	sek_hour

Q 42 - extra_cleaning_cost

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

Table S.142: Interviewee: I 3 Q 42

Answer	Score - low (given)	Score - up (given)	Unit
20–30 hours. Salmon traps: ~20 h each to clean at season end, sometimes more. Warm July brings more growth. In 2018 it was so warm that when we hauled a trap, with so much fouling, we nearly sank when a waterspout hit	20	30	hours_event

Q 43 - exytra_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?

[free text description]

Table S.143: Interviewee: I 3 Q 43

Answer	Score - low (given)	Score - up (given)	Unit
Mostly covered – repairs, gear blown ashore.	NA	NA	not_reported

Q 44 - unc_36_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

Table S.144: Interviewee: I 3 Q 44

Answer	Score
--------	-------

Q 45 - addition

Anything you want to add?

[free text description]

Table S.145: Interviewee: I 3 Q 45

Answer	Score
No, I think everything's covered. Biggest problems are seals and industrial fishing. With so few fish, seals get aggressive	- 999

Q 46 - substitution_capacity

If I lose one planned fishing day, I can usually make up for it later without it affecting what I achieve overall

[1–5; from 1 – strongly disagree, to 5 – strongly agree]

Table S.146: Interviewee: I 3 Q 46

Answer	Score
you work week by week. If you lose one day during the week, the entire sales for that week can be ruined. Can lead to a loss of about 150,000 SEK in income. If it is extreme weather for them, it is also the case for other fishermen from whom they might buy fish from.	1

Q 47 - unc_46

How much do you think the answers to these questions (question 46) vary between fishers in the group you represent?

Table S.147: Interviewee: I 3 Q 47

Answer	Score
Same answer if you ask other fishers with a similar business	1

Interviewee: I 4

id4 is a resident of the archipelago in the northern proper baltic (hasselÅ¶, tjust archipelago). archipelago fisherman cannot sell their catchers. in the past, there was a very diversified fishing community with various types of nets for coastal and freshwater species. nowadays they continue to fish with nets for perch, herring, whitefish. he mentions also the importances of being able to catch fish with rods, but in the second interview he only refers to fishing with nets. fishing mostly happens during summer (april to august), winter fishing is less common but can happen. he mentions that if the water is free from ice they can cast nets also in winter. how they adapt: id4 is not economically dependent on fishing and in general tend to do it only when there are good conditions. he can adapt to hw by performing shorter (not overnight) sets. can also adapt to hw, ab and storms by changing fishing location. to avoid hw and ab they can go further out, the opposite for storms. in any case they can move on very short distances. their spatial mobility is mainly affected by its assigned fishing lot.

Q 2 - go_fishing

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.148: Interviewee: I 4 Q 2

	Event	Answer	Score
a	Heatwave in the winter season	Quite likely, if the water is open we'll go out, 4	4
b	Heatwave in the summer season	2. Depends on how it affects winds and algal blooms; this year we had huge problems with algal blooms (1 month).	2
c	Severe icing (in the winter season)	No problem for us, too far south.	-999
d	Algal bloom (in the summer season)	1	1
e	Extreme cloudburst (in the summer season)	We can get to the boathouse, but from there out onto the lake it's more difficult; 1, if we don't have gear in the water. But we check the weather forecast very carefully before putting in the gear.	1
f	Gale	1	1
g	Storm	1	1
h	Are there any of these in combination that affect how likely it is that you go out and fish?	Strong winds and high water levels.	NA

Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better]

Table S.149: Interviewee: I 4 Q 3

	Event	Answer	Score
a	Heatwave in the winter season	NA	2
b	Heatwave in the summer season	Yes, in the past when perch existed, they liked warmth, but now there are no perch, so the effect is not so big on the species we fish for	3
c	Severe icing (in the winter season)	NA	-999

d	Algal bloom (in the summer season)	It's devastating, depressing to live around this algal soup, grandchildren can't swim, it's an imbalance in the Baltic Sea ecosystem, a sad new phenomenon. We try to deal with it when it drifts ashore. It becomes a white coating on the bottom when it sinks. We're worried about algal toxins (cows can break them down somewhat, but single-stomached animals can't; affects the liver). 1.	1
e	Extreme cloudburst (in the summer season)	No, 3	3
f	Gale	A little worried about strong winds since we don't have big boats.	2
g	Storm	Same with storms.	2

Multiple options identified

Table S.150: Interviewee: I 4 Q 3

	Event	target	Score (given)
b	Heatwave in the summer season	perch	4

Q 4 - catch_assemblage

Do you catch a different composition of species during such extreme weather events?

[1–5; from 1 – very similar, to 5 – very different species composition]

Table S.151: Interviewee: I 4 Q 4

	Event	Answer	Score
a	Heatwave in the winter season	Can't answer.	-998
b	Heatwave in the summer season	Can't answer.	-998
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	We don't really fish much then; would guess the fish flee from the algal bloom, 5	-998
e	Extreme cloudburst (in the summer season)	Can't answer	-998
f	Gale	Hard to say, we don't fish much then	-998
g	Storm	Hard to say, we don't fish much then	-998

Q 5 - catch_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

Table S.152: Interviewee: I 4 Q 5

	Event	Answer	Score
a	Heatwave in the winter season	Don't think it affects condition, 3.	3
b	Heatwave in the summer season	hen the water is warmer, fish flesh becomes looser (especially fattier fish), 2.	2
c	Severe icing (in the winter season)	we don't fish then	-999
d	Algal bloom (in the summer season)	we don't fish then	-998
e	Extreme cloudburst (in the summer season)	we don't fish then	-998
f	Gale	we don't fish then	-998
g	Storm	we don't fish then	-998

Q 6 - cope_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.153: Interviewee: I 4 Q 6

	Event	Answer	Score
a	Heatwave in the winter season	NA	-999
b	Heatwave in the summer season	If perch turn yellow and bad from the heat, we can do a day-set (not overnight). Have also set nets around a shoal and driven fish inside, which improved quality. We don't have ice machines, so we can influence it a little	3
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	NA	-999
e	Extreme cloudburst (in the summer season)	NA	-999
f	Gale	NA	-999

g	Storm	NA	-999
---	-------	----	------

Q 7 - lag_effect

Do this extreme weather events have long-term effects on how you fish or catch composition

Algal bloom doesn't need more comment, it's a big problem we've already discussed. Other factor is wind direction, NW is bad, then the fish disappear. Big impact. This spring it locked in at N & NW with near gale winds for several weeks

Table S.154: Interviewee: I 4 Q 7

	Event	Answer	Score
a	Heatwave in the winter season	NA	-999
b	Heatwave in the summer season	NA	-999
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	NA	5
e	Extreme cloudburst (in the summer season)	NA	-999
f	Gale	NA	5
g	Storm	NA	5

Q 8 - unc_2_7

How much do you think the answers to these questions (questions 2–7) vary between fishers in the group you represent? 1–5; from 1 – not at all, to 5 – very much

Table S.155: Interviewee: I 4 Q 8

Answer	Score
Among subsistence fishers there's quite strong agreement; we communicate fairly well with each other, 2	2

Q 9 - personal_safety

How much concern for your personal safety do these extreme weather events bring?

[0–2; 0 – no concern, 1 – personal injuries, 2 – life-threatening

With strong winds, you adapt as an islander. I also do forestry, and if weather is bad with fog and strong winds, you plan activities so as not to risk injury. You adapt to weather

Table S.156: Interviewee: I 4 Q 9

	Event	Answer	Score
a	Heatwave in the winter season	no worry	0
b	Heatwave in the summer season	no worry	0
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	avoid eating fish caught during bloom	0
e	Extreme cloudburst (in the summer season)	no worry, just wear the right clothes	0
f	Gale	slipping, breaking a leg from the rocking	1
g	Storm	slipping, breaking a leg from the rocking	1

Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0-3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

Table S.157: Interviewee: I 4 Q 10

	Event	Answer	Score
a	Heatwave in the winter season	0	0
b	Heatwave in the summer season	with low water (running aground), 1	1
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	0	0
e	Extreme cloudburst (in the summer season)	0	0
f	Gale	Gear can drag and be damaged, very rare; boat damage when docking can be critical, minor repairs, 1.	1
g	Storm	Gear can drag and be damaged, very rare; boat damage when docking can be critical, minor repairs, 1.	1

Q 11 - minor_damage_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

Table S.158: Interviewee: I 4 Q 11

Answer	Score - low (given)	Score - up (given)	Unit
Propeller replacement is quite heavy, costs 2,500 SEK. Hull repair we fix ourselves, a couple hundred SEK	200	2500	sek_unit

Q 12 - major_damage_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

Table S.159: Interviewee: I 4 Q 12

Answer	Score - low (given)	Score - up (given)	Unit
Engines and boats are insured; we must cover gear ourselves. A few thousand SEK more than the 2,500.	2500	5000	sek_unit

Q 13 - change_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0–2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

Table S.160: Interviewee: I 4 Q 13

	Event	Answer	Score
a	Heatwave in the winter season	0	0
b	Heatwave in the summer season	0 Only with algal bloom. If we want whitefish in summer during heatwave, they move deeper (15 m), =1. Move a few hundred meters	1
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	2 then we do not go fish	2
e	Extreme cloudburst (in the summer season)	Don't go out to set nets, but we do collect them if already out	1
f	Gale	We sometimes set in more sheltered bays less exposed, =2.	2
g	Storm	We sometimes set in more sheltered bays less exposed, =2.	2

Q 14 - unc_9_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

Table S.161: Interviewee: I 4 Q 14

Answer	Score
1, no	1

Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

Table S.162: Interviewee: I 4 Q 15

	Event	Answer	Score
a	Heatwave in the winter season	no need	-999
b	Heatwave in the summer season	Drive grandchildren to another bay; no, we don't change fishing spot, we stop fishing	1
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	Stop fishing	1
e	Extreme cloudburst (in the summer season)	1, no, we stop completely	1
f	Gale	4, switch bays	4
g	Storm	Don't go fishing	1

Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

Table S.163: Interviewee: I 4 Q 16

Answer	Score
1, somewhat	1

Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.164: Interviewee: I 4 Q 17

Answer	Score
Yes, but we don't, too much hassle. Since we're not economically dependent, we can skip fishing. =2.	2

Q 18 - management_space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.165: Interviewee: I 4 Q 18

Answer	Score
We have common fishing waters (a cooperative). Fishing lots have become permanent, so I stick to my lot. =0 (within cooperative).	0

Q 19 - management_species

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.166: Interviewee: I 4 Q 19

Answer	Score
Eel fishing ban; voluntary fishing stop Mar 15–May 15 in some spawning bays. We must restore predatory fish stocks. We think fish shelters (we've built 10 over 10 years) have limited effect at such low stock levels.	999

Q 20 - unc_15_19

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

Table S.167: Interviewee: I 4 Q 20

Answer	Score
Requires a vehicle/tractor, not all have it, but roughly consistent, =2.	2

Q 21 - stress

How much stress does the risk of the following events cause? A is for injuries, b for gear and c for condition [1–5; from 1 – really not stressful, to 5 – very stressful]

Table S.168: Interviewee: I 4 Q 21

	Type	Answer	Score
a	Injuries	not much, but in winter it's cold, windy, if motors fail then big problem, 3.	3
b	Damages to gear	1	1
c	Fish condition	1	1

Q 22 - unc_21

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

Table S.169: Interviewee: I 4 Q 22

Answer	Score
Probably fairly consistent, =2.	2

Q 23 - job_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.170: Interviewee: I 4 Q 23

Answer	Score
Not Asked	- 999

Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.171: Interviewee: I 4 Q 24

Answer	Score
Thinking about whether nets are best; with traps the fish are less affected. Just thoughts so far. =2.	2

Q 25 - credit

How easy is it to gain access to credits, financing or loans for investments

[1–5; from 1 – very difficult, to 5 – very easy]

Table S.172: Interviewee: I 4 Q 25

Answer	Score
Not Asked	-
	999

Q 26 - unc_23_25

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

Table S.173: Interviewee: I 4 Q 26

Answer	Score
Most are conservative, professionals are more innovative. In my group, low innovation.	2

Q 27 - knowledge_transmission

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1–5; from 1 – not important at all, to 5 – very important]

Table S.174: Interviewee: I 4 Q 27

Answer	Score
5. We’ve lost local competence when people die and take knowledge with them (e.g., neighbor who knew when/where to catch turbot, burbot).	5

Q 28 - benefit_local_community

How much does your local community benefit from your fishing

[1–5; from 1 – not important at all, to 5 – very important]

Table S.175: Interviewee: I 4 Q 28

Answer	Score
I can’t sell/deliver fish, only the professional fisher can. More cultural—nice to have fish on the table. =1.	1

Q 29 - benefit_personal_community

How much does your “community” benefit from your fishing? Explain!

[1–5; from 1 – not important at all, to 5 – very important]

Table S.176: Interviewee: I 4 Q 29

Answer	Score
--------	-------

Hard to judge. There have been big changes recently – many have passed away or disappeared, and some younger people have moved in who have the opportunity to fish but haven't started yet, they're still renovating. At our fishing meetings people do show up – they're not uninterested. People are interested in fishing, but to what extent they actually fish or communicate about it varies. It creates a unifying factor among us subsistence fishers who belong to the village association. Those who own a joint property with fishing rights and waters – and then there are the summer guests, but I have no idea about them.

3.
|3

Q 30 - traditional_food

How important is your fishing for providing traditional food

[1–5; from 1 – not important at all, to 5 – very important]

Table S.177: Interviewee: I 4 Q 30

Answer	Score
Generational gap. We like fried herring, but grandchildren want hamburgers. Son hates flounder (too much growing up). Daughter better. Have to teach them every time how to debone herring. 5. Also for preparedness—resilience groups forming on islands (root cellars, handling power cuts).	5

Q 31 - unc_26_30

How much do you think the answers to the questions about the contribution of fishing to society (questions 27–30) vary between fishers in the group you represent

Table S.178: Interviewee: I 4 Q 31

Answer	Score
Most people think roughly as we have just said.	2

Q 32 - outdoor_benefit

How important is it for you to be outdoors (when you fish) for your well-being

[1–5; from 1 – not important at all, to 5 – very important]

Table S.179: Interviewee: I 4 Q 32

Answer	Score
5. During test fishing we had wonderful mornings lifting nets—that's what you live on.	5

Q 33 - identity

How much does fishing contribute to your identity and self-image

[1–5; from 1 – very little, to 5 – very much]

Table S.180: Interviewee: I 4 Q 33

Answer	Score
Not so much the fishing, but managing it (chair of fishing association). =3.	3

Q 34 - sense_of_home

How much does fishing contribute to creating a bond to local places

[1–5; from 1 – very little, to 5 – very much]

Table S.181: Interviewee: I 4 Q 34

Answer	Score
There are several places we often go. Has some meaning, about 3	3

Q 35 - unc_31_34

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

Table S.182: Interviewee: I 4 Q 35

Answer	Score
Can't say, emotional things	2

Q 36 - fdays

Approximately how many days per year do you fish?

[free text description]

Table S.183: Interviewee: I 4 Q 36

Answer	Score - low (given)	Score - up (given)	Unit
Not much lately, maybe 10 days.	10	NA	days

Q 37 - income

Approximately how large is your gross income from a fishing day on average?

[free text description]

Table S.184: Interviewee: I 4 Q 37

Answer	Score - low (given)	Score - up (given)	Unit
NA	NA	NA	not_reported

Q 38 - season_maint

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

Table S.185: Interviewee: I 4 Q 38

Answer	Score - low (given)	Score - up (given)	Unit
5k SEK	NA	5000	sek_season

Q 39 - fuel_trip

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

Table S.186: Interviewee: I 4 Q 39

Answer	Score - low (given)	Score - up (given)	Unit
Significant; 20,000 SEK yearly, much for boats, transport to land, island transport. About 5,000 SEK/year.	5000	20000	sek_season

Q 40 - extra_fuel

What are the costs for extra fuel to travel further than normal?

[free text description]

Table S.187: Interviewee: I 4 Q 40

Answer	Score - low (given)	Score - up (given)	Unit
NA	NA	NA	not_reported

Q 41 - extra_crew

What are the costs for extra crew on board?

[free text description]

Table S.188: Interviewee: I 4 Q 41

Answer	Score - low (given)	Score - up (given)	Unit
NA	NA	NA	not_reported

Q 42 - extra_cleaning_cost

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

Table S.189: Interviewee: I 4 Q 42

Answer	Score - low (given)	Score - up (given)	Unit
When cleaning fish, takes a lot of time.	NA	NA	not_reported

Q 43 - exytra_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?

[free text description]

Table S.190: Interviewee: I 4 Q 43

Answer	Score - low (given)	Score - up (given)	Unit
I think we've gone through most of it. Water levels: your report mentions it, and many are concerned about low water levels, which indeed happened; but high water is just as problematic – wading out to the boathouse in waders.	NA	NA	not_reported

Q 44 - unc_36_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

Table S.191: Interviewee: I 4 Q 44

Answer	Score
There are some who have fishing rights but don't fish at all, and then some who fish like I do, just now and then. Then there are those who have leased out their fishing lot, and in those cases there is a subsistence fisher who fishes quite intensively	5

Q 45 - addition

Anything you want to add?

[free text description]

Table S.192: Interviewee: I 4 Q 45

Answer	Score
--------	-------

Regarding innovation: we're interested in all cooperation with experts who can help us restore ecological balance in the Baltic Sea.	-
There is interest moving forward. What I pick up is that we must do something about the algal blooms. Maybe find a way to collect them with boats; there is centrifuge competence in Sweden	999

Q 46 - substitution_capacity

If I lose one planned fishing day, I can usually make up for it later without it affecting what I achieve overall
[1-5; from 1 – strongly disagree, to 5 – strongly agree]

Table S.193: Interviewee: I 4 Q 46

Answer	Score
A lost fishing day does not constitute an impact because the fishing is not of economic importance.	5

Q 47 - unc_46

How much do you think the answers to these questions (question 46) vary between fishers in the group you represent?

Table S.194: Interviewee: I 4 Q 47

Answer	Score
Other subsistence fishermen will surely answer the same.	1

Interviewee: I 5

id5 represents the organization sportfiskarna, which is active in all sweden. he is based in stockholm. the members of sportfiskarna are anglers that mostly fish for pike, perch and zander. additionally, fishing for sea trout and trolling for salmon is common. they fish in lakes, streams and coastal areas. they can use boats in coastal areas and lakes, or just fish from the shore. in winter they also do ice fishing on lakes and generally they do not use boats until the ice is completely gone. the members practice both catch-and-release and keeping the catch. recreational fisherman can move around different fishing spots. how they adapt: sportfisherman are diverse but have in common to fish exclusively for fun. they tend to fish only when it is a pleasant experience. they can cope to some extent with extreme events by adapting their location if they want to fish on the coast (going deeper or finding sheltered places) or move to lakes. their spatial mobility has virtually no limits.

Q 2 - go_fishing

If the following type of extreme event occurs, how likely is it that you go out and fish?
[1-5; from 1 – very unlikely, to 5 – very likely]

Table S.195: Interviewee: I 5 Q 2

Event	Answer	Score
-------	--------	-------

a	Heatwave in the winter season	If it's otherwise a good winter, then it's just nice weather. But if it's a mild winter, then you can't go out. Is ice fishing the dominant form? Few go out in boats? On the big lakes. and at Simrishamn there is some trolling from boats. If we think about ice fishing? Absolutely, =1 if the ice isn't there or is dangerous to walk on.	1
b	Heatwave in the summer season	Yes, but in another way. When it's hot, fishing in running waters (salmon, trout, grayling) is closed. Perch and zander fishing only gets better in warm weather. Few target pike in a heatwave. But if you booked a salmon-fishing week, it simply gets canceled. Zander & perch = 5; salmon & trout = 1; pike = 2 (or 1, discussed with another angler present). But "vacation fishers" will probably go out regardless. Pike fishing is usually in spring & autumn; high temps affect survival.	NA
c	Severe icing (in the winter season)	Big impact.	1
d	Algal bloom (in the summer season)	Less attractive, but still possible to fish; =2.	2
e	Extreme cloudburst (in the summer season)	People hold back; it's a comfort factor. Most stay indoors	2
f	Gale	Yes, people hold back; most stay home, except the really keen ones who may go out in sheltered spots	2
g	Storm	No, impossible to fish.	1
h	Are there any of these in combination that affect how likely it is that you go out and fish?	Combination of gale/storm and heavy rain – Even more unpleasant, a negative synergy effect	NA

Multiple options identified

Table S.196: Interviewee: I 5 Q 2

	Event	target	Score (given)
b	Heatwave in the summer season	zander	5
b	Heatwave in the summer season	perch	5
b	Heatwave in the summer season	trout	1
b	Heatwave in the summer season	salmon	1
b	Heatwave in the summer season	pike	2

Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?
[1–5; from 1 – much worse, 3 – no difference, to 5 – much better]

Table S.197: Interviewee: I 5 Q 3

	Event	Answer	Score
a	Heatwave in the winter season	If there's no ice, you can't catch them with an ice rod, and you don't put in a boat just for a short ice-free spell.	-997
b	Heatwave in the summer season	species that dislike heat (salmonids) and even pike become harder to catch; perch and zander improve	NA
c	Severe icing (in the winter season)	Heavy icing is rare, and few troll in sub-zero temps – marginal effect on fish	2
d	Algal bloom (in the summer season)	Don't know	-998
e	Extreme cloudburst (in the summer season)	No.	3
f	Gale	Makes fishing more difficult, but doesn't affect catchability	2
g	Storm	Nobody goes out in storms, so we don't know if fish bite then	-998

Multiple options identified

Table S.198: Interviewee: I 5 Q 3

	Event	target	Score (given)
b	Heatwave in the summer season	zander	4
b	Heatwave in the summer season	perch	4
b	Heatwave in the summer season	trout	2
b	Heatwave in the summer season	salmon	2
b	Heatwave in the summer season	pike	2

Q 4 - catch_assemblage

Do you catch a different composition of species during such extreme weather events?
[1–5; from 1 – very similar, to 5 – very different species composition]

Table S.199: Interviewee: I 5 Q 4

	Event	Answer	Score
a	Heatwave in the winter season	No, very similar	1
b	Heatwave in the summer season	No, you fish where you fish, so you get the species present there. But in mountain areas, if you target trout or grayling but the system also has pike and perch, you might get more of those instead. From a boat it's usually small differences	2
c	Severe icing (in the winter season)	No, icing is rare	-998
d	Algal bloom (in the summer season)	No	1
e	Extreme cloudburst (in the summer season)	No	1
f	Gale	Not directly, but on the coast with prolonged winds you can get upwelling; perch may move closer inshore, and trout & whitefish may appear shallower than normal. Seen within a season	999
g	Storm	Not directly, but on the coast with prolonged winds you can get upwelling; perch may move closer inshore, and trout & whitefish may appear shallower than normal. Seen within a season	999

Q 5 - catch_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1-5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

Table S.200: Interviewee: I 5 Q 5

	Event	Answer	Score
a	Heatwave in the winter season	no	3
b	Heatwave in the summer season	Yes, species that thrive in cooler waters get poorer condition	2
c	Severe icing (in the winter season)	NA	-998
d	Algal bloom (in the summer season)	No (guess)	3
e	Extreme cloudburst (in the summer season)	no	3
f	Gale	no	3

Q 6 - cope_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.201: Interviewee: I 5 Q 6

	Event	Answer	Score
a	Heatwave in the winter season	NA	-999
b	Heatwave in the summer season	No, nothing can be done. If fish are in poor condition, there's nothing you can do. Doesn't really apply to sport fishing since fish aren't trapped in gear	1
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	NA	-999
e	Extreme cloudburst (in the summer season)	NA	-999
f	Gale	NA	-999
g	Storm	NA	-999

Q 7 - lag_effect

Do this extreme weather events have long-term effects on how you fish or catch composition

Greatest effect is probably on Arctic char (though not a major sportfishing species). Other species may move in: grayling, trout, whitefish, perch & pike. Biggest sportfishing impact from extreme weather is likely on char. On the coast, fish can always go deeper. Does it affect how you fish? Yes – e.g., trout anglers (spring & autumn from shore) may need to fish deeper or at different times. On Gotland, sea trout fishing is big – if warming lasts longer, the season gets shorter until it cools again.

Table S.202: Interviewee: I 5 Q 7

	Event	Answer	Score
a	Heatwave in the winter season	NA	-999
b	Heatwave in the summer season	NA	3
c	Severe icing (in the winter season)	NA	-999

d	Algal bloom (in the summer season)	NA	-999
e	Extreme cloudburst (in the summer season)	NA	-999
f	Gale	NA	-999
g	Storm	NA	-999

Q 8 - unc_2_7

How much do you think the answers to these questions (questions 2–7) vary between fishers in the group you represent? 1–5; from 1 – not at all, to 5 – very much

Table S.203: Interviewee: I 5 Q 8

Answer	Score
Depends on when/how you fish; varies a lot, =5.	5

Q 9 - personal_safety

How much concern for your personal safety do these extreme weather events bring?

[0–2; 0 – no concern, 1 – personal injuries, 2 – life-threatening

Table S.204: Interviewee: I 5 Q 9

	Event	Answer	Score
a	Heatwave in the winter season	for those fishing then, big concern – must assess ice conditions if a winter heatwave occurs	2
b	Heatwave in the summer season	0	0
c	Severe icing (in the winter season)	If someone is foolish enough to go out, some risk, but few do	1
d	Algal bloom (in the summer season)	0	0
e	Extreme cloudburst (in the summer season)	0	0
f	Gale	0	0
g	Storm	Surprise storms can be worrisome 2	2

Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0–3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

Table S.205: Interviewee: I 5 Q 10

	Event	Answer	Score
a	Heatwave in the winter season	0	0
b	Heatwave in the summer season	0	0
c	Severe icing (in the winter season)	1 (but most boats are stored then)	1
d	Algal bloom (in the summer season)	0	0
e	Extreme cloudburst (in the summer season)	could sink a boat if unlucky, 3	3
f	Gale	no	0
g	Storm	yes could sink a boat (3)	3

Q 11 - minor_damage_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

Table S.206: Interviewee: I 5 Q 11

Answer	Score - low (given)	Score - up (given)	Unit
Anything from 10,000 SEK to 1 million SEK – these are expensive toys; hard to say	10000	50000	sek_unit

Q 12 - major_damage_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

Table S.207: Interviewee: I 5 Q 12

Answer	Score - low (given)	Score - up (given)	Unit
Up to half a million SEK, not uncommon. For new trolling setups, even more. Sportfishing boats cost 50–300,000 SEK, sometimes double	50000	1000000	sek_unit

Q 13 - change_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0–2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

Table S.208: Interviewee: I 5 Q 13

	Event	Answer	Score
a	Heatwave in the winter season	2 – if the ice is unsafe, you don't go out.	2
b	Heatwave in the summer season	Mixed	NA
c	Severe icing (in the winter season)	NA	-998
d	Algal bloom (in the summer season)	Maybe 1 – switch to another bay; if bloom is everywhere, then skip.	1
e	Extreme cloudburst (in the summer season)	2 – you stay indoors	2
f	Gale	If sheltered, you can fish a little, =1, unless fully blown in.	1
g	Storm	Same	1

Multiple options identified

Table S.209: Interviewee: I 5 Q 13

	Event	target	Score (given)
b	Heatwave in the summer season	perch	0
b	Heatwave in the summer season	salmon/trout	2

Q 14 - unc_9_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

Table S.210: Interviewee: I 5 Q 14

Answer	Score
Depends on type of fishing – varies a lot, =5.	5

Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

Table S.211: Interviewee: I 5 Q 15

	Event	Answer	Score
a	Heatwave in the winter season	Hard, scale is large.	2
b	Heatwave in the summer season	Hard	2
c	Severe icing (in the winter season)	NA	-998
d	Algal bloom (in the summer season)	Easier – could switch to a lake if planning coastal fishing	4
e	Extreme cloudburst (in the summer season)	Hard	2
f	Gale	to some extent – find a sheltered spot	3
g	Storm	no	1

Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

Table S.212: Interviewee: I 5 Q 16

Answer	Score
Most can be somewhat mobile. Some are keen enough to travel far. =1.	1

Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.213: Interviewee: I 5 Q 17

Answer	Score
For mobile sport fishers, infrastructure is crucial. It exists, but not always accessible for everyone	1

Q 18 - management_space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.214: Interviewee: I 5 Q 18

Answer	Score
2. But you may need to buy a fishing license for new spots. On the coast it's free, which allows a lot of freedom.	2

Q 19 - management_species

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.215: Interviewee: I 5 Q 19

Answer	Score
Some restrictions. On the coast, fairly free. But salmon, trout & grayling are closed during spawning, or when it gets too warm (quite common). =1.	1

Q 20 - unc_15_19

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

Table S.216: Interviewee: I 5 Q 20

Answer	Score
Quite a lot – some don't fish from boats at all, some do.	4

Q 21 - stress

How much stress does the risk of the following events cause? A is for injuries, b for gear and c for condition [1–5; from 1 – really not stressful, to 5 – very stressful]

Table S.217: Interviewee: I 5 Q 21

Type	Answer	Score
a Injuries	Not really – only extreme cases (climbing steep riverbanks, pushing	1
b Damages to gear	no	1
c Fish condition	no	1

Q 22 - unc_21

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

Table S.218: Interviewee: I 5 Q 22

Answer	Score
some variation, but less than other questions	2

Q 23 - job_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.219: Interviewee: I 5 Q 23

Answer	Score
not asked	- 999

Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.220: Interviewee: I 5 Q 24

Answer	Score
I'm probably quite creative, I think so. =4.	4

Q 25 - credit

How easy is it to gain access to credits, financing or loans for investments

[1–5; from 1 – very difficult, to 5 – very easy]

Table S.221: Interviewee: I 5 Q 25

Answer	Score
not asked	- 999

Q 26 - unc_23_25

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

Table S.222: Interviewee: I 5 Q 26

Answer	Score
Varies greatly (only for Q24).	4

Q 27 - knowledge_transmission

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1–5; from 1 – not important at all, to 5 – very important]

Table S.223: Interviewee: I 5 Q 27

Answer	Score
Relatively important – nice if the next generation also enjoys fishing. Fairly important. =4.	4

Q 28 - benefit_local_community

How much does your local community benefit from your fishing

[1–5; from 1 – not important at all, to 5 – very important]

Table S.224: Interviewee: I 5 Q 28

Answer	Score
Generally, not a strong link. But in rural areas, sport fishing can keep the village shop open due to visiting guests in summer. In rural areas it has real significance	5

Q 29 - benefit_personal_community

How much does your “community” benefit from your fishing? Explain!

[1–5; from 1 – not important at all, to 5 – very important]

Table S.225: Interviewee: I 5 Q 29

Answer	Score
Some drive among those eager to show off successful fishing. But being part of that community doesn't really require fishing.	2

Q 30 - traditional_food

How important is your fishing for providing traditional food

[1–5; from 1 – not important at all, to 5 – very important]

Table S.226: Interviewee: I 5 Q 30

Answer	Score
In between. =3.	3

Q 31 - unc_26_30

How much do you think the answers to the questions about the contribution of fishing to society (questions 27–30) vary between fishers in the group you represent

Table S.227: Interviewee: I 5 Q 31

Answer	Score
Probably quite a lot.	4

Q 32 - outdoor_benefit

How important is it for you to be outdoors (when you fish) for your well-being
[1–5; from 1 – not important at all, to 5 – very important]

Table S.228: Interviewee: I 5 Q 32

Answer	Score
Quite significant – give it a 4.	4

Q 33 - identity

How much does fishing contribute to your identity and self-image
[1–5; from 1 – very little, to 5 – very much]

Table S.229: Interviewee: I 5 Q 33

Answer	Score
In between. =3.	3

Q 34 - sense_of_home

How much does fishing contribute to creating a bond to local places
[1–5; from 1 – very little, to 5 – very much]

Table S.230: Interviewee: I 5 Q 34

Answer	Score
Contributes strongly, at least a 4.	4

Q 35 - unc_31_34

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

Table S.231: Interviewee: I 5 Q 35

Answer	Score

Doesn't vary much. The place/environment matters, of course it varies somewhat, but everyone values getting out to beautiful places.
=2.

2

Q 36 - fdays

Approximately how many days per year do you fish?

[free text description]

Table S.232: Interviewee: I 5 Q 36

Answer	Score - low (given)	Score - up (given)	Unit
Not in SCB's survey. Average angler maybe 5–10 days, but some 50–100 days. More than 100? Yes, HC!	5	100	days

Q 37 - income

Approximately how large is your gross income from a fishing day on average?

[free text description]

Table S.233: Interviewee: I 5 Q 37

Answer	Score - low (given)	Score - up (given)	Unit
NA	NA	NA	not_reported

Q 38 - season_maint

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

Table S.234: Interviewee: I 5 Q 38

Answer	Score - low (given)	Score - up (given)	Unit
Servicing a boat: at least 5,000 SEK and up. Without a boat: maybe service a reel every couple of years, ~100 SEK	100	5000	sek_season

Q 39 - fuel_trip

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

Table S.235: Interviewee: I 5 Q 39

--	--	--	--

Answer	Score - low (given)	Score - up (given)	Unit
Varies hugely. Trolling anglers burn a lot. Ordinary perch & pike anglers maybe 30–50 L/trip on average	30	50	liter_trip

Q 40 - extra_fuel

What are the costs for extra fuel to travel further than normal?

[free text description]

Table S.236: Interviewee: I 5 Q 40

Answer	Score - low (given)	Score - up (given)	Unit
Maybe 50 L for a trolling angler. Boats are efficient, so 20–50 L/trip	20	50	liter_trip

Q 41 - extra_crew

What are the costs for extra crew on board?

[free text description]

Table S.237: Interviewee: I 5 Q 41

Answer	Score - low (given)	Score - up (given)	Unit
NA	NA	NA	not_reported

Q 42 - extra_cleaning_cost

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

Table S.238: Interviewee: I 5 Q 42

Answer	Score - low (given)	Score - up (given)	Unit
Some make their own lures/jigs/wobblers – for them, that’s half the fishing time, usually done in winter. Affected by algal bloom? No	999	999	hour

Q 43 - exytra_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?

[free text description]

Table S.239: Interviewee: I 5 Q 43

Answer	Score - low (given)	Score - up (given)	Unit
Yes – e.g., if you booked a salmon fishing week and they close fishing, you're stuck with costs for a cabin if it can't be rebooked. Usually you can rebook, but I don't know the cost.	NA	NA	not_reported

Q 44 - unc_36_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

Table S.240: Interviewee: I 5 Q 44

Answer	Score
5	5

Q 45 - addition

Anything you want to add?

[free text description]

Table S.241: Interviewee: I 5 Q 45

Answer	Score
I think you've captured most of it. The most likely major impacts are wind – if extreme weather is prolonged, it affects access. And if you fish salmonids, a summer heatwave affects both fish and fishing opportunities. Same with winter ice fishing.	- 999

Q 46 - substitution_capacity

If I lose one planned fishing day, I can usually make up for it later without it affecting what I achieve overall
[1–5; from 1 – strongly disagree, to 5 – strongly agree]

Table S.242: Interviewee: I 5 Q 46

Answer	Score
One fishing day might get lost, but you might be able to recover it later in the season	3

Q 47 - unc_46

How much do you think the answers to these questions (question 46) vary between fishers in the group you represent?

Table S.243: Interviewee: I 5 Q 47

Answer	Score
--------	-------

It might vary a lot among fishers. If you have booked a guide or an arrangement, a lost day might be difficult to recover. If you just go out fishing in your "home waters" it might be less of a loss. So substantial variation across fishers.

Interviewee: I 6

id6 fish pelagic fish both in the baltic sea and the north sea. in the baltic sea, the fishing targets herring/sprat and sharp-sprat, and the catch is sent to fishmeal. in the baltic sea, lars-uno fishes from the southern baltic up to the bothnian sea (at the level of sundsvall). they can occasionally fish in outermost regions (shetlands). the main fishing season is from august to april (but the season can extend longer into spring and summer), and the current law does not establish any limitation regarding time at sea (e.g.: seasonal ban, weekend ban). their activity is limited by the quota. in winter their activity in the baltic might be affected by ice coverage. how they adapt: pelagic trawlers have a great spatial mobility because of their large size vessels. their primary method to adapt to weather is to move around: they may switch from an area to another when they know the climatic conditions (i.e.: water temperature) are good. in the case of local weather is not good they stay in port. they have efficient cooling systems to deal with hot weather.

Q 2 - go_fishing

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.244: Interviewee: I 6 Q 2

	Event	Answer	Score
a	Heatwave in the winter season	5	5
b	Heatwave in the summer season	1; wait a minute, spring when the weather changes; in July when it's hottest, the fish are scattered, so there's no point in going out; say 2.	1.5
c	Severe icing (in the winter season)	we avoid going out, 1.	1
d	Algal bloom (in the summer season)	5	5
e	Extreme cloudburst (in the summer season)	5	5
f	Gale	4 in the Baltic Sea at least; if we are going to consume the fish in the North Sea, then we wait because the fish can spoil , but in the Baltic Sea we don't fish for human consumption	NA
g	Storm	1, we have the quota so we don't need to risk the crew.	1

h	Are there any of these in combination that affect how likely it is that you go out and fish?	Icing & high winds, then it's 1	NA
---	--	---------------------------------	----

Multiple options identified

Table S.245: Interviewee: I 6 Q 2

	Event	area	Score (given)
f	Gale	Baltic	4
f	Gale	NorthSea	1

Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better]

Table S.246: Interviewee: I 6 Q 3

	Event	Answer	Score
a	Heatwave in the winter season	It can go both better and worse, but it has an impact; for the same weather phenomenon, it can be both better and worse; weather changes affect catchability, as do moon phases and the time of year.	3
b	Heatwave in the summer season	in the North Sea, herring fishing is good in the summer; in August in Skagerrakk, we saw that when the water became warm, we got large quantities of herring at Skagen when the temperature was 20-21°C; But in the Baltic Sea, the fish spread out when it gets warm. 2	NA
c	Severe icing (in the winter season)	We can't load the boat when it's icy. 1.	1
d	Algal bloom (in the summer season)	1	1
e	Extreme cloudburst (in the summer season)	Doesn't affect 3.	3
f	Gale	3. It can sometimes, but no 3	3
g	Storm	it can be harder for us to handle the boat and get where we want to go, so it does have an impact. 2.	2

Multiple options identified

Table S.247: Interviewee: I 6 Q 3

--

	Event	area	Score (given)
b	Heatwave in the summer season	Baltic	2
b	Heatwave in the summer season	NorthSea	4

Q 4 - catch_assemblage

Do you catch a different composition of species during such extreme weather events?

[1–5; from 1 – very similar, to 5 – very different species composition]

Table S.248: Interviewee: I 6 Q 4

	Event	Answer	Score
a	Heatwave in the winter season	1	1
b	Heatwave in the summer season	NA	1
c	Severe icing (in the winter season)	1	1
d	Algal bloom (in the summer season)	1	1
e	Extreme cloudburst (in the summer season)	1	1
f	Gale	1	1
g	Storm	1	1

Q 5 - catch_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

Table S.249: Interviewee: I 6 Q 5

	Event	Answer	Score
a	Heatwave in the winter season	2, because the boat moves and the fish breaks	3
b	Heatwave in the summer season	1 same thing, in Lake Norske it becomes a problem because the boat moves and the fish breaks when it is warm	1
c	Severe icing (in the winter season)	3	3

d	Algal bloom (in the summer season)	3	3
e	Extreme cloudburst (in the summer season)	3	3
f	Gale	1	1
g	Storm	1	1

Q 6 - cope_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.250: Interviewee: I 6 Q 6

	Event	Answer	Score
a	Heatwave in the winter season	We cool with RSV systems, so it stays in good quality. 5	5
b	Heatwave in the summer season	We cool with RSV systems, so it stays in good quality. 5	5
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	NA	-999
e	Extreme cloudburst (in the summer season)	NA	-999
f	Gale	You can go ashore and unload what you have. 3	3
g	Storm	You can go ashore and unload what you have. 3	3

Q 7 - lag_effect

Do this extreme weather events have long-term effects on how you fish or catch composition

4, we saw that mackerel migrate, but the funny thing is that they are starting to come back again, as they did 20 years ago. I don't know if this has anything to do with the heat, or if it's a cycle; mackerel have been north of Shetland, but now it seems that they are back in Norway again; perhaps they will return to how it was 20-25 years ago; In the Baltic Sea, I think cod has been affected because it has gotten warmer. Thirty-five to forty years ago, the entire Baltic Sea was frozen, but sprat and herring have increased in the last five years. Smelt has also recovered because it has gotten warmer. Yes, it has affected us.

Table S.251: Interviewee: I 6 Q 7

Event	Answer	Score
-------	--------	-------

a	Heatwave in the winter season	NA	4
b	Heatwave in the summer season	NA	4
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	NA	-999
e	Extreme cloudburst (in the summer season)	NA	-999
f	Gale	NA	-999
g	Storm	NA	-999

Q 8 - unc_2_7

How much do you think the answers to these questions (questions 2–7) vary between fishers in the group you represent? 1–5; from 1 – not at all, to 5 – very much

Table S.252: Interviewee: I 6 Q 8

Answer	Score
I think they have roughly the same opinion, but there are probably some who think a little differently, so =3	3

Q 9 - personal_safety

How much concern for your personal safety do these extreme weather events bring?

[0–2; 0 – no concern, 1 – personal injuries, 2 – life-threatening

Table S.253: Interviewee: I 6 Q 9

	Event	Answer	Score
a	Heatwave in the winter season	1 if the heat causes high winds	0
b	Heatwave in the summer season	no concern, usually calm weather	0
c	Severe icing (in the winter season)	2 as the boat may capsize and sink	2
d	Algal bloom (in the summer season)	0	0
e	Extreme cloudburst (in the summer season)	1	1

f	Gale	1	1
g	Storm	2, perhaps even 3	2

Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0–3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

Table S.254: Interviewee: I 6 Q 10

	Event	Answer	Score
a	Heatwave in the winter season	3, [heat wave in winter, often accompanied by storms]	0
b	Heatwave in the summer season	0	0
c	Severe icing (in the winter season)	3 (but preferably without the trawl in, then we can get ashore)	3
d	Algal bloom (in the summer season)	0	0
e	Extreme cloudburst (in the summer season)	0	0
f	Gale	0	0
g	Storm	3 can even damage the trawl; This is because we have a 40 m boat, but those with 60-80 m boats in the Baltic Sea have no problem turning with trawls during storms in the Baltic Sea	3

Q 11 - minor_damage_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

Table S.255: Interviewee: I 6 Q 11

Answer	Score - low (given)	Score - up (given)	Unit
Can be repaired on board; 50,000 to repair a trawl, transport it 10,000 SEK; if the entire trawl were destroyed, our trawl costs 200,000 SEK, there are trawls that cost 800,000 SEK, if you lose it Repair costs 50,000-30,000 trawls The boat is rarely damaged; but a 60 m boat got a hole in the side so they had to replace a plate, which cost 20-30,000 SEK, but we go ash	re 10000	50000	sek_u

Q 12 - major_damage_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

Table S.256: Interviewee: I 6 Q 12

Answer	Score - low (given)	Score - up (given)	Unit
See 11	1e+05	5e+05	sek_unit

Q 13 - change_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0–2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

Table S.257: Interviewee: I 6 Q 13

	Event	Answer	Score
a	Heatwave in the winter season	2, stays upwind or goes ashore when the wind picks up to storm force, and waits 1-2 days until it calms down; when the waves are too high, the trawl does not fish as well.	2
b	Heatwave in the summer season	0	0
c	Severe icing (in the winter season)	2, does not go out	2
d	Algal bloom (in the summer season)	0	0
e	Extreme cloudburst (in the summer season)	0	0
f	Gale	0	0
g	Storm	2, does not go out	2

Q 14 - unc_9_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

Table S.258: Interviewee: I 6 Q 14

Answer	Score
Depends on the size of the boat, varies quite a lot, 4.	4

Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

Table S.259: Interviewee: I 6 Q 15

	Event	Answer	Score
a	Heatwave in the winter season	We avoid it, we can go down into the Baltic Sea on the other side of Åland; we can move, we go closer to Finland, 3-4.	3.5
b	Heatwave in the summer season	It has no effect	-999
c	Severe icing (in the winter season)	Yes, we can change course if we know in advance, but if the wind is too strong, it doesn't help to move closer to land; we can't go closer than 40 km	4
d	Algal bloom (in the summer season)	No effect	-999
e	Extreme cloudburst (in the summer season)	No effect	-999
f	Gale	No effect	-999
g	Storm	as we mentioned earlier	4

Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0-2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

Table S.260: Interviewee: I 6 Q 16

Answer	Score
Yes, 2.	2

Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0-2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.261: Interviewee: I 6 Q 17

Answer	Score
We have a few ports where we can land, but there's a small problem at the moment because there have been some ideological issues in Norrsundet, so now we're going to Oxelesund instead. 1	1

Q 18 - management_space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.262: Interviewee: I 6 Q 18

Answer	Score
Yes, I can choose the North Sea or Skagerrak instead if it's too windy in the Baltic Sea. 2	2

Q 19 - management_species

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.263: Interviewee: I 6 Q 19

Answer	Score
Not asked	- 999

Q 20 - unc_15_19

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

Table S.264: Interviewee: I 6 Q 20

Answer	Score
I think they think the same (=1 or 2?)	1.5

Q 21 - stress

How much stress does the risk of the following events cause? A is for injuries, b for gear and c for condition [1–5; from 1 – really not stressful, to 5 – very stressful]

Table S.265: Interviewee: I 6 Q 21

Type	Answer	Score
a Injuries	Quite high, it's always there lurking, you're always a little worried, 4, because my 15-year-old son has to go with us.	4
b Damages to gear	Also stressful	4
c Fish condition	yes, it's also stressful because we lose income, quite high stress = 4	4

Q 22 - unc_21

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

Table S.266: Interviewee: I 6 Q 22

Answer	Score
I think it's pretty similar, 2	2

Q 23 - job_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1-5; from 1 – very unlikely, to 5 – very likely]

Table S.267: Interviewee: I 6 Q 23

Answer	Score
no; some have tried to engage in wind power, with smaller boats. =1	1

Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1-5; from 1 – very unlikely, to 5 – very likely]

Table S.268: Interviewee: I 6 Q 24

Answer	Score
Yes, we are working on that, herring burgers, producing more products from the Baltic Sea for human consumption; we have to adapt, if it is to be more humane we have to adjust and go ashore after maybe 1-2 days; now we are out for a week; to maintain quality. (=4 or 5?).	4.5

Q 25 - credit

How easy is it to gain access to credits, financing or loans for investments

[1-5; from 1 – very difficult, to 5 – very easy]

Table S.269: Interviewee: I 6 Q 25

Answer	Score
Loans are no problem, we got help developing the equipment, it's much worse today; but you can get help to buy a new trawl =5	5

Q 26 - unc_23_25

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

Table S.270: Interviewee: I 6 Q 26

Answer	Score
large variation between different boats, 5	5

Q 27 - knowledge_transmission

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1–5; from 1 – not important at all, to 5 – very important]

Table S.271: Interviewee: I 6 Q 27

Answer	Score
very important, 5.	5

Q 28 - benefit_local_community

How much does your local community benefit from your fishing

[1–5; from 1 – not important at all, to 5 – very important]

Table S.272: Interviewee: I 6 Q 28

Answer	Score
we do a little repair work there; we don't land any fish in Fiskebäck; it could be better if we landed there; to a certain extent, but somewhat average, 3	3

Q 29 - benefit_personal_community

How much does your “community” benefit from your fishing? Explain!

[1–5; from 1 – not important at all, to 5 – very important]

Table S.273: Interviewee: I 6 Q 29

Answer	Score
For my colleagues? Well, you want to have your colleagues, because there are so few of us now, so you want to keep all your colleagues. =5	5

Q 30 - traditional_food

How important is your fishing for providing traditional food

[1–5; from 1 – not important at all, to 5 – very important]

Table S.274: Interviewee: I 6 Q 30

Answer	Score
I think it's very important, when we consume herring, canned herring, it's important, =5	5

Q 31 - unc_26_30

How much do you think the answers to the questions about the contribution of fishing to society (questions 27–30) vary between fishers in the group you represent

Table S.275: Interviewee: I 6 Q 31

Answer	Score
It varies, because we are pelagic, and then you have the demersal fishermen, who are a different species; among the pelagic fishermen, it is quite similar, it is important to everyone.	2

Q 32 - outdoor_benefit

How important is it for you to be outdoors (when you fish) for your well-being
[1–5; from 1 – not important at all, to 5 – very important]

Table S.276: Interviewee: I 6 Q 32

Answer	Score
It's good to get out on deck, get out on the sea? It's an important part!	4

Q 33 - identity

How much does fishing contribute to your identity and self-image
[1–5; from 1 – very little, to 5 – very much]

Table S.277: Interviewee: I 6 Q 33

Answer	Score
it's strong	5

Q 34 - sense_of_home

How much does fishing contribute to creating a bond to local places
[1–5; from 1 – very little, to 5 – very much]

Table S.278: Interviewee: I 6 Q 34

Answer	Score
Yes, Skagen is a really fantastic place, actually very important	5

Q 35 - unc_31_34

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

Table S.279: Interviewee: I 6 Q 35

Answer	Score
I think everyone thinks pretty much the same	2

Q 36 - fdays

Approximately how many days per year do you fish?

[free text description]

Table S.280: Interviewee: I 6 Q 36

Answer	Score - low (given)	Score - up (given)	Unit
In the Baltic Sea, from November to June, about half a year, 7 months, at the moment, but that can change; we fish about 80% of those days, are on board and have the trawl in and are fishing	140	170	days

Q 37 - income

Approximately how large is your gross income from a fishing day on average?

[free text description]

Table S.281: Interviewee: I 6 Q 37

Answer	Score - low (given)	Score - up (given)	Unit
It varies; if we're talking about all boats, we collaborate with another boat, and we've exchanged some fish between us; in the Bothnian Sea, [calculating] 11,000/hour, 25,000 per day in the Bothnian Sea; but off Shetland, we caught 400 tons of mackerel, the entire annual quota, in one hour! That was 8 million/hour	25	30	Million_sek_season

Q 38 - season_maint

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

Table S.282: Interviewee: I 6 Q 38

Answer	Score - low (given)	Score - up (given)	Unit

We are doing extensive renovations now, because we are doing a 10-year class when we go through the entire boat, so it could be 10% of the catch, in a normal year it is 1% of the catch (has an income of SEK 25-30 million per year), so 1% of that. If there are two trawls in a season, SEK 300-400. We have an extra trawl with us, as we have two drums, and we have 1-2 trawls in Norrsundet	1	10	% of income
---	---	----	-------------

Q 39 - fuel_trip

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

Table S.283: Interviewee: I 6 Q 39

Answer	Score - low (given)	Score - up (given)	Unit
if a fishing trip lasts about 1 week; in the Bothnian Sea; 50 m3/day * 6 days* = 240 kkr/week 2.5 tons * 4.5*22 hours; runs 22 hours per day 5000L*8h/day; for one week 240 kkr/week (in the Bothnian Sea) [talks about mackerel fishing]	600	650	liters_hour

Q 40 - extra_fuel

What are the costs for extra fuel to travel further than normal?

[free text description]

Table S.284: Interviewee: I 6 Q 40

Answer	Score - low (given)	Score - up (given)	Unit
transport by truck to Skagen from Norrsundet; do not want to drive the boat that far, it costs money; the Bothnian Sea to the Baltic Sea takes 1.5 days	NA	NA	not_reported

Q 41 - extra_crew

What are the costs for extra crew on board?

[free text description]

Table S.285: Interviewee: I 6 Q 41

Answer	Score - low (given)	Score - up (given)	Unit
Occasionally, if someone is injured or unavailable; there are four of us on the boat; previously, when we had the net, we needed nine people, but that was ten years ago; substitutes receive the same pay	NA	NA	not_reported

Q 42 - extra_cleaning_cost

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

Table S.286: Interviewee: I 6 Q 42

Answer	Score - low (given)	Score - up (given)	Unit
No, we don't need to clean the trawls; we lower them into the sea upside down so that they get clean; we both hose them down and dip them; no difference with algae blooms.	0	0	hours

Q 43 - exytra_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?

[free text description]

Table S.287: Interviewee: I 6 Q 43

Answer	Score - low (given)	Score - up (given)	Unit
Yes, a window can break if you hit a big wave. One boat had a compass on the roof that broke and made a hole right through the roof, so it can cause damage and break things. If water gets into the boat and damages the electronics, it becomes more expensive. 10,000 for a window	NA	NA	not_reported

Q 44 - unc_36_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

Table S.288: Interviewee: I 6 Q 44

Answer	Score
It varies a lot, many have a very new boat, so the costs are very low, and if you have a larger boat, it can withstand extreme weather better. Astrid loads > 2000 tons, while we load 500 tons, and we have to make 4 trips back and forth, while they only make 1 trip, so they probably have lower fuel costs per fish, even though they consume more when they drive	5

Q 45 - addition

Anything you want to add?

[free text description]

Table S.289: Interviewee: I 6 Q 45

--

Answer	Score
In terms of extreme weather with storms, I think we've covered everything; this large variation in temperature in the Baltic Sea is something you're looking at, I think, with temperature data from SMHI.	- 999

Q 46 - substitution_capacity

If I lose one planned fishing day, I can usually make up for it later without it affecting what I achieve overall [1–5; from 1 – strongly disagree, to 5 – strongly agree]

Table S.290: Interviewee: I 6 Q 46

Answer	Score
Because of the ITQ system, we can wait until the weather conditions improve. Before this system was implemented, people went out and fished despite the risks involved.	5

Q 47 - unc_46

How much do you think the answers to these questions (question 46) vary between fishers in the group you represent?

Table S.291: Interviewee: I 6 Q 47

Answer	Score
Same answer if you ask other pelagic fishers	1

Interviewee: I 7

id7 is a professional coastal fisherman in the southwestern bothnian sea (Älvsgrund area?). they fish with gillnets and traps. the main fishing season is summer. they can do some ice fishing. the main target is herring, whitefish, salmon, perch. they have more salmon traps than the average coastal fisherman. fishing for perch (nets) is done from a small boat close to home, while fishing for herring happens with a larger boat more far away. a trip for herring can cost up to 100 litres of diesel (approximately 10 l*hour at 10 knots means that they can go to cast at least 50-60 km from home). how they adapt: they primarily adapt when fishing with nets to hw by changing fishing target. when it's hot they aim for perch and not for herring. in addition, they can perform shorter sets and use cooling boxes. can also adapt to hw and storms by changing fishing location. they are not affected by ab. to avoid hw they can go further out in deeper waters, the opposite for storms. in any case they can move on short distances. their spatial mobility is mainly affected by the boat and infrastructure (they want to go back home) on second instance by management that makes possible to fish in proximity. there is not the same flexibility for salmon traps, which are fixed.

Q 2 - go_fishing

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.292: Interviewee: I 7 Q 2

	Event	Answer	Score
a	Heatwave in the winter season	I probably go fishing anyway, it's the wind that matters, I probably try anyway (=5? or =4?)	4.5
b	Heatwave in the summer season	I don't go because the fish won't keep as well, I don't fish for herring at all, in that case I fish for perch, 2	2
c	Severe icing (in the winter season)	I am not on the lake at that time of year when there is a risk of icing (because the ice forms so early	-999
d	Algal bloom (in the summer season)	I have not experienced it, but it probably has an impact, so then I don't fish	-998
e	Extreme cloudburst (in the summer season)	then I fish anyway	4
f	Gale	depends on the wind direction, if it's N then it's from the outside so then it's difficult, if it's S or SW, in gale force winds I don't fish anyway because there's a strong current in the lake so you get crap in your nets, but I do go out in gale force winds sometimes , depends on the wind direction	3
g	Storm	1	1
h	Are there any of these in combination that affect how likely it is that you go out and fish?	storms and weather like that	NA

Multiple options identified

Table S.293: Interviewee: I 7 Q 2

	Event	target	Score (given)
b	Heatwave in the summer season	herring	1
b	Heatwave in the summer season	perch	2

Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1-5; from 1 – much worse, 3 – no difference, to 5 – much better]

Table S.294: Interviewee: I 7 Q 3

	Event	Answer	Score
a	Heatwave in the winter season	difficult to say, haven't experienced it that much, am more limited when it's windy because seaweed gets caught in the nets (enough if it's gale force), f. fishing gets much worse	-999

b	Heatwave in the summer season	not when it comes to perch, but herring is much more sensitive to heat, so it gets much worse	NA
c	Severe icing (in the winter season)	We don't ask about that	-999
d	Algal bloom (in the summer season)	No, I've only heard that fishing gets worse; I've seen some green filamentous algae on the surface this year (I've seen a little in previous years; it never stays on the surface)	-998
e	Extreme cloudburst (in the summer season)	3	3
f	Gale	1	1
g	Storm	1	1

Multiple options identified

Table S.295: Interviewee: I 7 Q 3

	Event	target	Score (given)
b	Heatwave in the summer season	herring	1
b	Heatwave in the summer season	perch	3

Q 4 - catch_assemblage

Do you catch a different composition of species during such extreme weather events?

[1–5; from 1 – very similar, to 5 – very different species composition]

Table S.296: Interviewee: I 7 Q 4

	Event	Answer	Score
a	Heatwave in the winter season	difficult to answer	-999
b	Heatwave in the summer season	herring are fished a little further out, and perch closer in; always fish in different places, so the composition is not affected in each place	2
c	Severe icing (in the winter season)	We skip this one	-999
d	Algal bloom (in the summer season)	Haven't experienced it, but heard that it affects fishing	-999
e	Extreme cloudburst (in the summer season)	1	1

f	Gale	Yes, I think it affects the composition; seaweed gets caught in the nets and creates currents; can't say if other species are affected	-998
g	Storm	Yes, I think it affects the composition; seaweed gets caught in the nets and creates currents; can't say if other species are affected	-998

Q 5 - catch_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1-5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

Table S.297: Interviewee: I 7 Q 5

	Event	Answer	Score
a	Heatwave in the winter season	3	3
b	Heatwave in the summer season	Yes, the warmer it is, the shorter the shelf life of the fish, especially if it is fatty fish (herring, salmon, and whitefish), which is in the summer; herring is particularly sensitive 1	2
c	Severe icing (in the winter season)	Not asked	-999
d	Algal bloom (in the summer season)	Cannot answer	-999
e	Extreme cloudburst (in the summer season)	3	3
f	Gale	No, I don't think it's the quality, it's probably the catch volume 3	3
g	Storm	3	3

Multiple options identified

Table S.298: Interviewee: I 7 Q 5

	Event	target	Score (given)
b	Heatwave in the summer season	salmon	1.5
b	Heatwave in the summer season	whitefish	1.5
b	Heatwave in the summer season	herring	1

Q 6 - cope_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.299: Interviewee: I 7 Q 6

	Event	Answer	Score
a	Heatwave in the winter season	NA	-999
b	Heatwave in the summer season	Yes, a little (3), I can fish more often, for perch you can keep the nets in for shorter periods, the fish freeze as soon as I get in	3
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	NA	-999
e	Extreme cloudburst (in the summer season)	NA	-999
f	Gale	NA	-999
g	Storm	NA	-999

Q 7 - lag_effect

Do this extreme weather events have long-term effects on how you fish or catch composition

Of course it has changed, but we don't know why; more storms and such have a certain impact on how I fish, 3-4

Table S.300: Interviewee: I 7 Q 7

	Event	Answer	Score
a	Heatwave in the winter season	NA	-999
b	Heatwave in the summer season	NA	-999
c	Severe icing (in the winter season)	NA	-999
d	Algal bloom (in the summer season)	NA	-999
e	Extreme cloudburst (in the summer season)	NA	-999
f	Gale	NA	3.5

g	Storm	NA	3.5
---	-------	----	-----

Q 8 - unc_2_7

How much do you think the answers to these questions (questions 2–7) vary between fishers in the group you represent? 1–5; from 1 – not at all, to 5 – very much

Table S.301: Interviewee: I 7 Q 8

Answer	Score
roughly the same, roughly the same conditions, 2	2

Q 9 - personal_safety

How much concern for your personal safety do these extreme weather events bring?

[0–2; 0 – no concern, 1 – personal injuries, 2 – life-threatening

Table S.302: Interviewee: I 7 Q 9

	Event	Answer	Score
a	Heatwave in the winter season	if the ice becomes unsafe, there is a risk of falling into the water, which can be life-threatening; if you have equipment in the lake, you want to be able to access it, 2	NA
b	Heatwave in the summer season	0	0
c	Severe icing (in the winter season)	not asked	-999
d	Algal bloom (in the summer season)	0	0
e	Extreme cloudburst (in the summer season)	0	0
f	Gale	2	2
g	Storm	2	2

Multiple options identified

Table S.303: Interviewee: I 7 Q 9

	Event	gear	Score (given)
a	Heatwave in the winter season	icefishing	2

Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0–3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

Table S.304: Interviewee: I 7 Q 10

	Event	Answer	Score
a	Heatwave in the winter season	0	0
b	Heatwave in the summer season	0	0
c	Severe icing (in the winter season)	not asked	-999
d	Algal bloom (in the summer season)	Yes, it could be on the gear in the nets, getting clogged, so 1	1
e	Extreme cloudburst (in the summer season)	0 , the boat could fill with water, but that would have to be extreme; if you have an open boat, it could happen	1
f	Gale	On gear and on the boat; 1	1
g	Storm	. 2, gear could be completely destroyed, such as salmon traps and so on	2

Q 11 - minor_damage_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

Table S.305: Interviewee: I 7 Q 11

Answer	Score - low (given)	Score - up (given)	Unit
You know about nets, that's minor damage, but salmon traps cost more to repair because they're special equipment; a few thousand for nets, but salmon traps can cost tens of thousands to repair. Yes, on the boat, it's probably if you drive in bad weather, and there is greater damage to the propeller and gearbox on smaller boats, and on large boats it's also the propeller, then it's tens of thousands of kronor.	1000	50000	sek_unit

Q 12 - major_damage_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

Table S.306: Interviewee: I 7 Q 12

Answer	Score - low (given)	Score - up (given)	Unit
see 11	50000	100000	sek_unit

Q 13 - change_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0–2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

Table S.307: Interviewee: I 7 Q 13

	Event	Answer	Score
a	Heatwave in the winter season	Yes, if there is no ice, I cannot fish, so I have to fish from a boat, but that is usually not very productive; 2	2
b	Heatwave in the summer season	1 changes how much perch I fish compared to herring	1
c	Severe icing (in the winter season)	I don't know, I haven't experienced it (maybe I'll skip it then),	-999
d	Algal bloom (in the summer season)	NA	-999
e	Extreme cloudburst (in the summer season)	0	0
f	Gale	Changes the fishing spot	1
g	Storm	2; Do you go out in a storm? It has happened that you make a mistake and go out, but you have to put the nets in a sheltered spot, you have your spots	2

Q 14 - unc_9_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

Table S.308: Interviewee: I 7 Q 14

Answer	Score
They probably have similar conditions, so they probably answer quite similarly; =2	2

Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

Table S.309: Interviewee: I 7 Q 15

	Event	Answer	Score
a	Heatwave in the winter season	1 (if the ice is not strong enough, it's pretty much the same, then you have to go very far),	1
b	Heatwave in the summer season	I don't know, but you can still fish for perch; the roach moves further out, and I can go a little further out, but I don't have a very big boat, so I can't go that far out, 2	2
c	Severe icing (in the winter season)	I think so , it depends on how it accumulates in certain places	4
d	Algal bloom (in the summer season)	NA	-999
e	Extreme cloudburst (in the summer season)	1 (but it doesn't matter)	1
f	Gale	4, depends on the wind direction, and sometimes the fishing spots don't match where it's sheltered.	4
g	Storm	same there, but you rarely fish in a storm if you know about it, but sometimes you have to go out as I said	4

Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0-2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

Table S.310: Interviewee: I 7 Q 16

Answer	Score (self)	Score (given)
I have mobile gear (perch can move distance =1), herring =1 (somewhat limited by boat), but I cannot choose the location for fixed gear (= 0)	M	NA

Multiple options identified

Table S.311: Interviewee: I 7 Q 16

gear	Score (given)
NA	1
NA	0

Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.312: Interviewee: I 7 Q 17

Answer	Score
Extremely, there are other places to go ashore; fixed with the salmon because then you have to report the landing port before you unload them (and when you unload); for the herring, I have the shaking machine in the boathouse, so it is difficult to find out the catch elsewhere, but I can go to an emergency port, i.e. =0	0

Q 18 - management_space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.313: Interviewee: I 7 Q 18

Answer	Score
yes, in a limited area; the license determines where you can fish; we have quotas and fixed gear in fixed locations for salmon, but with mobile gear you can move around in the local area at least; =1	1

Q 19 - management_species

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Table S.314: Interviewee: I 7 Q 19

Answer	Score
There isn't much else to fish for; we catch other fish species, but in much smaller quantities, such as pike and pikeperch; salmon is subject to quotas; herring too, but we don't fill the coastal quota, and haven't done so for several years.	999

Q 20 - unc_15_19

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

Table S.315: Interviewee: I 7 Q 20

Answer	Score
Not everyone has salmon traps, but otherwise it's pretty similar; =2.	2

Q 21 - stress

How much stress does the risk of the following events cause? A is for injuries, b for gear and c for condition [1–5; from 1 – really not stressful, to 5 – very stressful]

Table S.316: Interviewee: I 7 Q 21

	Type	Answer	Score
a	Injuries	1 no, I don't think about it much	1
b	Damages to gear	sometimes, when the weather is bad, windy, stormy, that's when you feel like you have stuff out there, it's mostly the salmon traps you think about; you usually have time to remove the movable gear if you know bad weather is coming; 4	4
c	Fish condition	I never think about it, 1	1

Q 22 - unc_21

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

Table S.317: Interviewee: I 7 Q 22

Answer	Score
I would think they would answer the same, =1	1

Q 23 - job_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.318: Interviewee: I 7 Q 23

Answer	Score
a little; 3	3

Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1–5; from 1 – very unlikely, to 5 – very likely]

Table S.319: Interviewee: I 7 Q 24

Answer	Score
I try to develop fishing so that it is as easy as possible to do certain things, and I try a little bit to find new ways to sell. =4	4

Q 25 - credit

How easy is it to gain access to credits, financing or loans for investments

[1–5; from 1 – very difficult, to 5 – very easy]

Table S.320: Interviewee: I 7 Q 25

Answer	Score
It probably depends on how much money and what for. You need to have good ideas. So, to buy a salmon trap? It's a bit uncertain because salmon stocks are low at the moment. A filleting machine? It's probably not entirely impossible.	2

Q 26 - unc_23_25

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

Table S.321: Interviewee: I 7 Q 26

Answer	Score
I think it would vary a bit; some don't do much processing and mostly want to fish, so 4	4

Q 27 - knowledge_transmission

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1–5; from 1 – not important at all, to 5 – very important]

Table S.322: Interviewee: I 7 Q 27

Answer	Score
I've given up. I wish it could continue, but it's so difficult, there are so many problems, with the authorities, among other things. Yes, I think it's important. I'm interested, but the financial aspect is difficult for a new start. =5	5

Q 28 - benefit_local_community

How much does your local community benefit from your fishing

[1–5; from 1 – not important at all, to 5 – very important]

Table S.323: Interviewee: I 7 Q 28

Answer	Score
They get fresh fish nearby; quite a few people shop there, people come to his home to buy fish, and not just when he's at the market; =5	5

Q 29 - benefit_personal_community

How much does your “community” benefit from your fishing? Explain!

[1–5; from 1 – not important at all, to 5 – very important]

Table S.324: Interviewee: I 7 Q 29

Answer	Score
No, there are so few who fish. Do you share experiences? Yes, we benefit from each other, in the past they couldn't talk to each other. Sometimes it was a matter of life and death, you've heard about that, it was when there were several in each village, but now there are so few of us that it's easy to get along. =4	4

Q 30 - traditional_food

How important is your fishing for providing traditional food

[1–5; from 1 – not important at all, to 5 – very important]

Table S.325: Interviewee: I 7 Q 30

Answer	Score
5, that's what I do.	5

Q 31 - unc_26_30

How much do you think the answers to the questions about the contribution of fishing to society (questions 27–30) vary between fishers in the group you represent

Table S.326: Interviewee: I 7 Q 31

Answer	Score
Some, and some would say differently, =3	3

Q 32 - outdoor_benefit

How important is it for you to be outdoors (when you fish) for your well-being

[1–5; from 1 – not important at all, to 5 – very important]

Table S.327: Interviewee: I 7 Q 32

Answer	Score
yes, it is quite important, it is very important =5	5

Q 33 - identity

How much does fishing contribute to your identity and self-image

[1–5; from 1 – very little, to 5 – very much]

Table S.328: Interviewee: I 7 Q 33

Answer	Score
quite a lot (=4 or 5? [He doesn't use any superlatives!])	4.5

Q 34 - sense_of_home

How much does fishing contribute to creating a bond to local places

[1–5; from 1 – very little, to 5 – very much]

Table S.329: Interviewee: I 7 Q 34

Answer	Score
yes, but I usually go to the same place, and stupidly enough, I don't try out many different places; some places are special to visit =4?	4

Q 35 - unc_31_34

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

Table S.330: Interviewee: I 7 Q 35

Answer	Score
They would probably be similar; =2.	2

Q 36 - fdays

Approximately how many days per year do you fish?

[free text description]

Table S.331: Interviewee: I 7 Q 36

Answer	Score - low (given)	Score - up (given)	Unit
around 200 days	NA	200	days

Q 37 - income

Approximately how large is your gross income from a fishing day on average?

[free text description]

Table S.332: Interviewee: I 7 Q 37

Answer	Score - low (given)	Score - up (given)	Unit

It depends on processing, fishing, and smoking; it varies at different times of the year. Spring and early summer are quite intense, then it slows down. You don't get your money back for surströmming until you've sold the cans. Varies from SEK 10,000-50,000/day, depending on the catch and salmon and such; then it is processed. If you get 100 salmon in a trap in one day, that's worth quite a lot just raw, then you smoke it and save it, so you can spread it out over the

year | 50000 | se
| 1000

Q 38 - season_maint

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

Table S.333: Interviewee: I 7 Q 38

Answer	Score - low (given)	Score - up (given)	Unit
Nets require quite a lot of work, and salmon traps in the spring; say SEK 10,000-35,000/year. And in terms of time? About 30 days, depending on whether you encounter breakdowns and such.	10000	35000	sek_season

Q 39 - fuel_trip

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

Table S.334: Interviewee: I 7 Q 39

Answer	Score - low (given)	Score - up (given)	Unit
30-50 liters of diesel for herring; if fishing for perch, the costs are not so high because it is quite close, then 4-5 liters of gasoline (because it is a small boat).	5	50	liter_trip

Q 40 - extra_fuel

What are the costs for extra fuel to travel further than normal?

[free text description]

Table S.335: Interviewee: I 7 Q 40

Answer	Score - low (given)	Score - up (given)	Unit
100L of diesel for herring	50	100	liter_trip

Q 41 - extra_crew

What are the costs for extra crew on board?

[free text description]

Table S.336: Interviewee: I 7 Q 41

Answer	Score - low (given)	Score - up (given)	Unit
has one specifically for herring fishing; he does not fish that many days; costs SEK 60,000-100,000/year	60000	100000	sek_year

Q 42 - extra_cleaning_cost

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

Table S.337: Interviewee: I 7 Q 42

Answer	Score - low (given)	Score - up (given)	Unit
Rinses the salmon traps, takes a few hours in both spring and fall, approx. 20 hours; extra work if there is algae blooms; sometimes there is growth and sometimes there is moss floating around, which is difficult to rinse off, so it has to lie there and rot first; it grows on what is on the beaches and if it is windy, it comes loose and gets stuck in the trap	20	NA	hours

Q 43 - exytra_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?

[free text description]

Table S.338: Interviewee: I 7 Q 43

Answer	Score - low (given)	Score - up (given)	Unit
No, it's usually repairs if there is any damage, etc	NA	NA	not_reported

Q 44 - unc_36_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

Table S.339: Interviewee: I 7 Q 44

Answer	Score
Yes, I think so, 5	5

Q 45 - addition

Anything you want to add?

[free text description]

Table S.340: Interviewee: I 7 Q 45

Answer	Score
I don't have much experience with it yet, but what we've seen is milder winters and more wind; things have changed, but we don't know what the cause is.	- 999

Q 46 - substitution_capacity

If I lose one planned fishing day, I can usually make up for it later without it affecting what I achieve overall

[1-5; from 1 – strongly disagree, to 5 – strongly agree]

Table S.341: Interviewee: I 7 Q 46

Answer	Score
If it comes on the wrong day of the week, an entire week could be ruined as I always sell fish on Fridays. So if Tuesdays and Wednesdays are lost during a week, it is not certain that it will be possible to get that week back.	3

Q 47 - unc_46

How much do you think the answers to these questions (question 46) vary between fishers in the group you represent?

Table S.342: Interviewee: I 7 Q 47

Answer	Score
Colleagues who work in a similar way to Lars-Ivan have the same challenges	1

Treatment of fisheries statistics available in European and Swedish repositories

Data sources by fishing style

Commercial fishers

Data for fishery dependent information for professional commercial fishers are available through the European Data Collection Framework, of which most recent update at the time of writing includes data until 2024. Our case study area spans over four ICES statistical divisions (northern part of 27, 29, 30, 31; Fig. B1). The majority of the commercial fishers included in our dialogues operate in the area 30 (Bothnian Sea), which is chosen as reference area for statistical information. The dataset relevant for this study includes:

- *Fishery dependent information (FDI) for EU Fleets* [jrc]: provides fishing effort (dataset D1; fishing days) and catch (dataset D2; weight and value) data by country, area, fleet segment and target species
- *Annual Economic report on EU fishing fleet* [jrc2025]: provides economic overview (dataset D3; number of vessels, income, expenses) of european fishing fleets by country and fleet segment

The FDI data resolution allow us to obtain information for the study area (Bothnian Sea - ICES area 27.3.D.30), while the AER is only available at country (Sweden) level. Different aggregation of the data requires to implement some simplifications or assumptions, hereby explained. Terminology adopted is consistent with definitions in use in the Data collection framework of the European Union [gear2026].

Small scale fishers Small scale fishers catch and effort data correspond to vessels of size $< 12\text{m}$, deploying passive gears and operating in the sub-region 27.3.D.30. This region includes the majority of small scale coastal fishers deploying gillnets in Sweden, especially for those targeting Herring (more than 50% of total fishers with GNS-SPF metier are registered in the study area). Economic data for the fleet segments, although refers to values at country level, are considered to be representative of the study area.

Pelagic trawlers Pelagic correspond to vessels of size $> 24\text{m}$ operating pelagic nets (gear code: Trawl Midwater - TM) in the sub-region 27.3.D.30. Pelagic trawlers move between Swedish areas, and the same fleet can report catches in several divisions. Economic data at country level are therefore an appropriate aggregation. However, given the very small number of active vessels, Sweden does not provide economic data at fleet segment level for confidentiality reasons [jrc2025]. Economic data are provided for the aggregated Large Scale fleet, which includes vessels $> 24\text{m}$ operating demersal trawlers, pelagic trawlers and purse seiners. On average, over the period 2020-2025 40 % of landed value.

Recreational fishers

Dataset for recreational fishers is available through Sweden statistic reports [swam2024], which most recent update at the time of writing includes data until 2024. The dataset relevant for this study are the following tables (table number is reported as it appears in SwAM and SCB, 2024):

- Table 5: target species in the sea in 2024
- Table 6: primary and secondary gear used in recreational fishing in 2024
- Table 8: Catches in recreational fishing in the sea in 2024
- Table 9: Expenditures in recreational fishing in 2024, million SEK

Dataset ID	Dataset_name	Description
D1	fdi_effort_SWE.csv	Fishing effort for commercial fisheries from FDI data call [@jrc]
D2	fdi_catch_SWE.csv	Catches for commercial fisheries from FDI data call [@jrc]
D3	fdi_SWE.csv	Economic overview (number of vessels, income, expenses) of european fishing fleets [@jrc2025]
D4	rec_gears_SWE.csv	Thousand of practitioners reported for recreational fishing technique in Sweden
D5	rec_catch_SWE.csv	Thousand of practitioners reported for target species in Sweden
D6	rec_economics_SWE.csv	Economic and fishing effort data

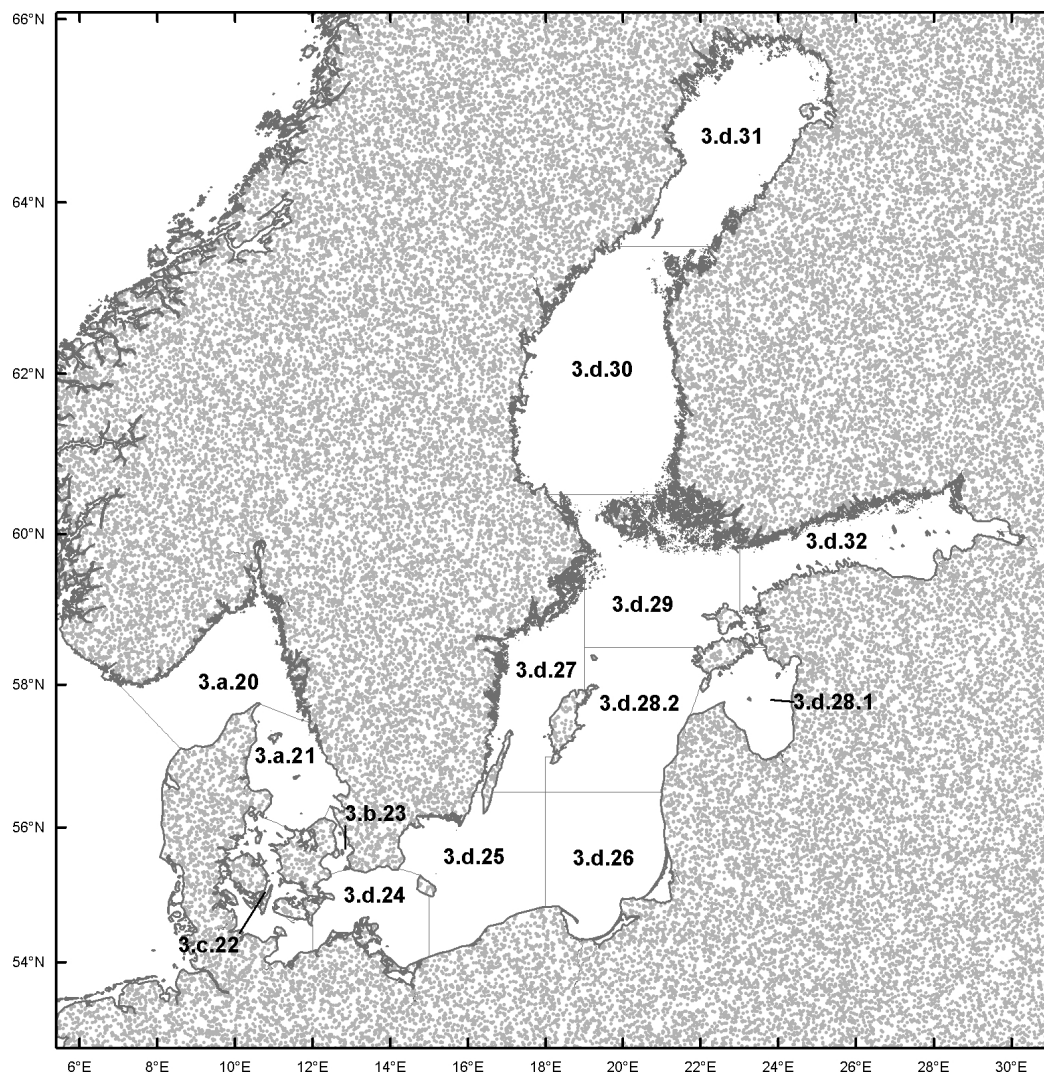


Figure S.1: Map of the Baltic Sea (numbers indicate the statistical divisions for fisheries data collection). The study area spans across Swedish fisheries in areas 3.d.31, 3.d.30, 3.d.29 and the northern part of 3.d.27.

Fishing gears

Exploration of fishery statistics data

Small-scale coastal fishing We analysed dataset D1, subset to ICES area 27.3.D.30, Swedish country code, vessels categories VL0812 and VL0008 to identify the mostly used métiers by the Small-scale coastal fishing (ssf). A **métier** is defined as group of fishing activities targeting a similar species or assemblage of species, using similar gear [dcf2021]. Figure B.2 shows average total fishing days by métier over the available data (years 2008-2024). We select for consideration in our study the métiers (1) gillnets-freshwater species (GNS-FWS), (2) gillnets-small pelagic species (GNS-SPF), (3) pound nets- anadromous species (FPN-ANA). The sum of average fishing days for the three selected métiers accounts for 85% of total fishing days for ssf in our study area.

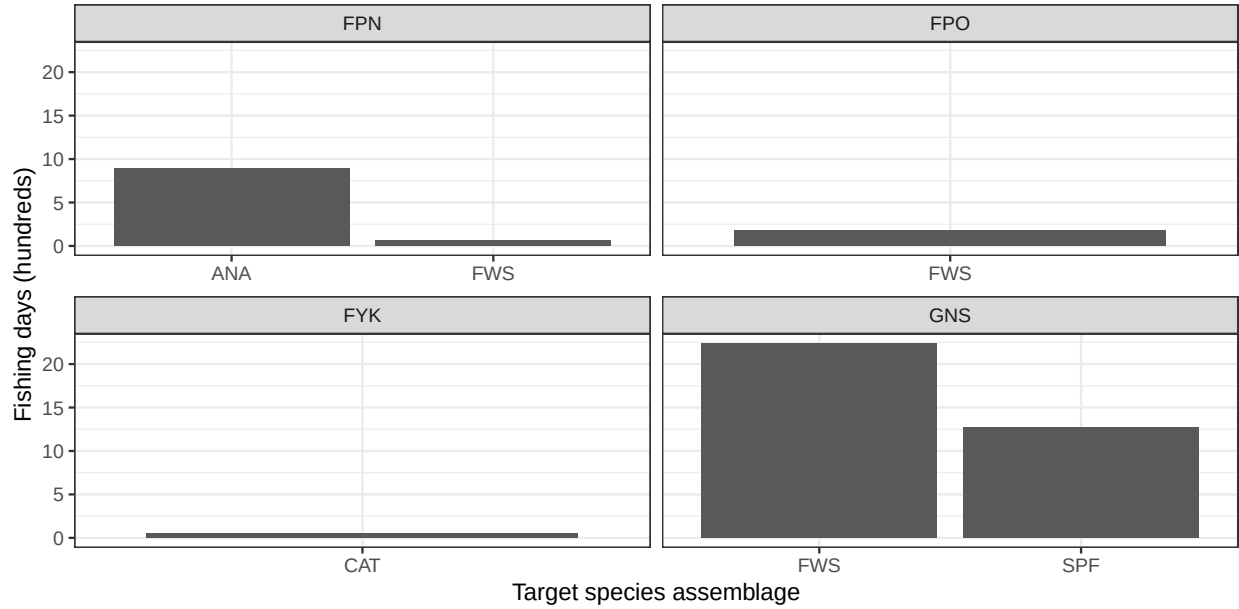


Figure S.2: Fishing days by metier, swedish small scale coastal fleet in Bothnian Sea. Panels represents gear type.

Pelagic trawling and large scale fleet We analysed dataset D1, filtered to Swedish fleet, vessels categories VL2440 and VL40XX, to assess the fishing effort distribution by fishing technique in the Swedish large scale fleet. This information supports the interpretation of economic data, for which data are aggregated across fishing techniques. Figure B.3 shows that the fishing technique *Demersal Trawl Shelf* (DTS) and *Trawl Midwater* (TM) includes similar amount of fishing days, and together represents the vast majority of Swedish fishing effort. In the DTS category, pelagic trawling gears (OTM, OTT, PTM) dominates the fishing activity.

Recreational fishing Dataset D4 shows the number of practitioners by recreational fishing method. Fishing methods based on hook and lines are dominating the distribution reported in Fig. B.4. Cages and nets are the two main techniques that do not base on hook and line. Cages are predominantly used in crustaceans fishery on the west coast of Sweden. The use of nets is popular in the Baltic Sea. For the purpose of this study we consider that only hook and line methods apply for the recreational fishing style, with no distinction among methods. Nets applies to the household fishing style.

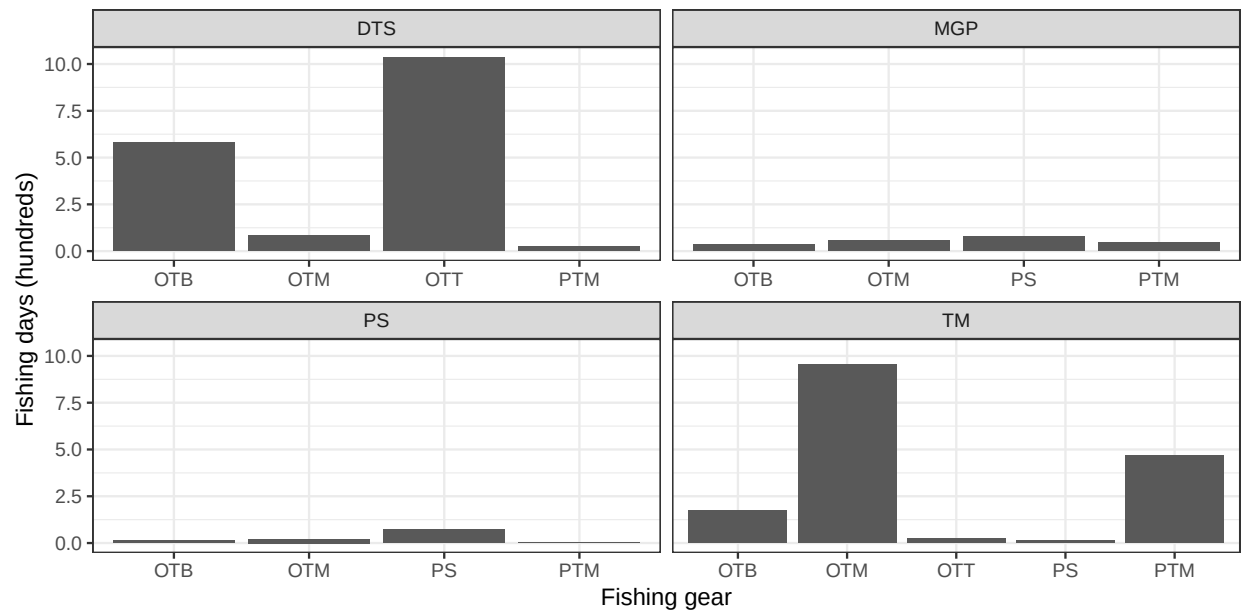


Figure S.3: Fishing days by gear and fishing technique, swedish large scale fleet. Panels represents fishing technique.

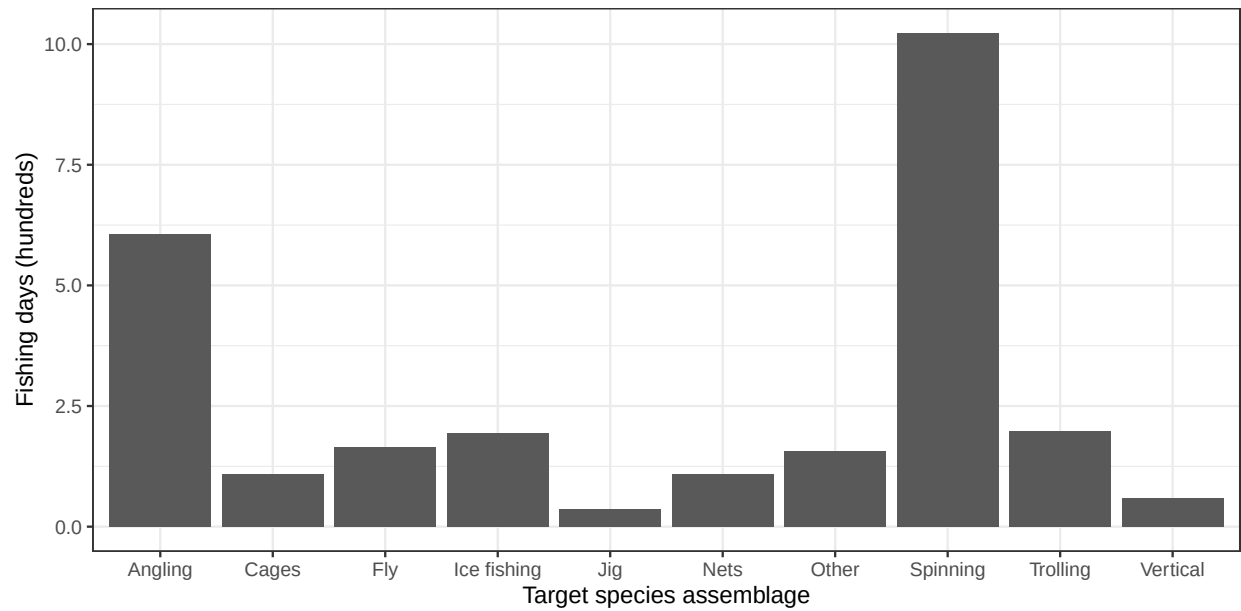


Figure S.4: Fishing days by fishing technique applied by swedish recreational fishers

Assignations

Table S.344: Baseline fishing gear assignation

gear	recreational	small_scale	household	trawler	guide
otm	0	0.0	0.0	1	0
gns_fws	0	0.5	0.5	0	0
gns_spf	0	0.3	0.5	0	0
fpn_ana	0	0.2	0.0	0	0
lhp	1	0.0	0.0	0	1
icefishing	0	0.0	0.0	0	0
inland	0	0.0	0.0	0	0

Fishing target

Exploration of fishery statistics

Small-scale coastal fishing We analysed dataset D2 to identify the catch composition associated to ssf métiers selected in the previous section. The dataset, filtered to Swedish fleet in ICES area 27.3.D.30, vessels categories VL0812 and VL0008, was subset to match the ssf métiers (1) gillnets-freshwater species (GNS-FWS), (2) gillnets-small pelagic species (GNS-SPF), (3) pound nets- anadromous species (FPN-ANA). Species scientific name were assigned according to ASFIS List of Species for Fishery Statistics Purposes [fao2026].

Catch data were aggregated by metier and averaged over years. We calculated the cumulative contribution of each species to a metier and we excluded species falling beyond the 95th percentile. Figure B.5 shows average tons by species and metier over the available data (years 2008-2024). The species mostly represented are Atlantic herring (target of gillnets for small pelagic fish), Atlantic salmon (main target of pound nets for anadromous species), European perch (main target of gillnets for freshwater species). Whitefish is also considered a relevant species as it significantly contribute to catches for FPN-ANA and GNS-FWS.

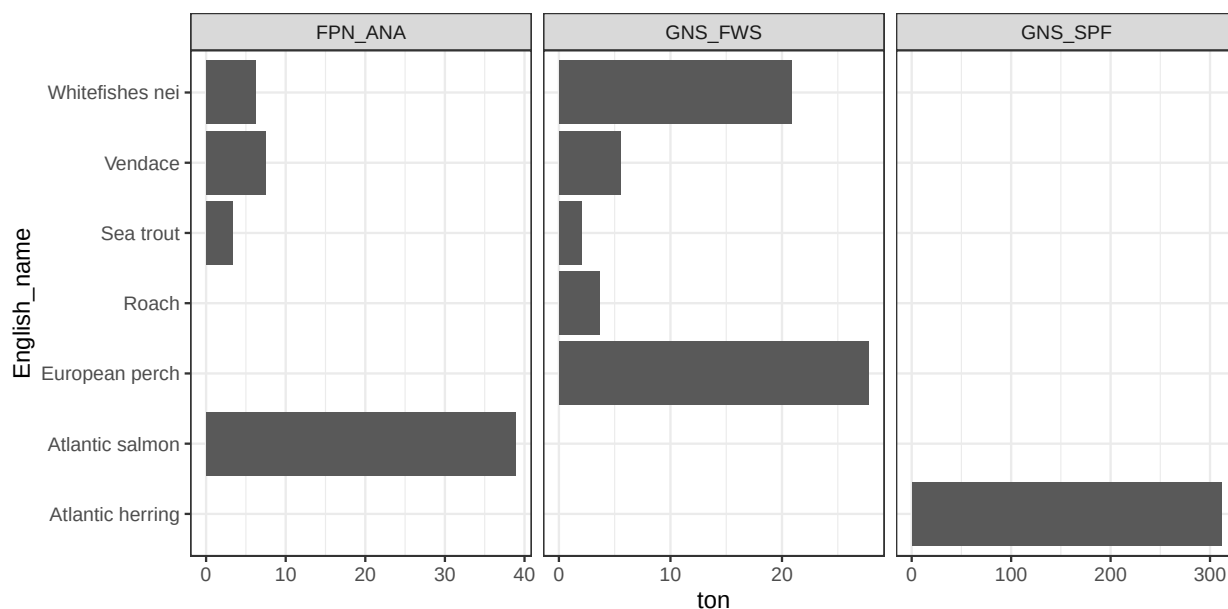


Figure S.5: Catches by metier, swedish small scale coastal fleet in Bothnian Sea (selection of metiers)

Pelagic trawling and large scale fleet We analysed dataset D2 to evaluate the catch value associated with fishing gears deployed by Swedish large scale fleet, and then the catch composition of pelagic trawlers. The economic value of catches, filtered to Swedish fleet in ICES area 27.3.D.30, and vessel categories ‘VL2440’, ‘VL40XX’ was aggregated by metier and averaged over years. Figure B.6 shows that pelagic fleet (TM) dominates the distribution (58% of total), and the fishing gears Otter Trawl Midwater (OTM) contributes nearly to 37% of the entire landed value.

To explore fishing target of the pelagic trawl fleet in the study area, dataset D2 was subset to match the gear type “OTM”, Swedish fleet in ICES area 27.3.D.30, and vessel categories ‘VL2440’, ‘VL40XX’. Species scientific name were assigned according to ASFIS List of Species for Fishery Statistics Purposes [fao2026]. We calculated the cumulative contribution of each species to a metier and we excluded species falling beyond the 99th percentile. Figure B.7 shows that Atlantic herring contributes for more than 97% of average catches. For pelagic trawlers we therefore consider Herring as the only relevant species.

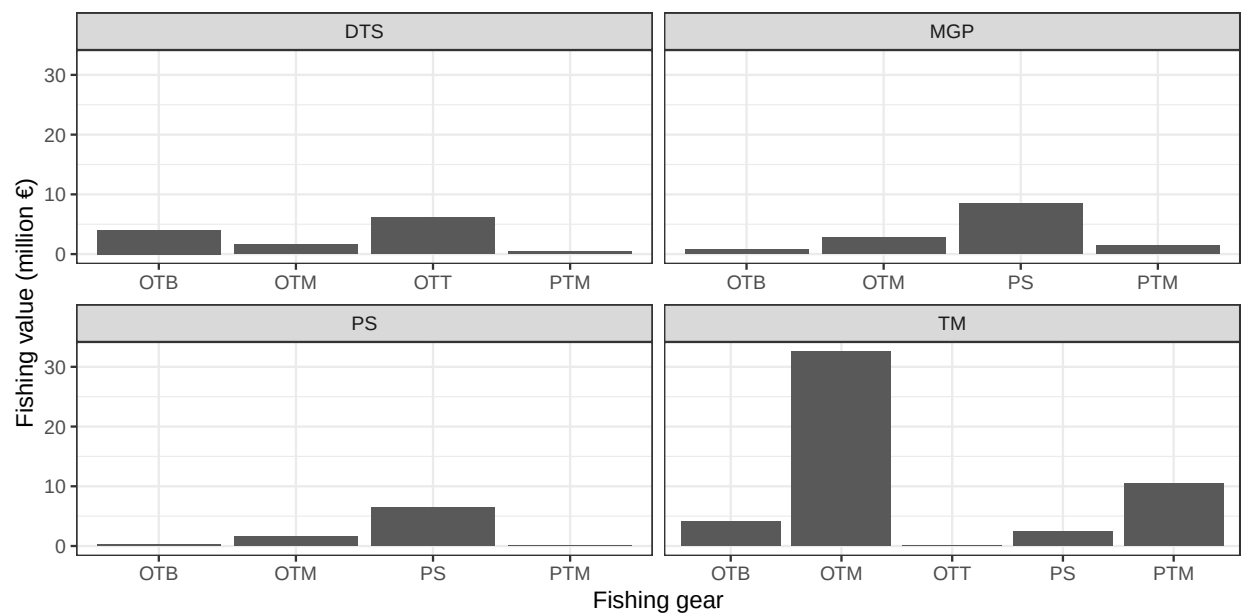


Figure S.6: Catches by gear and fishing method, swedish large scale fleet

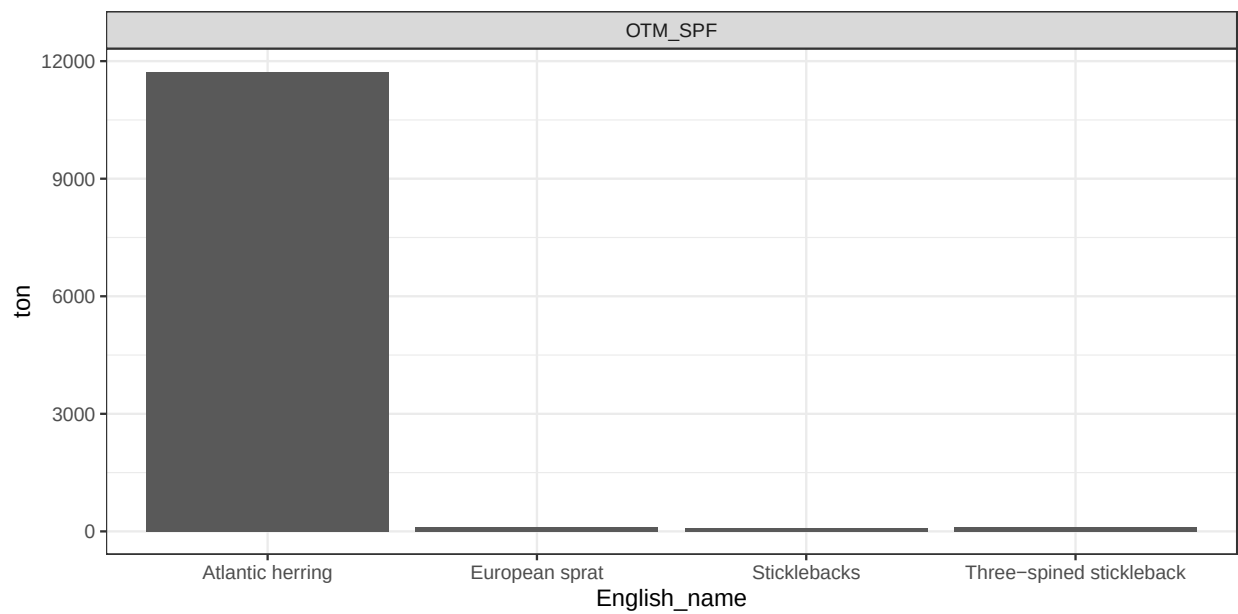


Figure S.7: Catches for pelagic trawlers, swedish fleet in Bothnian Sea

Recreational fishing For recreational fishers we are interested in the distribution of number of practitioners by target species, in the Sea. Dataset D5 was subset to exclude species that are not available in the Baltic Sea. Fig. B.8 shows that Perch and pike dominates the distribution. We excluded some relevant species (Cod, Zander) for simplifying the model. Whitefish was included, although not contributing substantially to recreational fishing target, because it is a shared resource with small scale fishers.

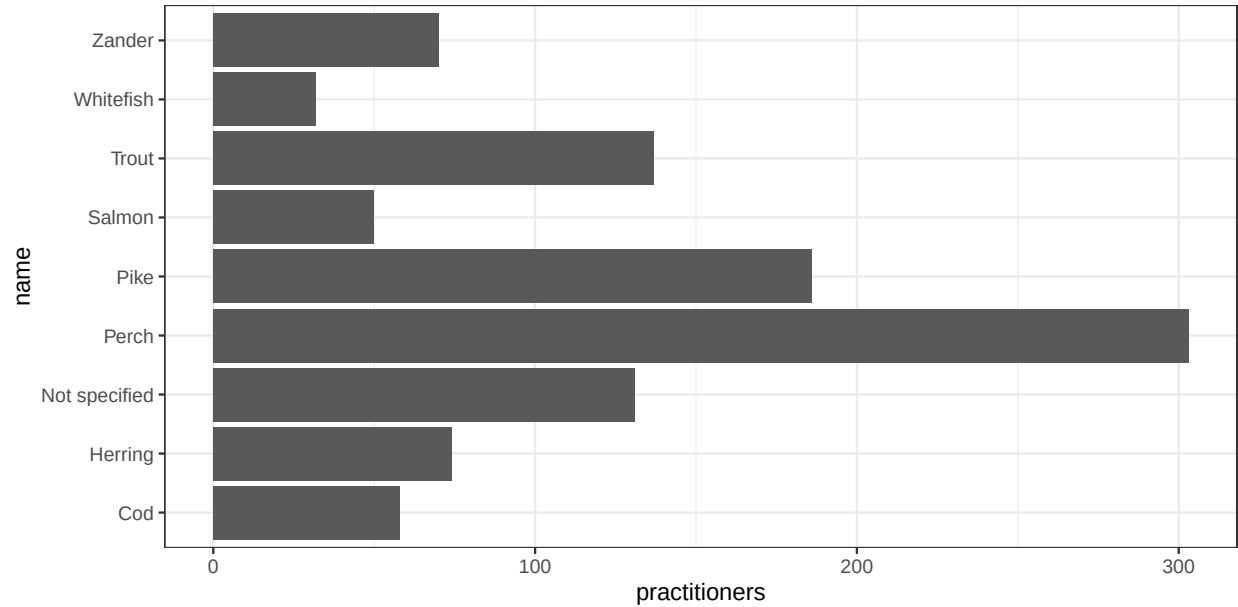


Figure S.8: Catches for recreational fishers

Assignations

Table S.345: Baseline fishing gear assignation

target	otm	gns_fws	gns_spf	fpn_ana	lhp	icefishing	inland
herring	1	0.00	0.99	0.00	0.07	0.00	0
perch	0	0.44	0.00	0.00	0.29	0.33	0
pike	0	0.00	0.00	0.00	0.18	0.33	0
salmon	0	0.00	0.00	0.66	0.05	0.00	0
sea_trout	0	0.03	0.00	0.06	0.13	0.00	0
whitefish	0	0.42	0.00	0.23	0.03	0.33	0
not_specified	0	0.10	0.01	0.05	0.25	0.00	1

Economic data

Methods

The cost node reflects the economic impact of one single extreme weather event compared to annual average expenses. Cost categories to be used in the conditional probability table are: *negligible* (cost < 0.05%); *low* (0.05% < cost < 2.5%); *medium* (2.5% < cost < 5%); *high* (> 5%).

The conditional probability tables are estimated through the function `cost.cpt`, here below reported. The input needed are:

- expenses (mu and sd): annual total expenses, including maintenance, fuel, personnel. This value is anchored in literature, when available;
- fuel.cost (mu and sd): average cost for single fishing trip. This value is anchored in literature, when available;
- damage.small (lo and hi): self declared cost for repairing a minor damage;
- damage.large (lo and hi): self declared cost for repairing a large damage;
- damage.unc: self declared uncertainty on the cost for repairing damages;
- fuel.avg (lo and hi): self declared fuel consumption for single average fishing trip;
- fuel.large (lo and hi): self declared fuel consumption for single longer-than-average fishing trip;
- miss.catch (mu and sd): average gross economic value of catches from one fishing day
- economic.unc: self declared uncertainty associated with answer regarding expenses;
- fishing style: label describing the fishing style.

```
cost.cpt=function(expenses.mu, expenses.sd,
                  fuel.cost.mu,fuel.cost.sd,
                  damage.small.lo, damage.small.hi,damage.large.lo, damage.large.hi,
                  damage.unc,
                  fuel.avg.lo, fuel.avg.hi,
                  fuel.large.lo, fuel.large.hi,
                  miss.catch.mu,miss.catch.sd,
                  economic.unc,
                  f.style){

  # creating distributions
  ## distribution for annual expenses
  cost.annual=rtruncnorm(a=0, b=Inf,
                        mean=expenses.mu,
                        sd=expenses.sd, n=1000)

  ## distribution for minor damage
  cost.lo=rtruncnorm(a=damage.small.lo, b=damage.small.hi,
                    mean=(damage.small.hi-damage.small.lo)/2,
                    sd=damage.small.hi*damage.unc, n=1000)

  ## distribution for major damage
  cost.hi=rtruncnorm(a=damage.large.lo, b=damage.large.hi,
                    mean=(damage.large.hi-damage.large.lo)/2,
```

```

sd=damage.large.hi*damage.unc, n=1000)

## Distribution for avg travel
avg.fuel=(fuel.avg.hi+fuel.avg.lo)/2
fuel.lo=rtruncnorm(a=fuel.avg.lo, b=fuel.avg.hi,
                  mean=avg.fuel,
                  sd=avg.fuel*economic.unc, n=1000)

## Distribution for large travel
lrg.fuel=(fuel.large.hi+fuel.large.lo)/2
fuel.hi=rtruncnorm(a=fuel.large.lo, b=fuel.large.hi,
                  mean=lrg.fuel,
                  sd=lrg.fuel*economic.unc, n=1000)

# Cost for fuel in extra travel
extra.fuel.cost=(fuel.hi/fuel.lo)*rtruncnorm(a=0, b=Inf,
                                           mean=fuel.cost.mu,
                                           sd=fuel.cost.sd, n=1000)

# missing catches
miss.catch=rtruncnorm(a=0, b=Inf, mean=miss.catch.mu, sd=miss.catch.sd, n=1000)

# create data frame
cost.df=data.frame(minor.no=cost.lo, major.no=cost.hi, no.travel=extra.fuel.cost)
cost.df$minor.travel=cost.df$minor.no+cost.df$no.travel
cost.df$major.travel=cost.df$major.no+cost.df$no.travel
cost.df.2=cost.df+miss.catch
names(cost.df.2)=paste(names(cost.df.2), 'miss', sep='.')
cost.df.2$no.no.miss=miss.catch
cost.df=cbind(cost.df, cost.df.2)
cost.df[,]=apply(cost.df[,],2,function(x) x/cost.annual)
cost.df.disc=cost.df
cost.df.disc[,]=apply(cost.df.disc[,],2,
                     function(x) cut(x, c(-Inf, 0.005,0.025,0.05,Inf),
                                     labels=c('negligible','low','medium','high'))))

cost.df.disc$fishing_style=f.style
cost.cpt=cost.df.disc%>%
pivot_longer(-fishing_style, names_to = 'parent', values_to = 'state')%>%
dplyr::group_by(fishing_style,parent, state)%>%
tally()%>%
dplyr::mutate(prob=n/sum(n))

return(cost.cpt)
}

error.propagation=function(x.mu, x.er, y.mu, y.er){
  z.mu=x.mu/y.mu
  z.er=sqrt(((x.er/x.mu)^2)+((y.er/y.mu)^2))*z.mu
  return(data.frame(mu=z.mu, sd=z.er))
}

```

Small-scale coastal fishing

Economic information for small scale coastal fishers, available in dataset D3 (Table C.1), filtered for vessel length categories VL0010 and VL0812, fishing technologies FPO and DFN and Swedish country code. Cost of fuel per trip is calculated by dividing total fuel expenses by number of fishing trips. Annual spending is estimated by dividing total annual expenses by number of fishing vessels. Gross value of catches in a single fishing trip is obtained by dividing total annual value of catches by number of fishing trips. Other values are provided in the interviews.

Table S.346: Variables from interviews

id_I	short_description	range.lwr	range.upr	unit_range
2	minor_damage_cost	10000	50000	sek_unit
3	minor_damage_cost	1	50000	sek_year
7	minor_damage_cost	1000	50000	sek_unit
2	major_damage_cost	100000	500000	sek_unit
3	major_damage_cost	50000	200000	sek
7	major_damage_cost	50000	100000	sek_unit
3	fuel_trip	1500	2500	sek_day
3	extra_fuel	3000	5000	sek_day
7	extra_fuel	50	100	B liter_trip

Table S.347: Variables from literature

variable	value_mean	value_sd	unit	source
Annual expenses	1702	1567	thousand SEK	STECF_AER
Fuel cost	6670	8668	SEK	STECF_AER

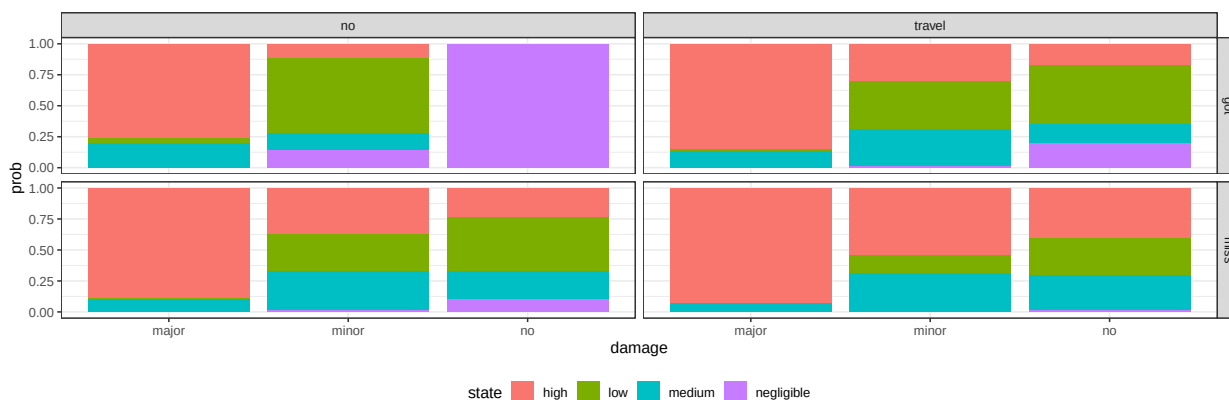


Figure S.9: CPT estimated for coastal fisherman

Pelagic trawling

Pelagic economic information is derived from dataset D3, filtered for vessel length categories VL2440 and VL40XX, and Sweden country code. Cost of fuel per trip is calculated by dividing total fuel expenses by number of fishing trips. Annual spending is estimated by dividing total annual expenses by number of fishing vessels. The cost of longer travel was not available, and it is based on the assumption that an average travel costs between 80% and 120% of the average fuel cost, while a longer travel is between 100% and 300% of average fuel cost. Other values are provided in the interviews.

[1] 4981303 5279201 5837720 5749069 5529277 6694086

Table S.348: Variables from interviews

id_I	short_description	range.lwr	range.upr	unit_range
6	minor_damage_cost	10000	50000	sek_unit
6	major_damage_cost	100000	500000	sek_unit
6	fuel_trip	600	650	liters_hour
6	extra_fuel	NA	NA	not_reported

Table S.349: Variables from literature

variable	value_mean	value_sd	unit	source
Annual expenses	50404	5643	thousand SEK	STECF_AER
Fuel cost	259142	117662	SEK	STECF_AER

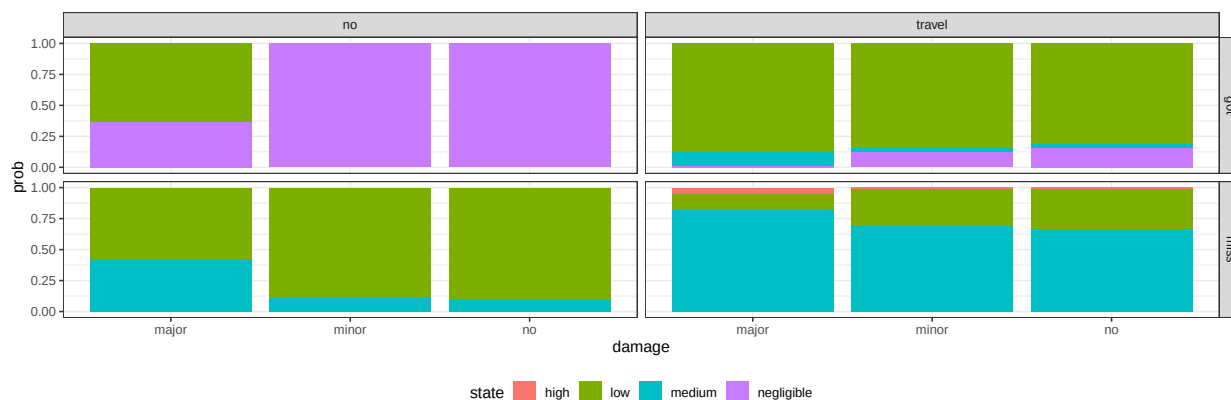


Figure S.10: CPT estimated for pelagic trawlers

Recreational fishing

Economic information is derived from dataset D6, representing the totality of recreational fisherman in Sweden with no distinction between region or inland vs coastal fishing. Cost of fuel per trip is separately calculated for travel and for boat fuel. These two costs are then averaged. All the uncertainties (on costs and on number of practitioners) are accounted for by propagating errors. Annual spending is estimated by dividing total annual expenses by number of practitioners. Other values are provided in the interviews.

id_I	short_description	range.lwr	range.upr	unit_range
5	minor_damage_cost	10000	50000	sek_unit
5	major_damage_cost	50000	1000000	sek_unit
5	fuel_trip	30	50	B liter_trip
5	extra_fuel	20	50	B liter_trip

Table S.351: Variables from literature

variable	value_mean	value_sd	unit	source
Annual expenses	9213	3943	SEK	SWAM
Fuel cost	101	31	SEK	SWAM

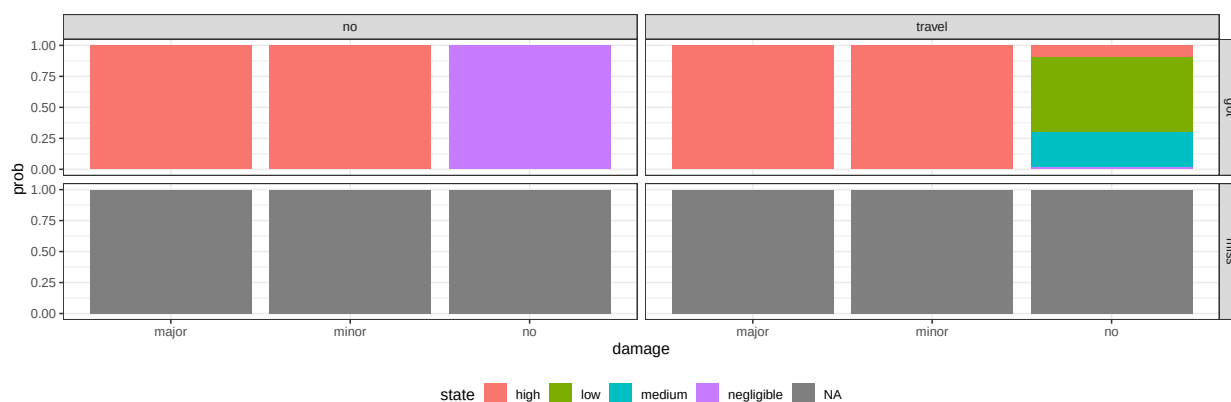


Figure S.11: CPT estimated for recreational fisherman

Fishing guides

No updated public information is available for fishing guide economic aspects in Sweden. All estimations are based on interview results and our assumptions. Seasonal expenses are estimated by summing the declared maintenance costs and an estimation of fuel costs.

id_I	short_description	range.lwr	range.upr	unit_range
1	season_maint	50000	10000	sek_season
1	minor_damage_cost	10000	20000	sek_unit
1	major_damage_cost	20000	6000000	sek_unit
1	fuel_trip	1000	1000	sek_day
1	extra_fuel	500	500	sek_day

Table S.353: Variables from literature

variable	value_mean	value_sd	unit	source
Annual expenses	175770	14456	SEK	Our estimation based on interviews

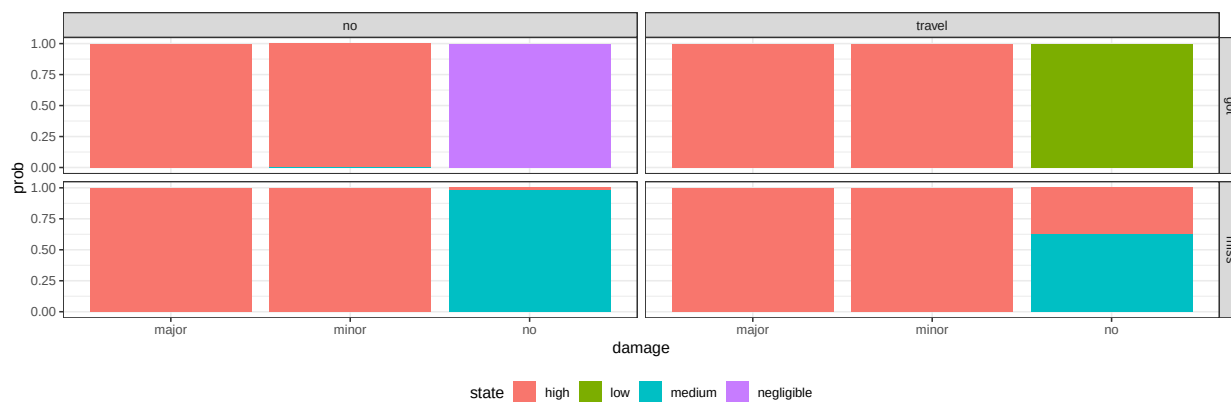


Figure S.12: CPT estimated for fishing guide

Household fishing

No updated public information is available for archipelago fishers economic aspects in Sweden. All estimations are based on interview results and our assumptions. Seasonal expenses are estimated by summing the declared maintenance costs and an estimation of fuel costs.

id_I	short_description	range.lwr	range.upr	unit_range
4	season_maint	NA	5000	sek_season
4	minor_damage_cost	200	2500	sek_unit
4	major_damage_cost	2500	5000	sek_unit
4	fuel_trip	1000	6000	sek_season
4	extra_fuel	NA	NA	not_reported

Table S.355: Variables from literature

variable	value_mean	value_sd	unit	source
Annual expenses	8979	1740	SEK	Our estimation based on interviews

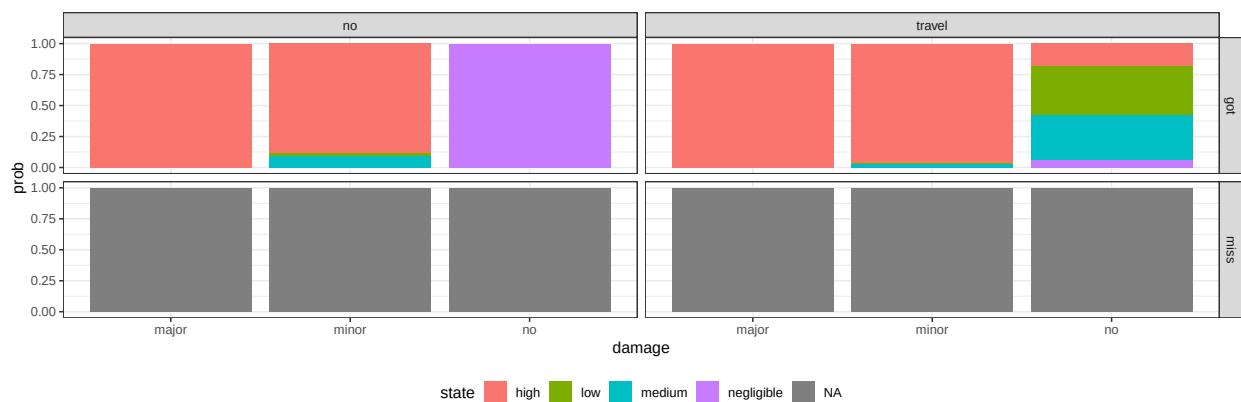


Figure S.13: CPT estimated for archipelago fisher

Conditional probability tables

Network structure

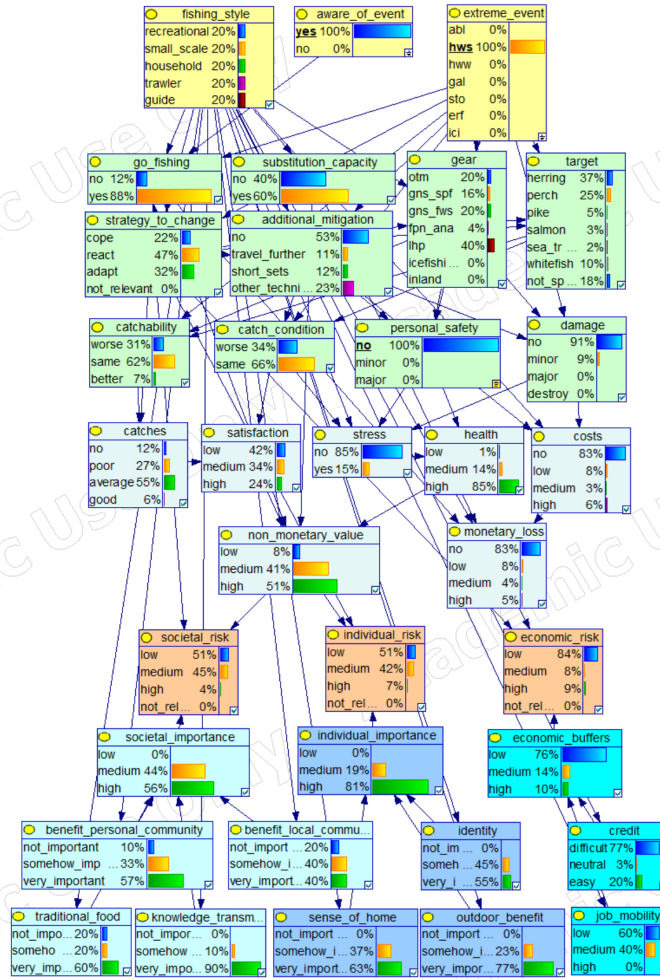


Figure S.14: Bayesian Network representing risks to Swedish fisheries posed by Marine Extreme Weather events (abl algal bloom, hws summer heatwave, hww winter heatwave, gal gale, sto storm, erf extreme rainfall, ici icing) effects on fishing operations, as identified at the conclusion of the third cycle of stakeholder (fishers) dialogues. The probability distributions shown in the boxes represents the example of how summer heatwave affects fishing operations across all fishing styles. Conditional probability distribution details and terminology are reported in Supp. Info. C. Colors are associated to node types and group Yellow: decision nodes; lime green: chance nodes – operational group; grey: chance nodes - user group; navy blue: chance nodes - individual group; light blue: chance nodes - societal group; cyan: chance nodes - economic group; orange: risk nodes

Algebraic methods for Ranked nodes

Ranked nodes have order-preserving meaning states (e.g.: low/medium/high) that can be mapped to a continuous numerical scale bounded to $[0,1]$, assuming equal intervals [fenton2019]. For instance, in a node having three states (low, medium, high), the “low” state takes the interval $[0-0.33)$. The probability distribution of a ranked node in the numerical scale is described as $TNormal(\mu, \sigma^2, 0, 1)$. The use of a numerical scale allows us to use linear algebra to perform operations among variables, a technique that is particularly useful to build a CPT from a reduced number of questions [fenton2007]. When a child has multiple parents with multiple states (binary or more), the distribution of a child node can be calculated by implementing algorithms based on the state, direction (positive or negative) and weight of their parents. The algorithms most commonly applied to ranked nodes gives equal (Eq. B1) or unequal contribution to states. Weighted max (Eq. B2) assigns more contribution to large states, while Weighted min (Eq. B3) assigns more contribution to small states.

$$\mu = \frac{\sum_{i=1}^n w_i X_i}{\sum_{i=1}^n w_i}$$

[Eq.C1]

$$\mu = \min\left[\frac{w_i X_i + \sum_{i \neq j}^n X_j}{w_i + (n - 1)}\right]$$

[Eq.C2]

$$\mu = \max\left[\frac{w_i X_i + \sum_{i \neq j}^n X_j}{w_i + (n - 1)}\right]$$

[Eq.C3]

Decision nodes

Extreme weather events (node: `extreme_event`)

The node *extreme_event* is used to wrap all the extreme weather events we are considering.

Levels (and acronyms)

- **Storms** (sto). Strong wind events, with intensity above 24.5 m/s. Season: summer.
- **Gales** (gal), or moderately strong wind events, with intensity between 15 m/s and 24.5 m/s;
- **Extreme precipitation** (erf) – rainstorm (summer), or deviation from average precipitation regime;
- **Algal bloom** (abl): intense algal bloom;
- **Summer heatwaves** (hws): prolonged discrete anomalously warm water event occurring in summer (Hobday et al., 2016);
- **Winter heatwaves** (hww): prolonged discrete anomalously warm water event occurring in winter (Hobday et al., 2016);
- **Icing** (ici): strong wind events associated to ice, which then forms icing pile up on gears and boats.

Fishing style (and acronyms)

The node *fishing_style* is used to wrap all the fishing styles and seasons.

Levels (and acronyms)

- **Recreational fisher (recreational - rcf)**: fishers without a commercial fishing license, deploying hand-held gears for recreational purposes;
- **Fishing guides (guides - fg)**: commercial actors who provide recreational fishers with fishing experience by sharing expert and local knowledge, giving access to angling concession and offering boat chartering ;
- **Small-scale coastal fishing (small_scale - ssf)**, fishers operating small boats (<12 m) deploying species-specific passive gears.
- **Pelagic trawling (trawlers - trw)**, fishers operating medium to large size vessels (24-60 m) deploying pelagic trawl nets;
- **Household fishing (household - hou)**, fishing by residents of the Swedish archipelago, using coastal-type passive gears, without commercial licences and without legal access to markets.

Aware of the event

The node *aware_of_event* controls the fisherman awareness of the event occurrence. In the case when fisher is not aware, they always goes out fishing. The fisher can still adopt react or adapt str

Levels (and acronyms)

- **yes**: the fisher is aware of the extreme event. Decisions whether to go out fishing or not are taken according to interviews
- **no**: the fisher is not aware of the extreme event. The probability of going out fishing are 100%

Chance nodes - Operational

Strategy to_change

Parents: fishing_style, extreme_event.

Levels: cope, react, adapt, not_relevant.

Description: Fishers strategy is coded as adapt, react or cope (Green et al., 2021). Adapt is proactive planning, it implies large changes in fishing strategy (i.e.: changing fishing target and/or gear). React is unplanned response, or small modification to fishing practice (fishing in a diferent location or in different time of the day). Cope is passively accepting consequences, it implies no modifications at all to the fishing strategy.

Method: Case C1. We interpreted narrative (especially questions Q 2, 4, 13 and 15) to associate one or more among Adapt, React or Cope to each combination of fisher and MEW. The question chosen to fill the CPT depends on the narrative.

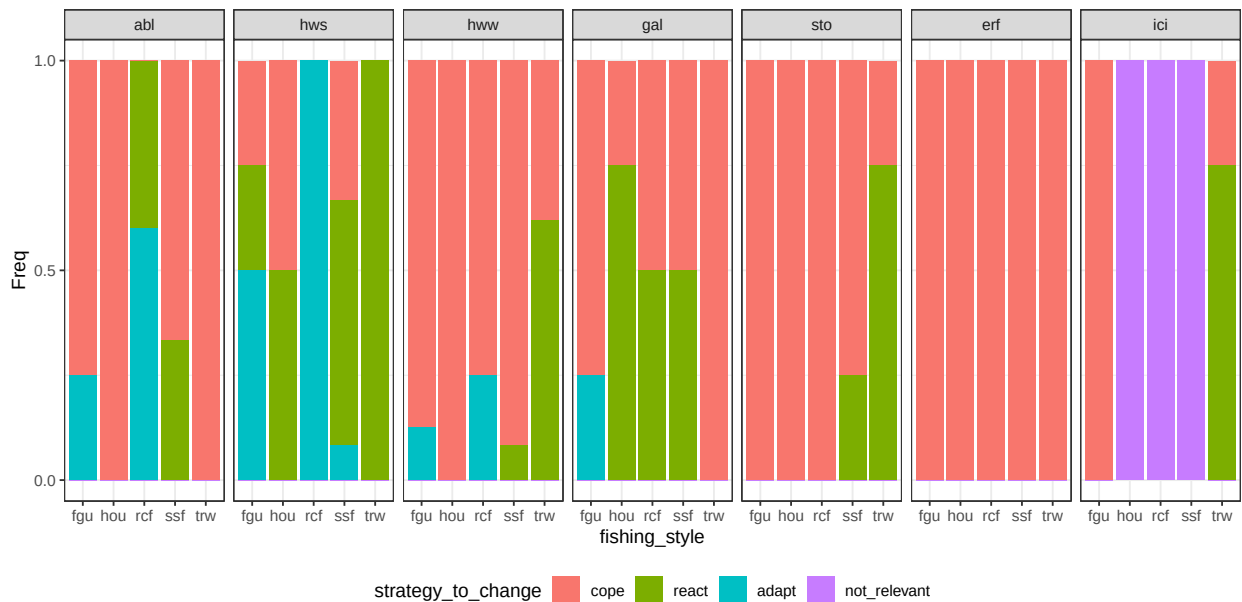


Figure S.15: conditional distribution for strategy_to_change. Columns represents levels of the node extreme_event.

Additional mitigation

Parents: strategy_to_change, fishing_style, extreme_event.

Levels: no, travel_further, short_sets, other_technical.

Description: Measures implemented in the case when the strategy is React. The different solutions work differently and have different cost: travel further implies fishers change fishing location to seek a sheltered place or to find deeper waters; short sets mean that the fisher perform shorter sessions than average (i.e.: reduced soak time for gillnets); other technical includes strategies that are mentioned sporadically (i.e.: use of ice machines).

Method: Case C4. We interpreted narrative to associate one or more solution to each combination of fisher and MEW. In case only one strategy is mentioned, this is assigned probability = 1. When two or more strategies are mentioned, they are given equal probability unless explicitly indicated in the interviews.



Figure S.16: conditional distribution for additional_mitigation. Columns represents levels of the node fishing_style; rows are levels of the node extreme_event.

Gear

Parents: fishing_style, extreme_event, strategy_to_change.

Levels: otm, gns_spf, gns_fws, fpn_ana, lhp, icefishing, inland.

Description: Most probable gear used by each fisher. Terminology is mostly inspired by codes used in the European Data Collection Framework (https://dcf.ec.europa.eu/data-calls/definitions-and-terminology/g/gear_en). Otm is pelagic (midwater) otter-trawl; gns_spf is gillnets targeting small pelagic fish; gns_fws is gillnets targeting freshwater species; fpn_ana is traps (or pots) targeting anadromous species; lhp is Handlines and pole-lines (hand-operated); icefishing is fishing with lines by carving holes on iced surfaces of coastal areas and lakes; inland is a general description for any fishing activity performed in lakes and rivers.

Method: Case C4. In the case when the strategy is cope or react, the CPT reflects literature as described in Annex C. When the strategy is React, it is assigned expert based probability based on interviews interpretation.

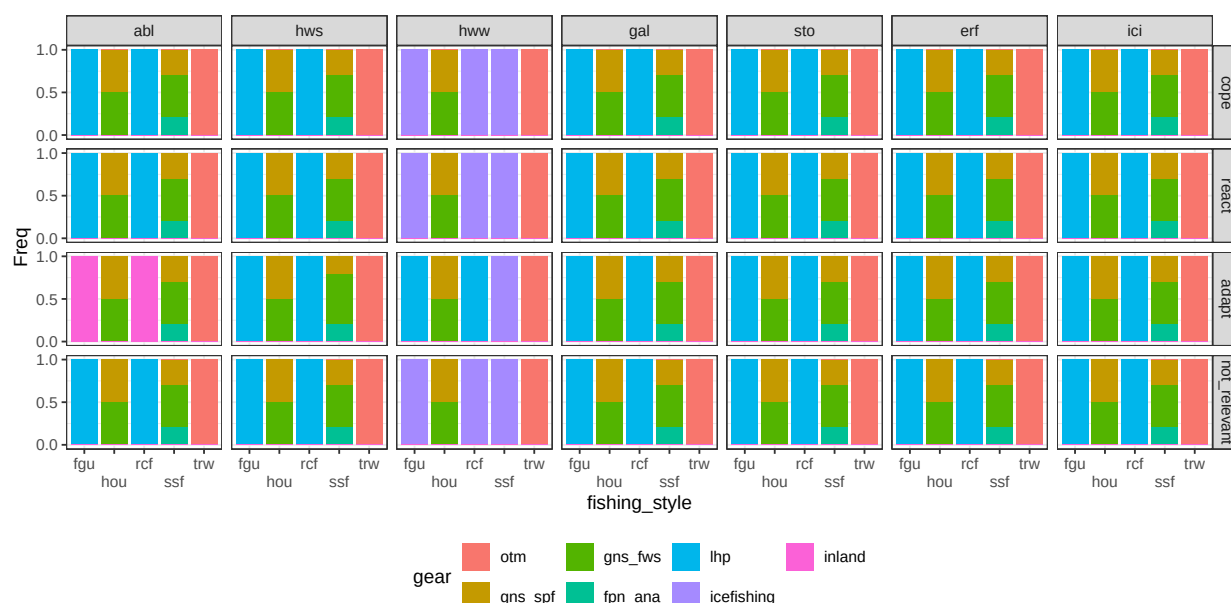


Figure S.17: conditional distribution for gear. Columns represents levels of the node `extreme_event`; rows are levels of the node `strategy_to_change`.

Catchability

Parents: target, extreme_event, gear.

Levels: worse, same, better.

Description: The ability to catch fish depending on gear-fish interactions. For example, a gillnet clogged in algae is more visible and less efficient (less catchability).

Method: Case C3 based on question Q 3.

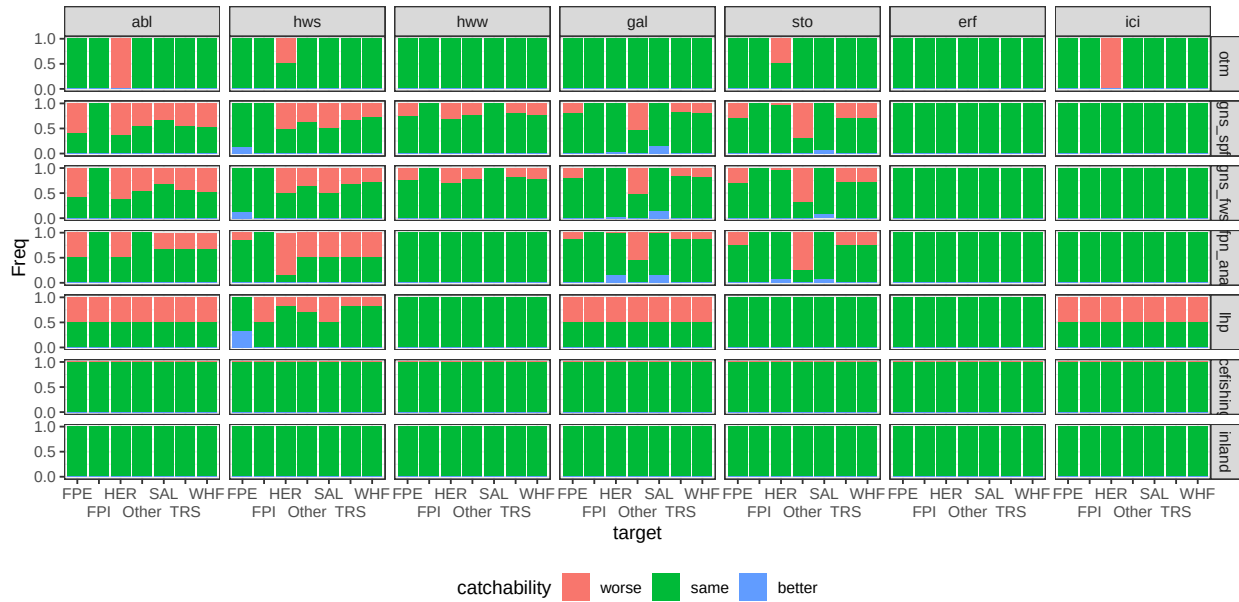


Figure S.18: conditional distribution for catchability. Columns represents levels of the node extreme_event; rows are levels of the node gear.

Catch condition

Parents: target, extreme_event, additional_mitigation.

Levels: worse, same.

Description: The real or perceived quality of catches in terms of aspect/health status. Does NOT refer to species composition or distribution. There are some “technical solutions” that balance for negative effects of MEW.

Method: Case C3 based on question Q5. The CPT is changed to 100% probability of level *same* when appropriate technical solutions (i.e.: travel further, ice machine) are mentioned in the narrative.

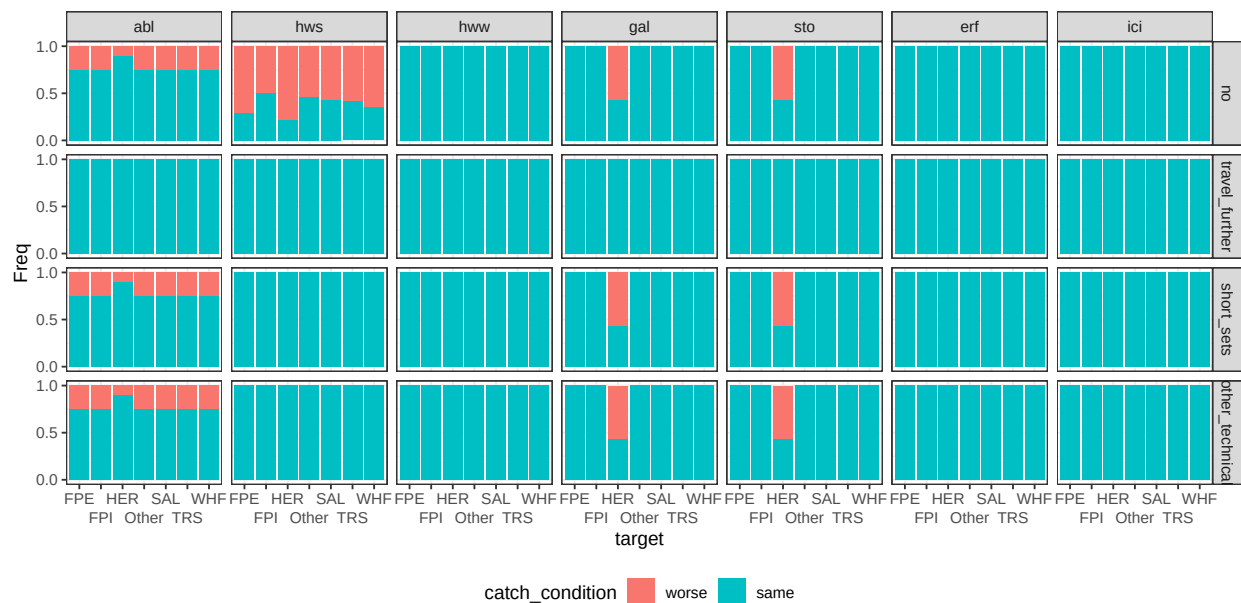


Figure S.19: conditional distribution for `catch_condition`. Columns represents levels of the node `extreme_event`; rows are levels of the node `additional_mitigation`.

Personal safety

Parents: extreme_event, gear, additional_mitigation.

Levels: no, minor, major.

Description: Any variation to ordinary working conditions that can increase the potential physical damage for operators (i.e.: slipping; a violent storm is life threatening). There are some “technical solutions” that balance for negative effects of MEW.

Method: Case C3 based on question Q9. The CPT is changed to 100% probability of level *no* when appropriate technical solutions (i.e.: travel further) are mentioned in the narrative.

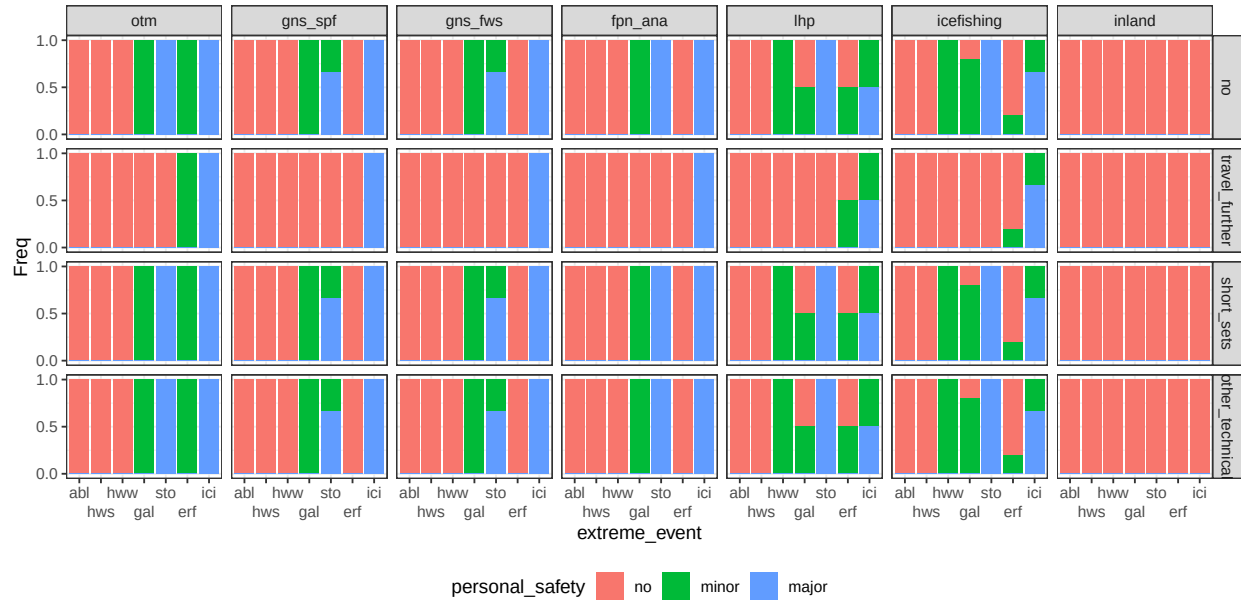


Figure S.20: conditional distribution for *personal_safety*. Columns represents levels of the node *gear*; rows are levels of the node *additional_mitigation*.

Damage

Parents: extreme_event, gear, additional_mitigation.

Levels: no, minor, major, destroy.

Description: Damage to the fishing gear ranging from minor to completely losing the fishing gear or part of it because it was lost or irreparably damaged. There are some “technical solutions” that balance for negative effects of MEW.

Method: Case C3 based on question Q10. The CPT is changed to 100% probability of level *no* when appropriate technical solutions (i.e.: travel further) are mentioned in the narrative.

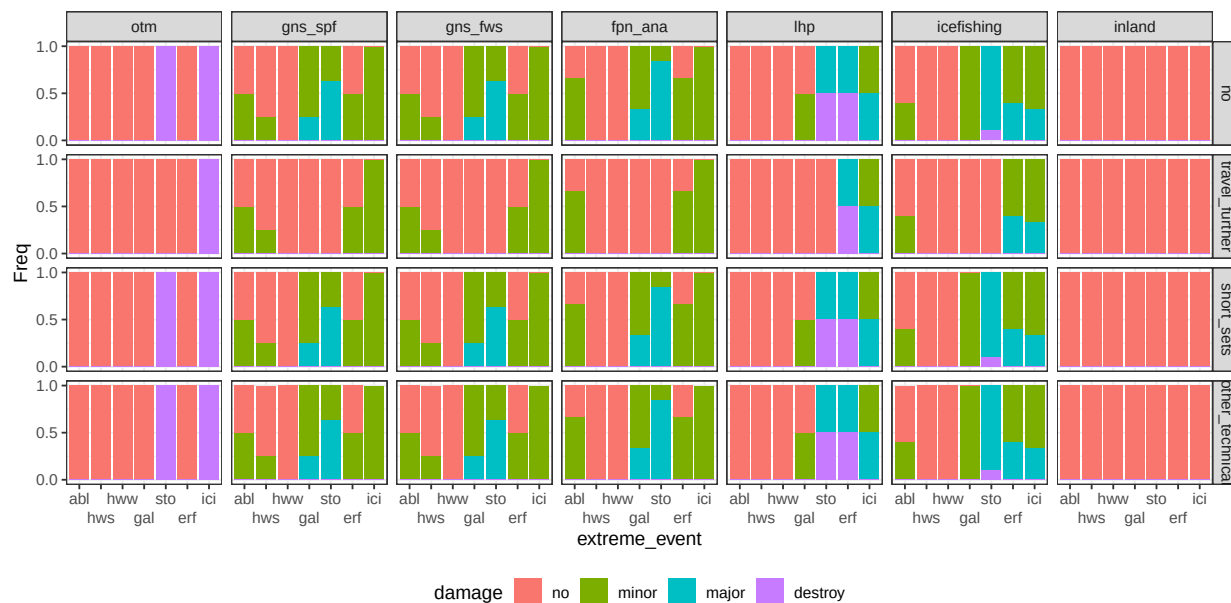


Figure S.21: conditional distribution for damage. Columns represents levels of the node *gear*; rows are levels of the node *additional_mitigation*.

Target

Parents: gear, strategy_to_change, extreme_event.

Levels: herring, perch, pike, salmon, sea_trout, whitefish, not_specified.

Description: The most probable target species associated to each fishing gear. The levels describe the 5 most common species that are relevant across the fishing styles, one aggregation for whitefish (including Common whitefish and vendace), and one category to group any other species. The category not_specified is also assigned to inland and not relevant.

Method: Case C4. In the case when the strategy is cope or react, the CPT reflects literature as described in Annex C. When the strategy is React, it is assigned expert based probability based on interviews interpretation.

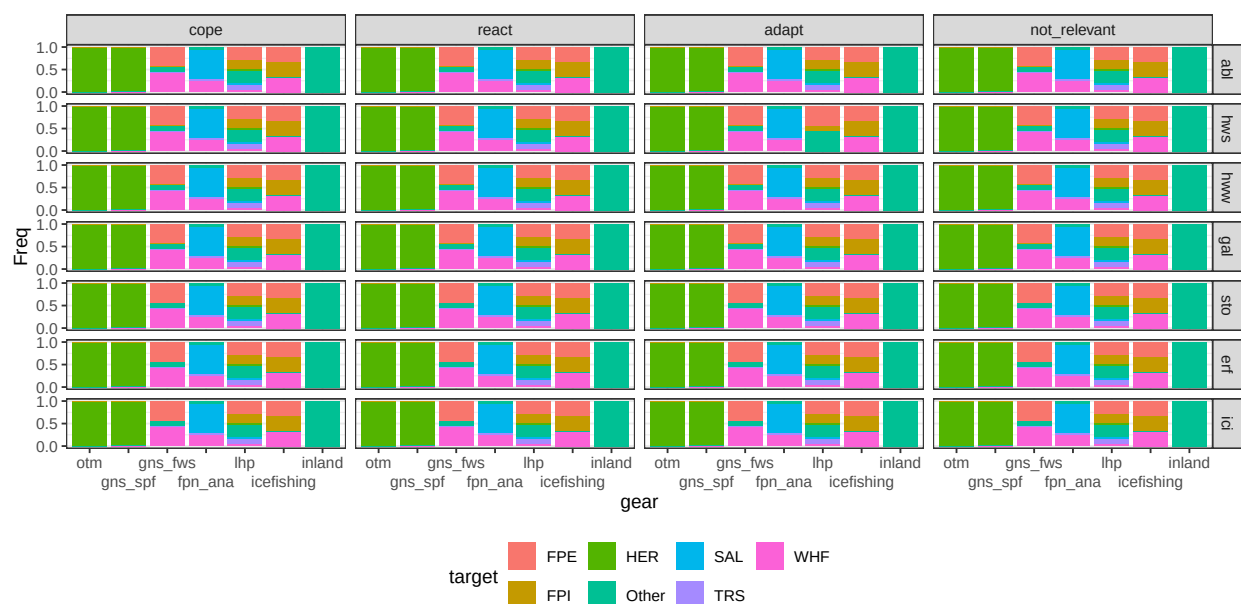


Figure S.22: conditional distribution for target. Columns represents levels of the node strategy_to_change; rows are levels of the node extreme_event.

Go fishing

Parents: extreme_event, fishing_style, strategy_to_change, aware_of_event.

Levels: no, yes.

Description: The probability to go out fishing.

Method: Case C3 based on question Q2 when the level of the parent node strategy is *cope*. The CPT is changed to 100% probability of level *yes* when parent node strategy is *adapt* or *react*.

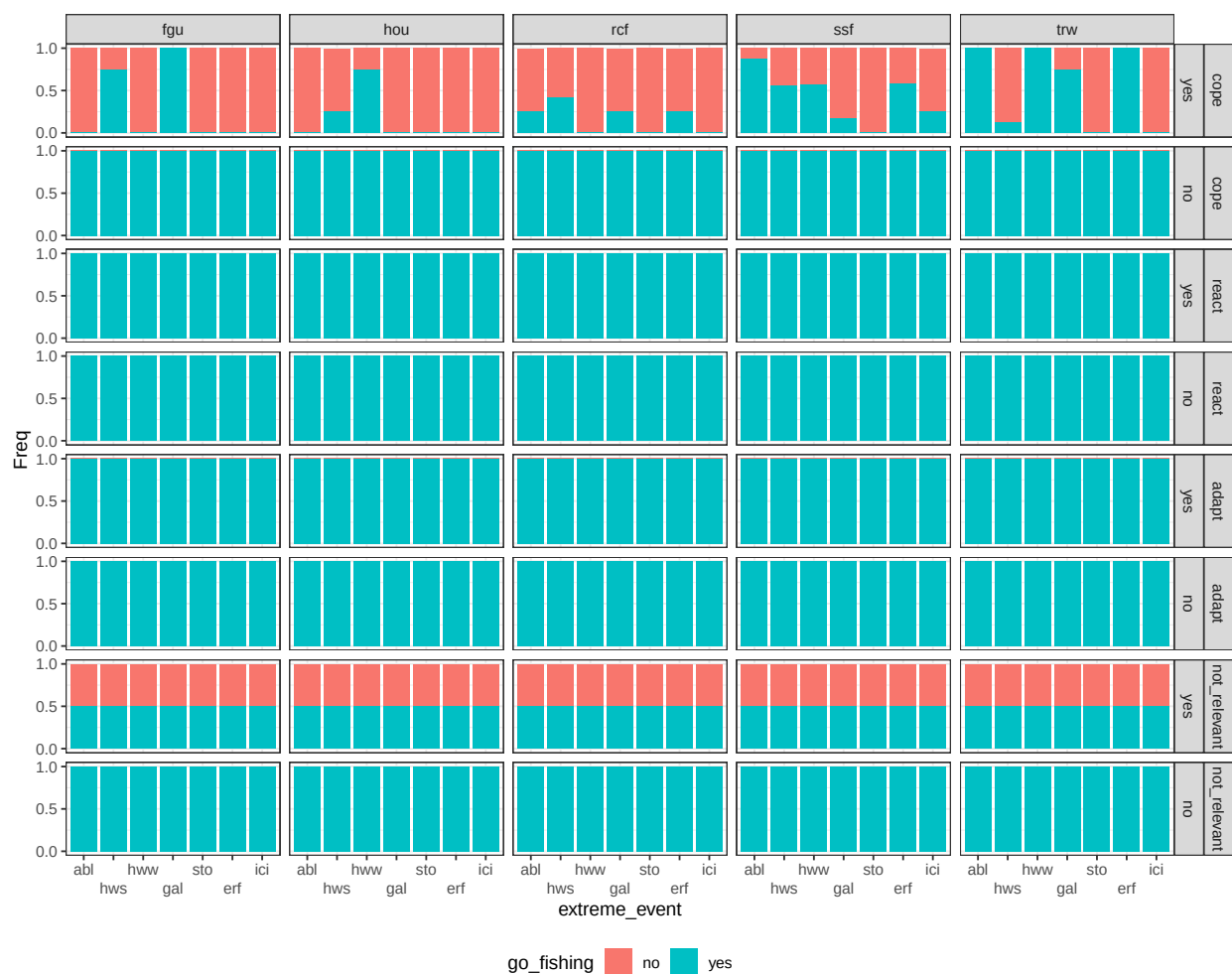


Figure S.23: conditional distribution for go_fishing. Columns represents levels of the node fishing_style; rows are combinations levels of the nodes strategy_to_change and aware_of_event.

Chance nodes - User

Stress

Parents: personal_safety, damage, catch_condition, fishing_style.

Levels: no, yes.

Description: Psychological aspect, elements such as anxiety and sense of fatigue (Chatterjee and and Bolar, 2019). The stressful events considered are having damages to the gear, being injured, being the catch in poor condition.

Method: Case C2. Weight based on questions Q21 a, b and c. Algorithm is wmin.



Figure S.24: conditional distribution for stress. Columns represents levels of the node damage; rows are combinations levels of the nodes catch_condition and fishing_style.

Health

Parents: stress, personal_safety.

Levels: low, medium, high.

Description: Summarizes variables related to physical and mental health. The wmin algorithm implies that all the conditions are necessary to obtain a large value, however the physical injuries are more relevant than stress.

Method: Case C2. Weight ratio for the parent *stress* and *personal safety* is 1:3. Algorithm is wmin.

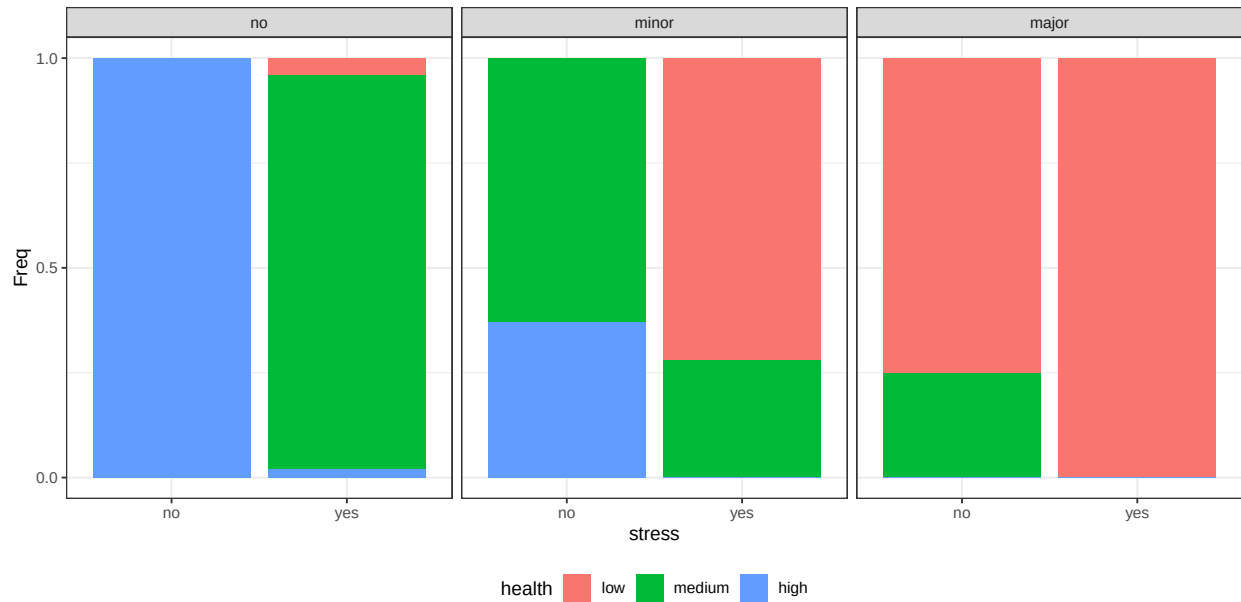


Figure S.25: conditional distribution for health. Columns represents levels of the node personal_safety.

Satisfaction

Parents: catches, catch_condition.

Levels: low, medium, high.

Description: Summarizes variables related to the quality and amount of catches. The wmin algorithm implies that all the conditions are necessary to obtain a large value.

Method: Case C2 with equal weight. Algorithm is wmin.

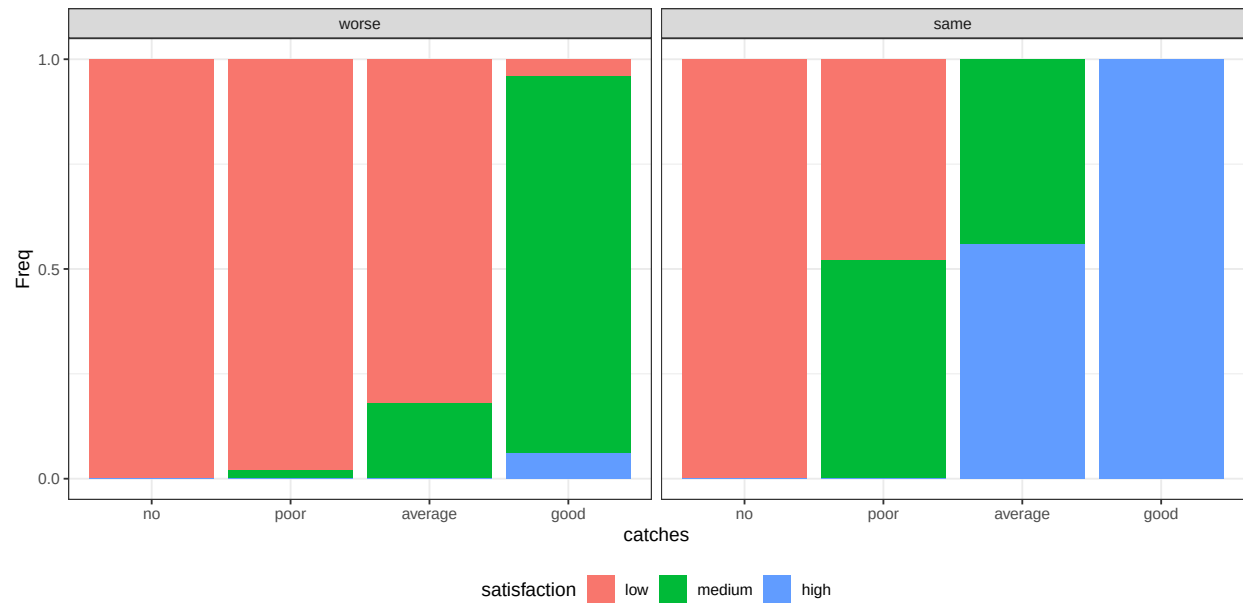


Figure S.26: conditional distribution for satisfaction. Columns represents levels of the node catch_condition.

Substitution capacity

Parents: fishing_style.

Levels: no, yes.

Description: The impact of a lost day is not determined by the loss itself, but by its recoverability. When value is tied to outcomes, time can be lost and later absorbed. When value is tied to time itself, loss becomes irreversible.

Method: Case C1 based on extra question.

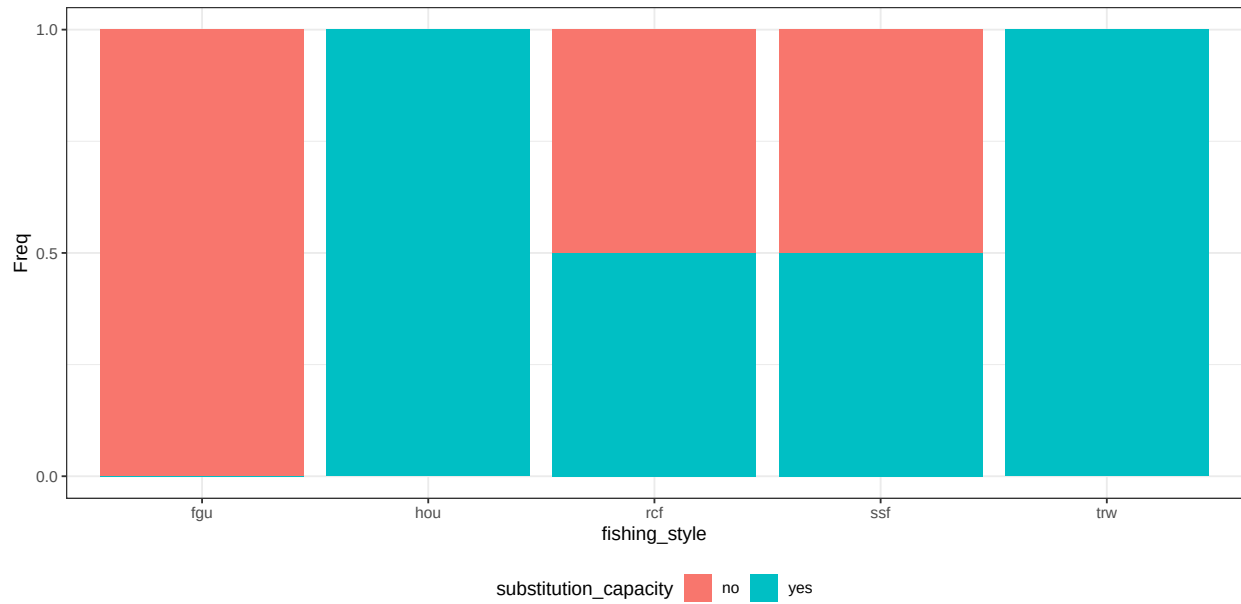


Figure S.27: conditional distribution for substitution_capacity.

Non monetary_value

Parents: health, satisfaction, substitution_capacity, go_fishing.

Levels: low, medium, high.

Description: Summarizes the non-monetary outcomes associated to the fishing experience. It depends on the possibility to go out fishing and to obtain satisfying catches, while staying safe. The wmin algorithm implies that all the conditions are necessary to obtain a large value.

Method: Case C2 with equal weight. Algorithm is wmin.

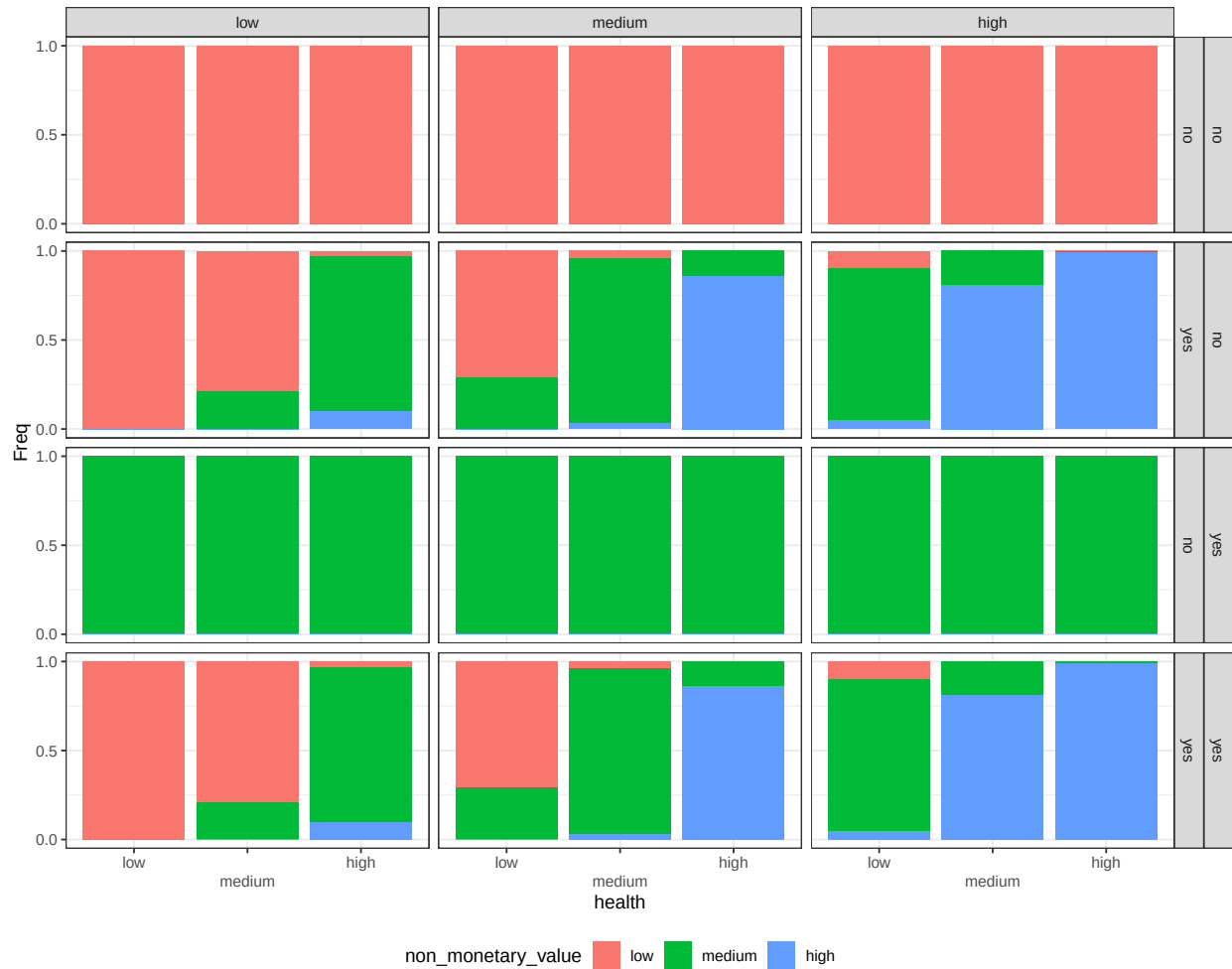


Figure S.28: conditional distribution for non_monetary_value. Columns represents levels of the node satisfaction; rows are combinations levels of the nodes substitution_capacity and go_fishing.

Chance nodes - Economic

Costs

Parents: additional_mitigation, damage, fishing_style.

Levels: no, low, medium, high.

Description: Represent how large is the costs associated with an extreme weather event compared to the annual expected costs. No is $< 0.5\%$; Low is between 0.5% and 2.5% ; Medium is between 2.5% and 5% ; high is larger than 5% .

Method: Case C4. The CPT reflects literature and interviews as described in Annex C..

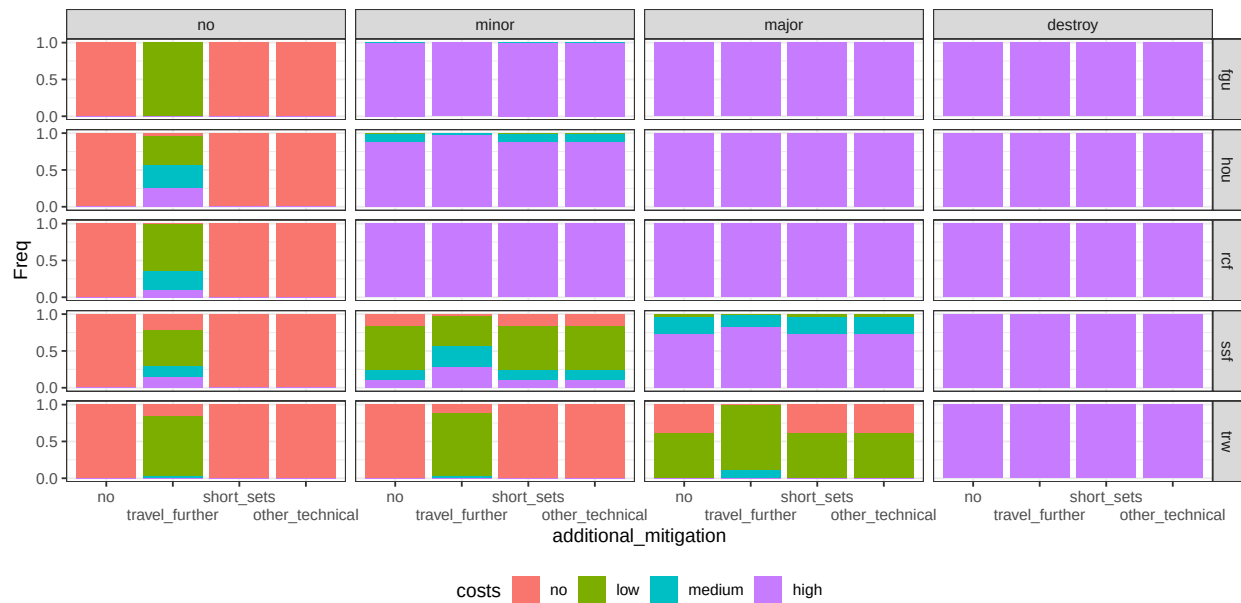


Figure S.29: conditional distribution for costs. Columns represents levels of the node damage; rows are levels of the node fishing_style.

Monetary loss

Parents: substitution_capacity, costs, fishing_style, go_fishing.

Levels: no, low, medium, high.

Description: NA.

Method: Case C3. In the case of recreational fishers, if the node go_out is no then the cost is no. For commercial fishers, if the node go_out is no the cost depend on the estimated cost of a missed fishing day balanced on objectives type. When the node go out is yes, loss equal costs.



Figure S.30: conditional distribution for monetary_loss. Columns represents levels of the node costs; rows are combinations levels of the nodes fishing_style and go_fishing.

Credit

Parents: fishing_style.

Levels: difficult, neutral, easy.

Description: The possibility to obtain loans for investments.

Method: Case C1 based on question Q 25.

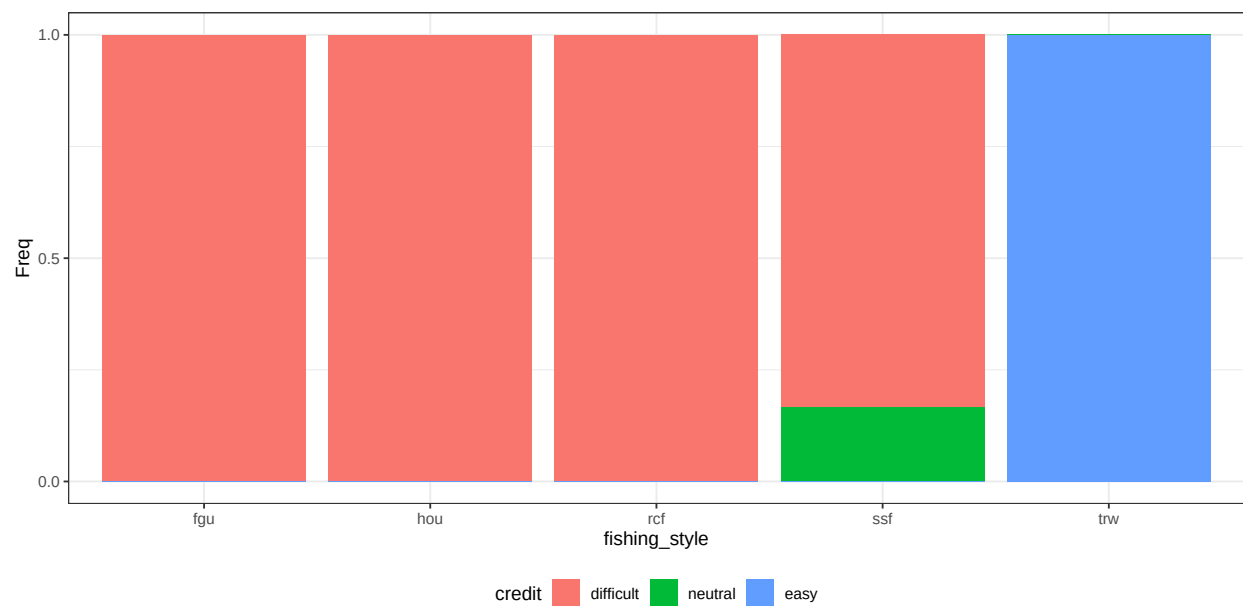


Figure S.31: conditional distribution for credit.

Job mobility

Parents: fishing_style.

Levels: low, medium, high.

Description: The extent to which non-fishing income-generating activities are accessible to an individual or household.

Method: Case C1 based on question Q 23.

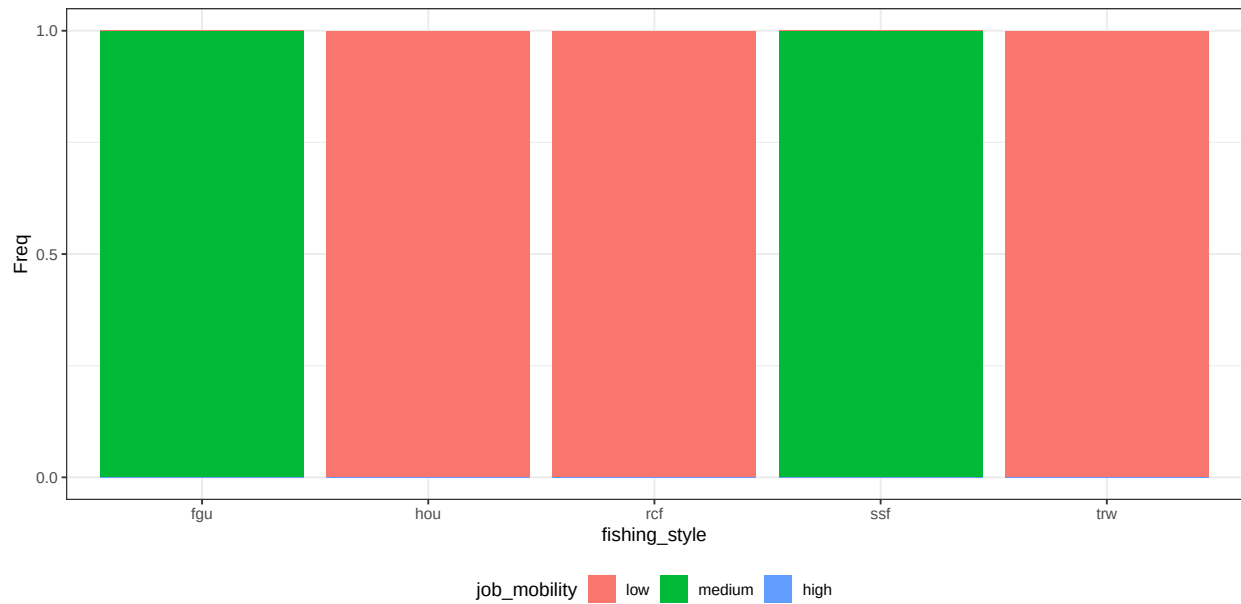


Figure S.32: conditional distribution for job_mobility.

Economic buffers

Parents: job_mobility, credit.

Levels: low, medium, high.

Description: Describes the capability to buffer economic loss by obtaining credits or by replacing with other jobs. Only professionals are capable to obtain economic buffers.

Method: Case C2 with unequal weight, job mobility is 50% less important than credit. Algorithm is wmax.

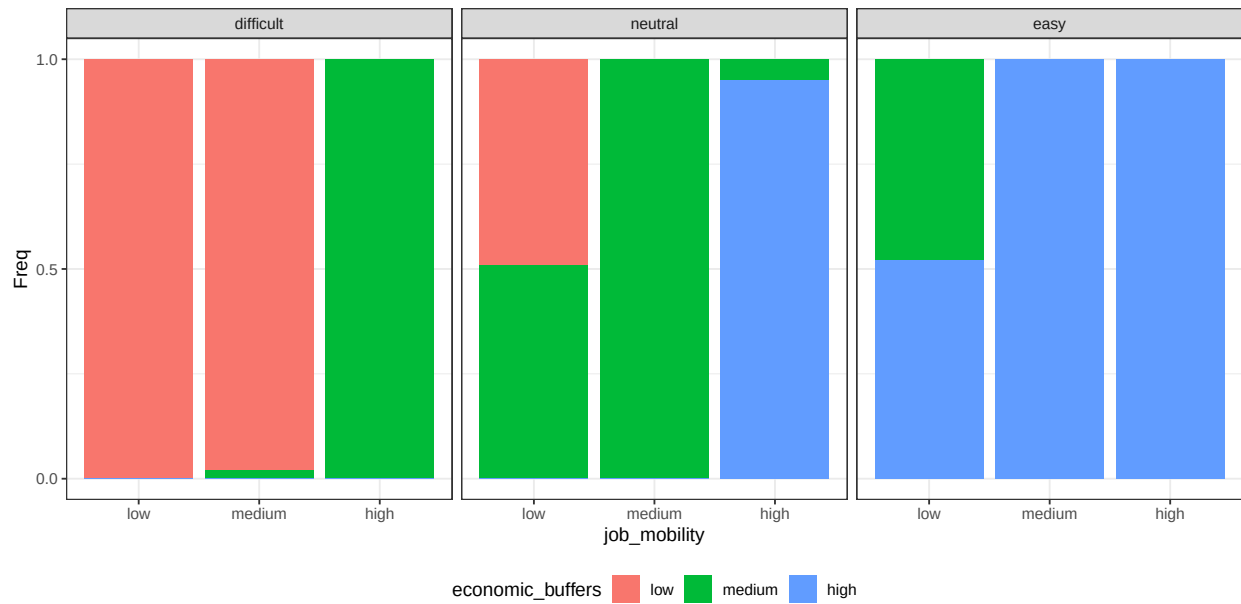


Figure S.33: conditional distribution for economic_buffers. Columns represents levels of the node credit.

Next figure represent the conditional probability distribution of the node economic buffers by fishing style

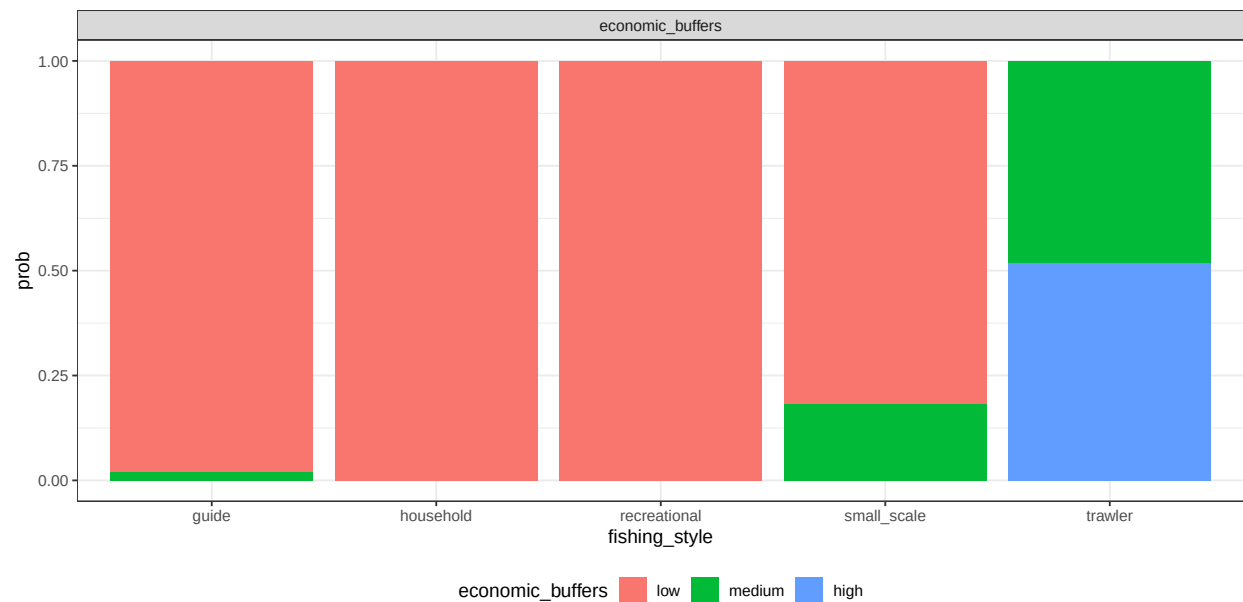


Figure S.34: conditional probability distribution of the node individual importance by fishing style

Chance nodes - Individual

Identity

Parents: fishing_style.

Levels: not_important, somehow_important, very_important.

Description: The degree to which fishing contributes to an individual's sense of identity, including self-perception.

Method: Case C1 based on question Q 33.

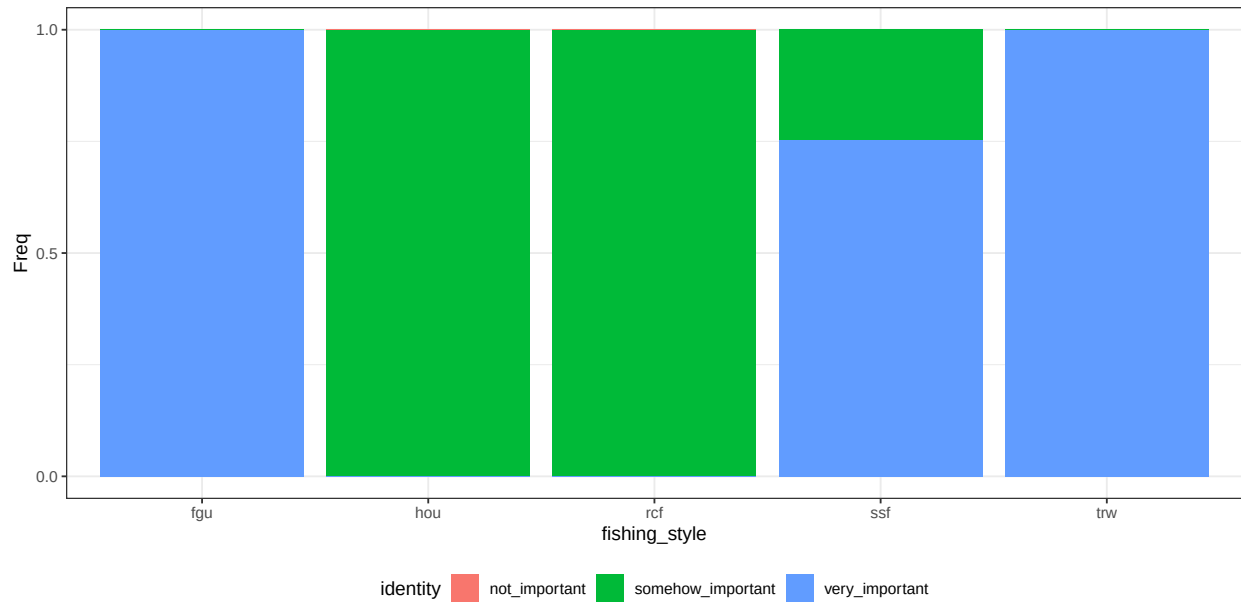


Figure S.35: conditional distribution for identity.

Sense of_home

Parents: fishing_style.

Levels: not_important, somehow_important, very_important.

Description: The degree to which fishing contributes to build a sense of place/home to a specific location. It can either be at sea or at land.

Method: Case C1 based on question Q 34.

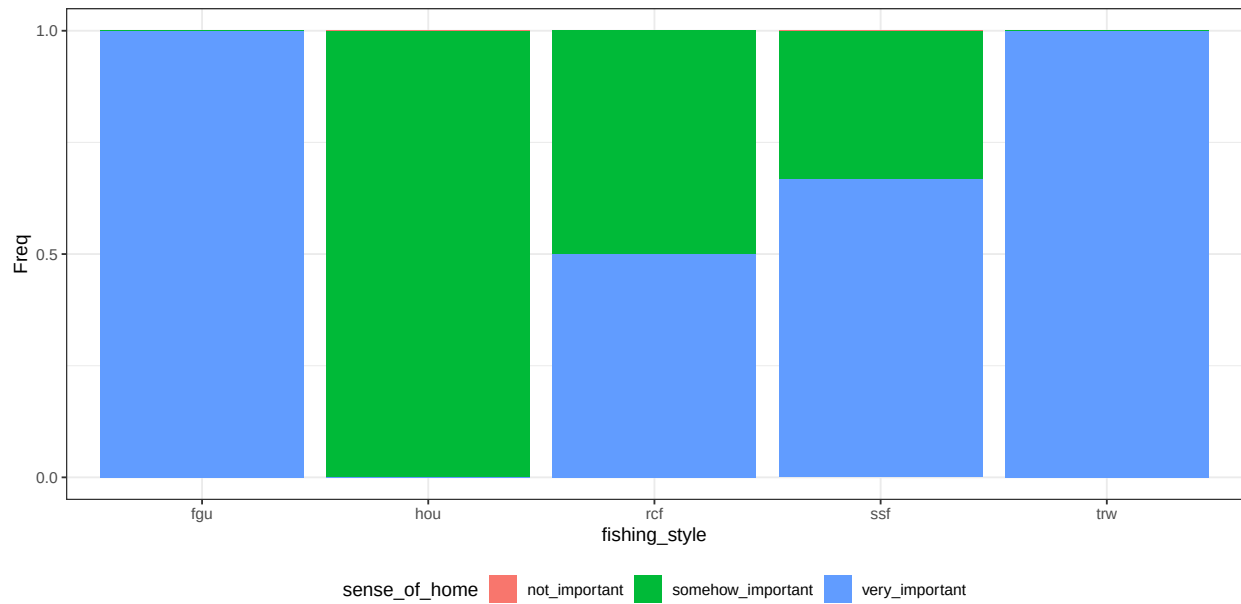


Figure S.36: conditional distribution for sense_of_home.

Outdoor benefit

Parents: fishing_style.

Levels: not_important, somehow_important, very_important.

Description: The extent which an individual value the time spent outdoor (while fishing) to his personal well-being.

Method: Case C1 based on question Q 32.

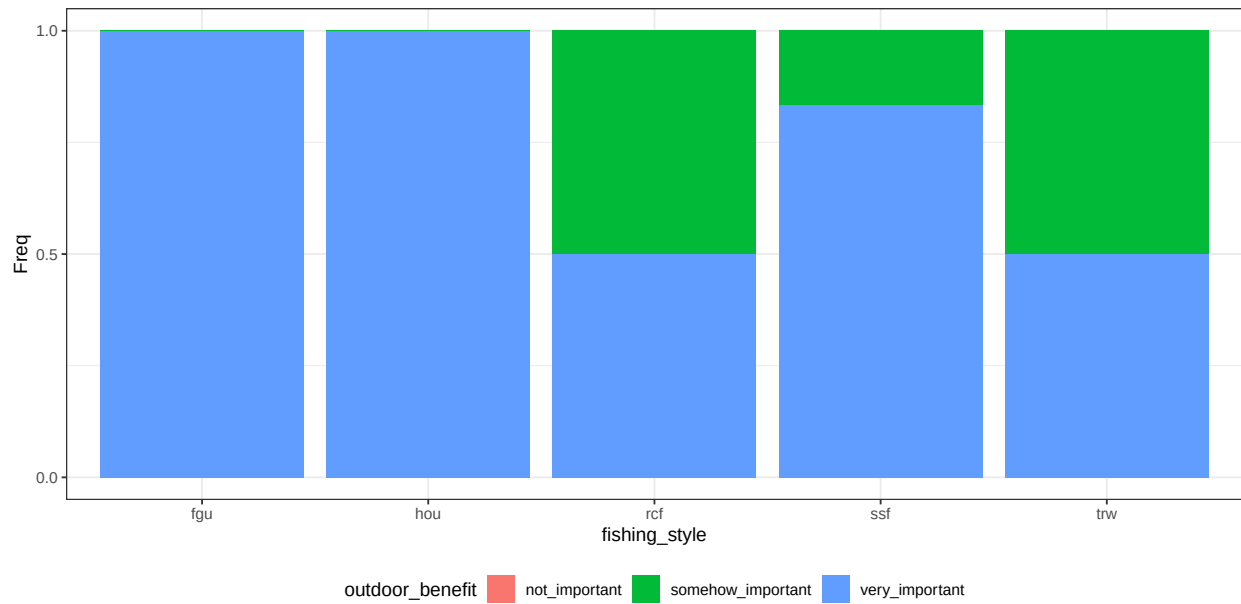


Figure S.37: conditional distribution for outdoor_benefit.

Individual importance

Parents: outdoor_benefit, identity, sense_of_home.

Levels: low, medium, high.

Description: Summarizes the importance of the variables defining individual features to different fishers. The algorithm wmax gives more contribution to large states.

Method: Case C2 with equal weight. Algorithm is wmax.

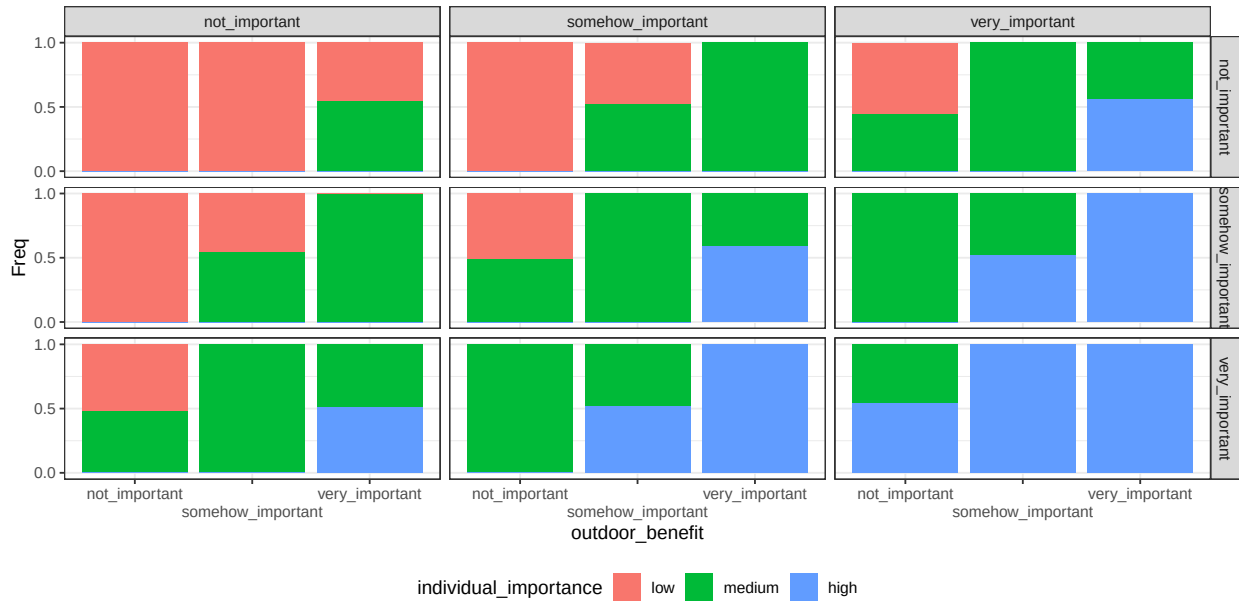


Figure S.38: conditional distribution for individual_importance. Columns represents levels of the node identity; rows are levels of the node sense_of_home.

Next figure represent the conditional probability distribution of the node individual importance by fishing style

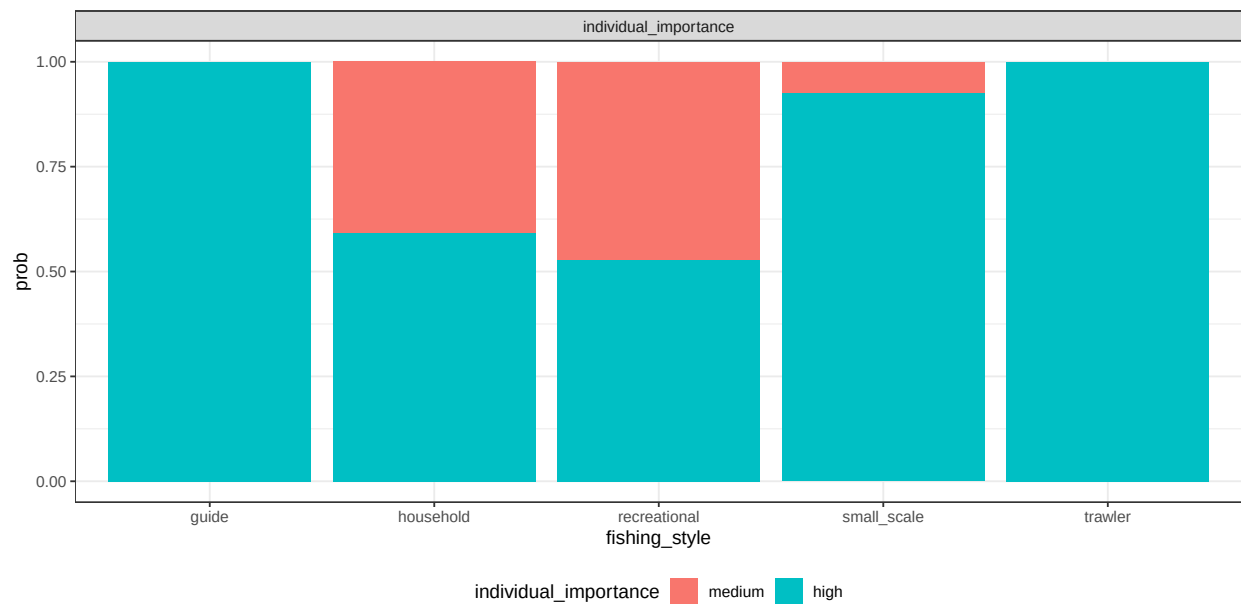


Figure S.39: conditional probability distribution of the node individual importance by fishing style

Chance nodes - Societal

Knowledge transmission

Parents: fishing_style.

Levels: not_important, somehow_important, very_important.

Description: The extent which an individual value the transmission of fishing knowledge and culture to future generations.

Method: Case C1 based on question Q 27.

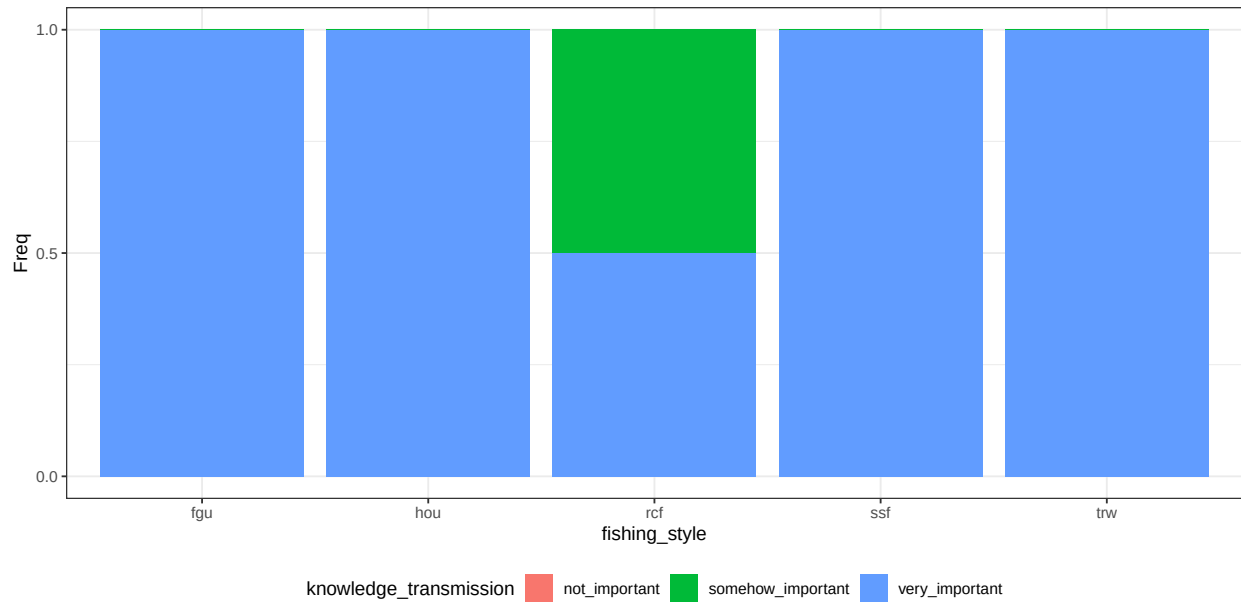


Figure S.40: conditional distribution for knowledge_transmission.

Benefit local_community

Parents: fishing_style.

Levels: not_important, somehow_important, very_important.

Description: The extent which an individual value fishing to build/maintain social relations within an extended local community, such as a town or neighborhood.

Method: Case C1 based on question Q 28.

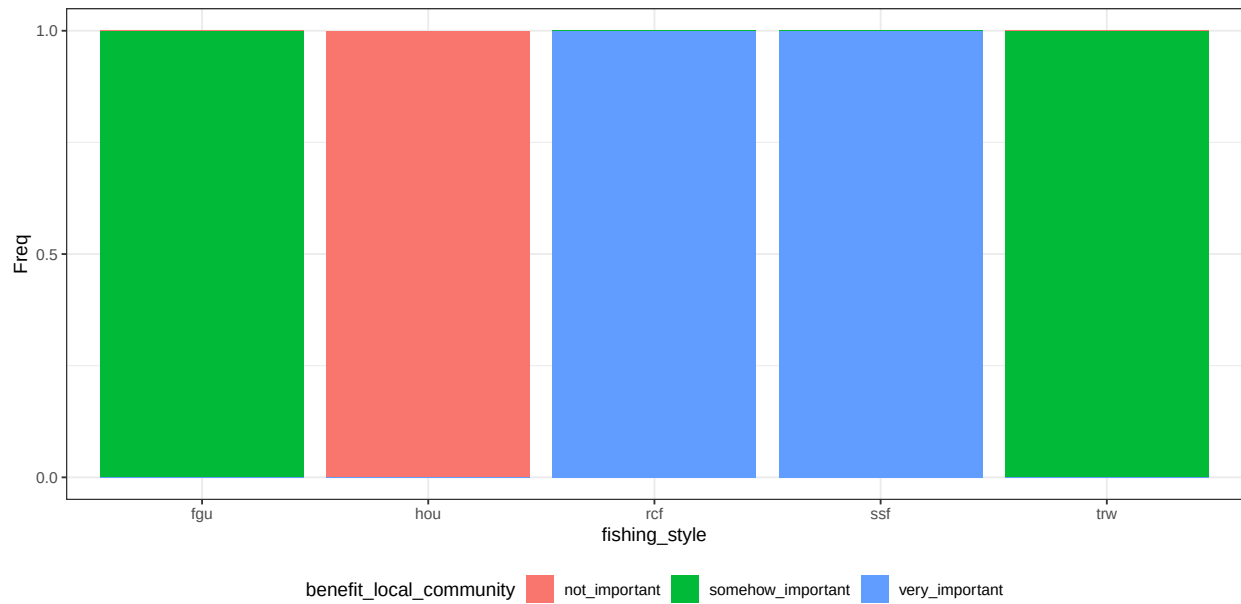


Figure S.41: conditional distribution for benefit_local_community.

Traditional food

Parents: fishing_style.

Levels: not_important, somehow_important, very_important.

Description: The extent which fishing contributes to provide for traditional food.

Method: Case C1 based on question Q 30.

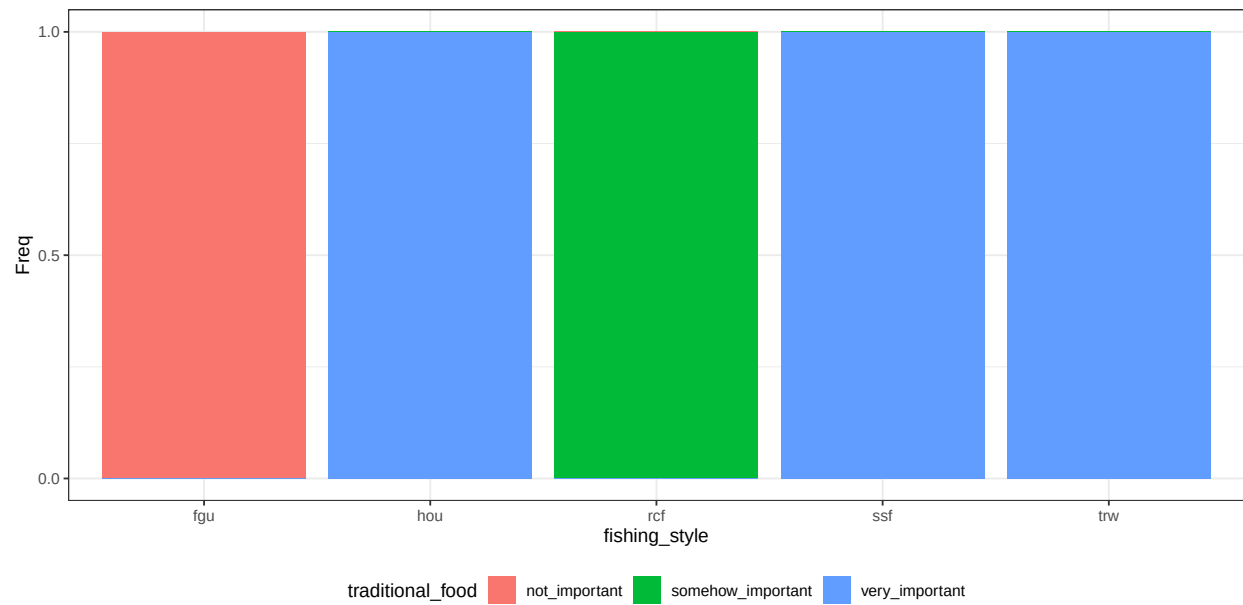


Figure S.42: conditional distribution for traditional_food.

Benefit personal_community

Parents: fishing_style.

Levels: not_important, somehow_important, very_important.

Description: The extent which an individual value fishing to build/maintain social relations within a small community, such as groups of friends or associations.

Method: Case C1 based on question Q 29.

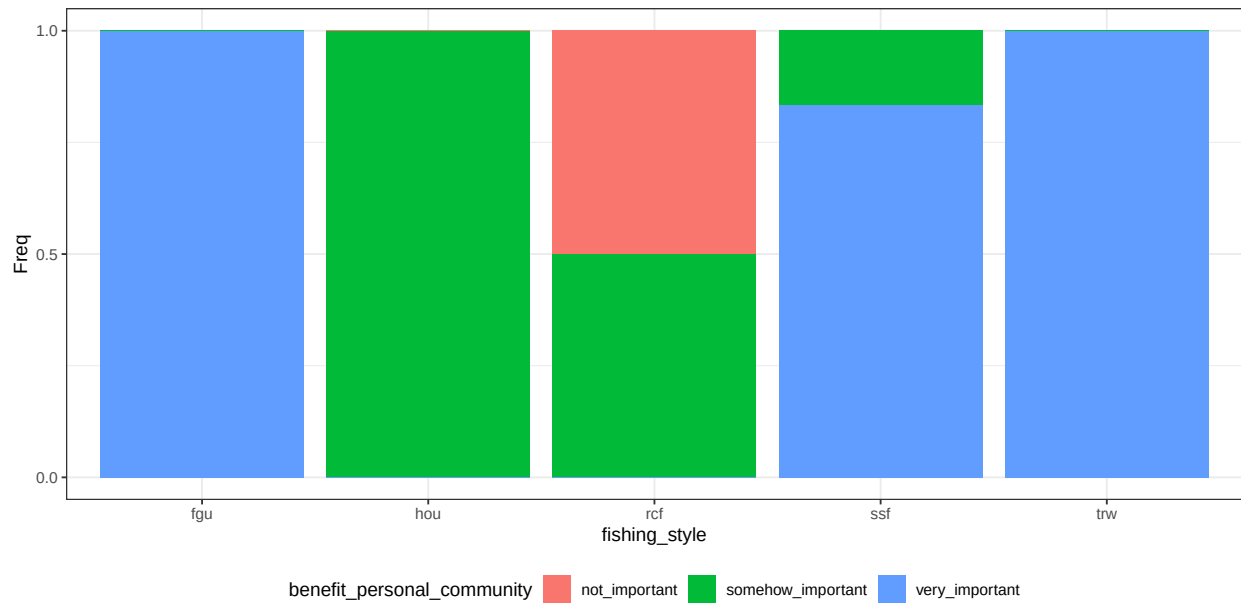


Figure S.43: conditional distribution for benefit_personal_community.

Societal importance

Parents: knowledge_transmission, benefit_local_community, traditional_food, benefit_personal_community.

Levels: low, medium, high.

Description: Summarizes the importance of the variables defining societal features to different fishers. The algorithm wmax gives more contribution to large states.

Method: Case C2 with equal weight. Algorithm is wmax.

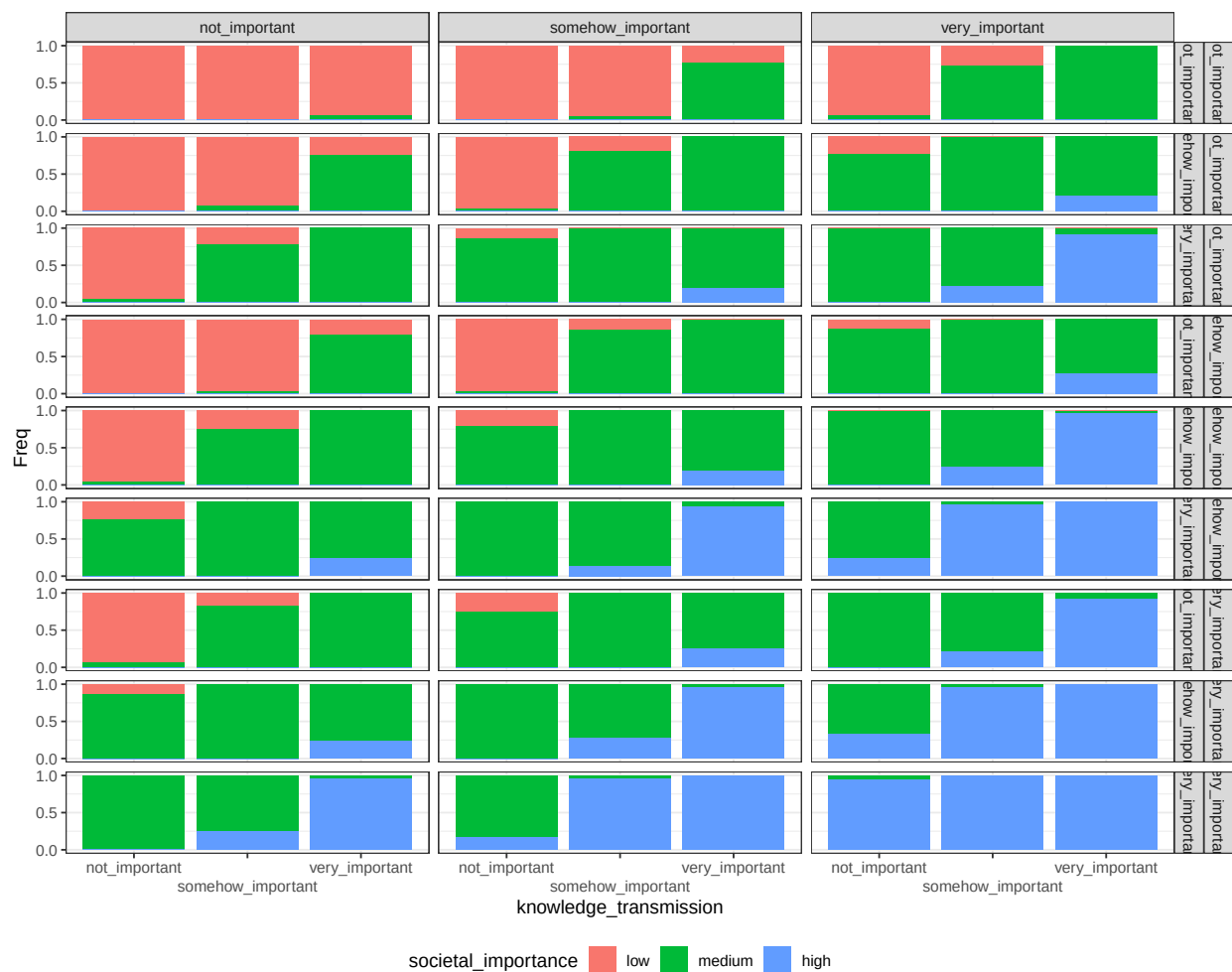


Figure S.44: conditional distribution for societal_importance. Columns represents levels of the node benefit_local_community; rows are combinations levels of the nodes traditional_food and benefit_personal_community.

Next figure represent the conditional probability distribution of the node societal importance by fishing style

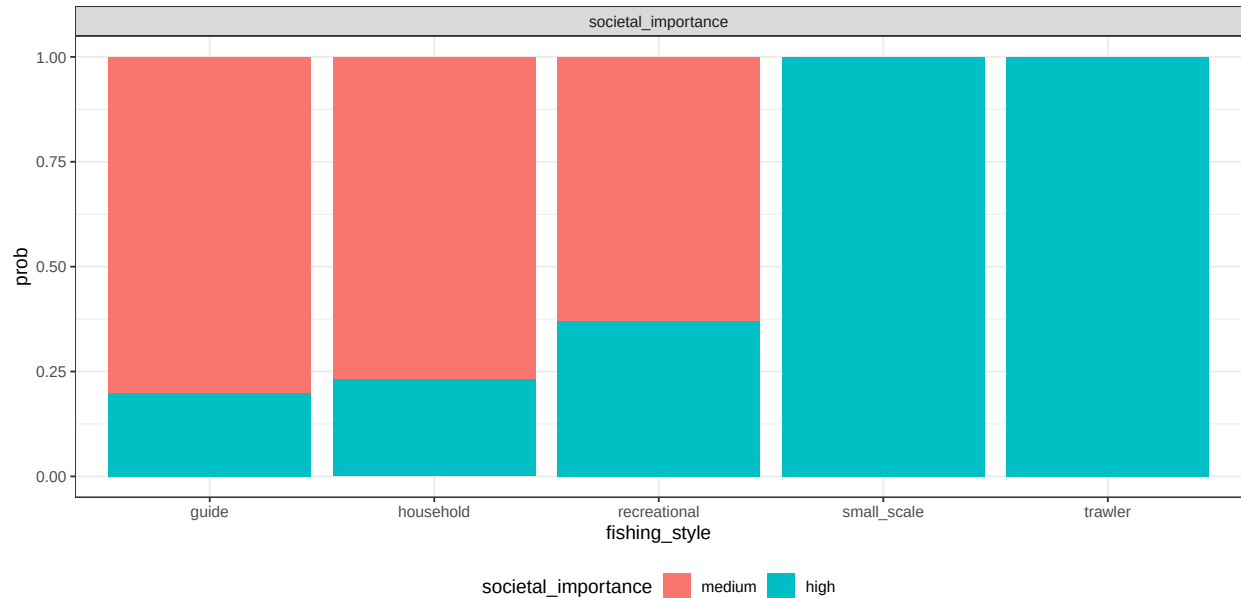


Figure S.45: conditional probability distribution of the node individual importance by fishing style

Risk

Shape how different fisheries will perceive risk. To each of these variables, the effect of the human community variables is to be interpreted as “how much this variable is important”? The CPT table combine the concepts of ranked nodes and the classic definition of $\text{Risk} = \text{Hazard} * \text{Vulnerability}$. In the case of individual and societal importance, Risk is exactly calculated as $\text{Hazard} * \text{Vulnerability}$. Parent categories (Low, medium and high; as well as not important, somehow important and very important) are mapped to 0, 0.5 and 1 values. The parent value are multiplied.

Individual risk

Parents: individual_importance, non_monetary_value, strategy_to_change.

Levels: low, medium, high, not_relevant.

Description: The risk of having a mismatch between the outcomes that fishing provides and the importance that individual well-being receives from fishing.

Method: Case C5.

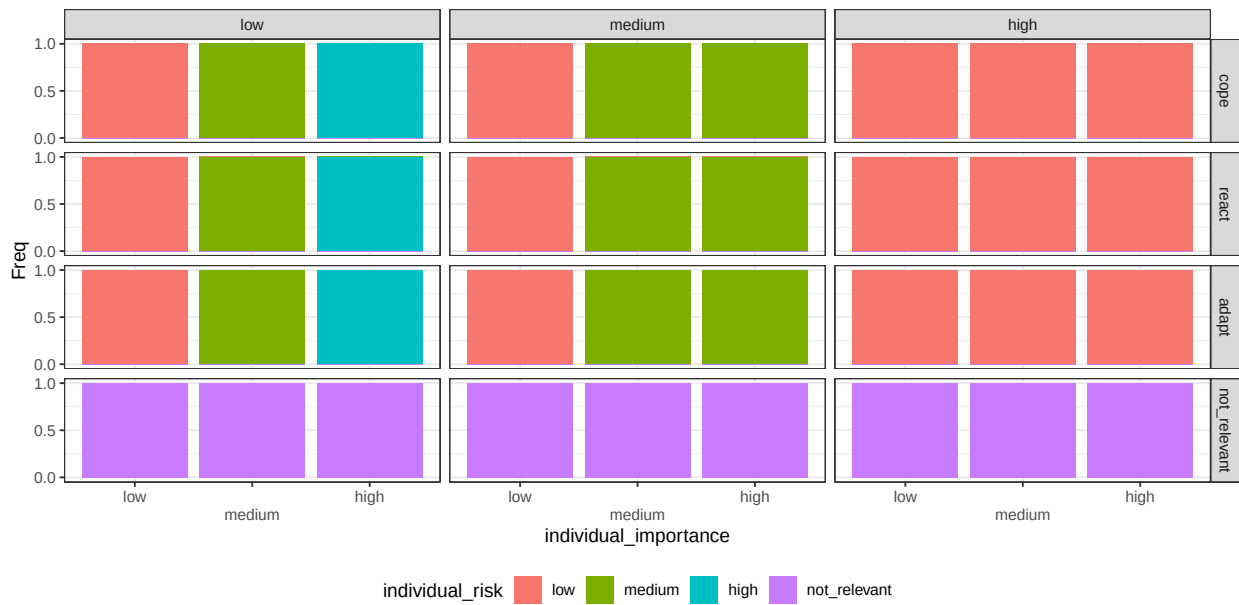


Figure S.46: conditional distribution for individual_risk. Columns represents levels of the node non_monetary_value; rows are levels of the node strategy_to_change.

Societal risk

Parents: societal_importance, non_monetary_value, strategy_to_change.

Levels: low, medium, high, not_relevant.

Description: The risk of having a mismatch between the outcomes that fishing provides and the importance that fishing provides to society.

Method: Case C5.

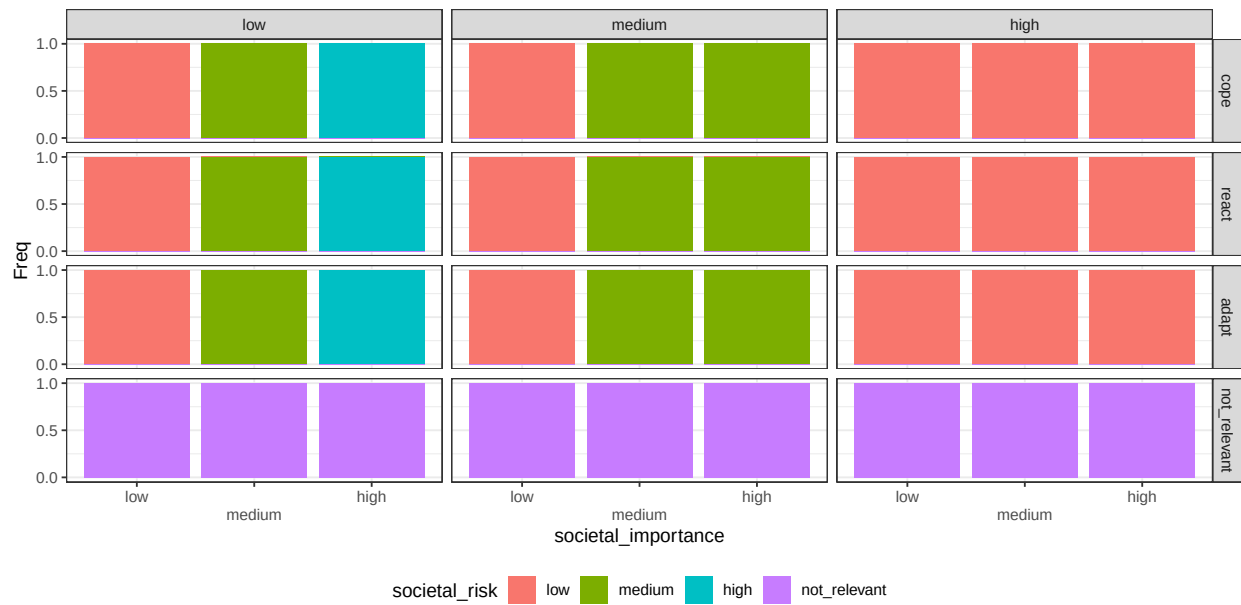


Figure S.47: conditional distribution for societal_risk. Columns represents levels of the node non_monetary_value; rows are levels of the node strategy_to_change.

Economic risk

Parents: economic_buffers, monetary_loss, strategy_to_change.

Levels: low, medium, high, not_relevant.

Description: The risk of having economic fluctuations balanced for how economy is considered an important value.

Method: Case C2. Weight ratio for the parent *importance* and *cost* is 1:2. Algorithm is wmax.

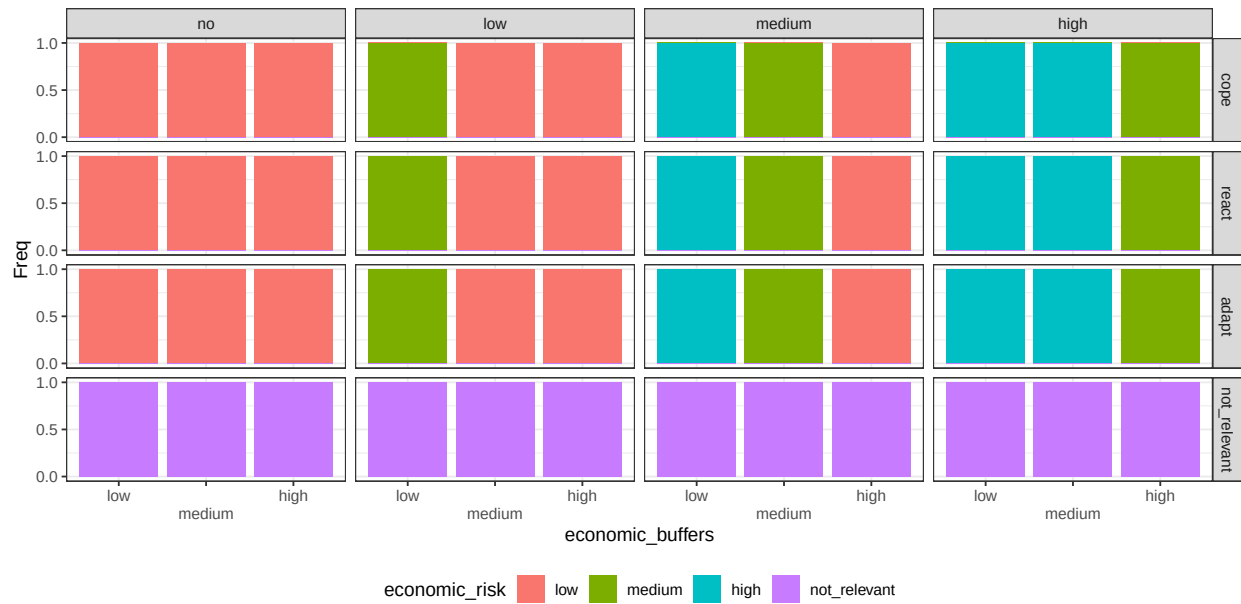


Figure S.48: conditional distribution for economic_risk. Columns represents levels of the node monetary_loss; rows are levels of the node strategy_to_change.